

Engineering Mathematics

Detailed Solutions

PROBABILITY AND STATISTICS

(EC, EE, IN, CS, ME, CE, CH, PI)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Probability and Statistics (EC/EE/IN/CS/ME/CE/CH/PI), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

Himanshu Agarwal

- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience



Anish Saha

- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
- 3+ years in teaching & the VLSI industry



> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Branch Snapshot

A Maths unit can span up to eight branches (EC, EE, IN, CS, ME, CE, CH, PI), each weighted very differently in the real exam. Right after every Unit divider, a **Branch Snapshot** page lists every branch that unit covers, in the same order the solutions themselves follow, each in its own compact box:

Questions	the real total for that branch.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy branch showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Branch Snapshot*

Branch: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real EC questions on this unit; on average EC has drawn about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** for your branch points to a consistently high-yield topic worth prioritizing; a low one is largely someone else's problem. A wide **Range** usually means the topic's importance has shifted over the years, so it is worth checking *which* years drove that before deciding how much weight to give it today.

Important Links & Resources

Visit our Website

prepfusion.in

Solutions, practice & report

pyq.prepfusion.in

Subscribe: English Channel

youtube.com/@PrepFusion_GATE

Subscribe: Hindi Channel

youtube.com/@prepfusion-hindi

Join our Telegram group

t.me/All_About_Learning

Follow us on LinkedIn

linkedin.com/company/prepfusionee

Subscribe: Himanshu Agarwal

youtube.com/@HimanshuAgarwal_

Subscribe: Anish Saha

youtube.com/@AnishSaha_

Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

This book exists because of the people below, who built, reviewed and typeset its content.

Design & Layout Architect — Nawras Ahamed.

Technical Reviewers — Ayush Prakash Singh, Abhishek Bhattacharjee, Shujaut Yaseen, Yuva Kishore Gottapu, Manepalli Pavan Vignesh Kumar, Palak Agarwal, Devmalya Ghatak, Kasi Viswanath A, Suresh Ravanam, Aabhas Anuraag Suman and Piyush Raj.

Content Developers — A S Srimithra, Chinthada Lakshmi Prasad, Gantasala Ravi Teja, Guduru Vinayak Varma, Harshaditya Punera, Katikala Sarath, Pranitha Musunuru, Safdar Nazim, Sai Mahalakshmi Janjanam, Shaila Daniels, Sharvari Patwardhan, Sumedha Ghosh, Vinay K Gowda, Aayushi Ghosh, Chevuri Siva Naga Sai Abhi Chandar, Govind Sharma, Kasina Sai Surya Vamsi, Machanpally Karthik Kumar, Seeku Vamsi Krishna and Suprio Dutta.

Contents

Unit	Topic	Pages
III.	**Probability and Statistics**	**1**
	I. EC	3
	II. EE	37
	III. IN	59
	IV. CS	77
	V. ME	107
	VI. CE	135
	VII. CH	162
	VIII. PI	178

SOLUTIONS — UNIT

III

Probability and Statistics

Topics

1. Basic Probability - Definitions and Axioms
2. Conditional Probability and Baye's Theorem
3. Random Variables - Discrete and Continuous
4. Exception, Variance and Standard Deviation
5. Binomial and Poisson Distribution
6. Uniform and Exponential Distribution
7. Gaussian (Normal) Distribution
8. Multivariable Random Variables
9. Correlation and Regression

Probability and Statistics

Engineering Mathematics

Branches: EC,EE,IN,CS,ME,CE,CH,PI

Branch Snapshots*

EC — Electronics & Communication Engineering

Questions: 53 **MCQ:** 29 **MSQ:** 1 **NAT:** 23
1M: 24 **2M:** 29 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2024)–4M(2013)

EE — Electrical Engineering

Questions: 34 **MCQ:** 23 **MSQ:** 1 **NAT:** 10
1M: 16 **2M:** 18 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2013)–3M(2025)

IN — Instrumentation Engineering

Questions: 33 **MCQ:** 21 **MSQ:** 0 **NAT:** 12
1M: 16 **2M:** 17 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2022)–4M(2018)

CS — Computer Science & IT

Questions: 58 **MCQ:** 31 **MSQ:** 2 **NAT:** 25
1M: 25 **2M:** 33 **Desc:** 0 **Avg Weightage** ~4M **Range:** 1M(2013)–5M(2026)

ME — Mechanical Engineering

Questions: 67 **MCQ:** 50 **MSQ:** 0 **NAT:** 17
1M: 29 **2M:** 38 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2025)–5M(2013)

CE — Civil Engineering

Questions: 58 **MCQ:** 32 **MSQ:** 1 **NAT:** 25
1M: 31 **2M:** 27 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2020)–6M(2016)

CH — Chemical Engineering

Questions: 26 **MCQ:** 18 **MSQ:** 1 **NAT:** 7
1M: 11 **2M:** 15 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2026)–4M(2014)

PI — Production & Industrial Engineering

Questions: 42 **MCQ:** 29 **MSQ:** 0 **NAT:** 13
1M: 20 **2M:** 22 **Desc:** 0 **Avg Weightage** ~3M **Range:** 1M(2025)–5M(2026)

**See preface for how Avg Weightage and Range are calculated.*

SOLUTIONS — Probability and Statistics

EC

Electronics & Communication Engineering

PrepFusion

Q3.26.1 MCQ | 2026 | 2M | Ans: A | ● ● ○

Since $Y = X + N$ and $Z = X^2 + N$,the covariance is evaluated using the independence of X and N. Given:

$$X \sim U(-1, 1), \qquad N \sim \mathcal{N}(0, 1)$$

For a uniform random variable over $(-1, 1)$,

$$E[X] = 0, \qquad E[X^2] = \frac{1}{3}, \qquad E[X^3] = 0$$

Also, for the Gaussian random variable,

$$E[N] = 0$$

Correlation between X and Y:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$\text{Cov}(X, Y) = E[X(X + N)] - 0 \qquad (\because Y = X + N)$$

$$= E[X^2] + E[XN] = \frac{1}{3} + 0$$

$$E[XN] = E[X]E[N] = 0 \qquad (\because X \text{ and } N \text{ are independent})]$$

$$\Rightarrow Cov(X, Y) = \frac{1}{3}$$

Correlation between X and Z:

$$\text{Cov}(X, Z) = E[XZ] - E[X]E[Z]$$

$$\text{Cov}(X, Z) = E[X(X^2 + N)] \qquad (\because Z = X^2 + N)$$

$$= E[X^3] + E[XN] = 0 + 0$$

$$\Rightarrow Cov(X, Z) = 0$$

Here the requested correlations are the unnormalized cross-moments $R_{XY} = E[XY]$ and $R_{XZ} = E[XZ]$. Since $E[X] = 0$, these happen to equal the corresponding covariances, giving option (A): $1/3$ and 0.

$$\text{Var}(X) = \frac{1}{3}, \qquad \text{Var}(Y) = \frac{4}{3}, \qquad \rho_{XY} = \frac{1/3}{\sqrt{(1/3)(4/3)}} = \frac{1}{2}, \qquad \rho_{XZ} = 0.$$

The normalized Pearson coefficient ρ is a different quantity; it must not be confused with the unnormalized correlation used in this question.

$$\boxed{\text{(a) } \frac{1}{3} \text{ and } 0}$$

Key Points

- If two random variables are independent, then $E[XY] = E[X]E[Y]$.
- Odd moments of a symmetric uniform distribution about zero are zero.
- For any continuous uniform distribution across an interval from a to b, the variance ($\text{Var}(X)$) is found using the following formula:$\text{Var}(X) = \frac{(b-a)^2}{12}$

Q3.25.1 MCQ 2025 1M Ans: A ● ○ ○

Two balls are drawn randomly without replacement from a pot containing 2 red balls and 2 blue balls. The probability that the two balls are of different colors is obtained using favorable outcomes divided by total outcomes.

$$\text{Total ways of choosing 2 balls from 4 balls} = {}^4C_2 = \frac{4!}{2!\,2!} = 6$$

To obtain balls of different colors:

$$\text{Choose 1 red ball from 2 red balls} = {}^2C_1 = 2$$

$$\text{Choose 1 blue ball from 2 blue balls} = {}^2C_1 = 2$$

Hence, total favorable cases are

$${}^2C_1 \times {}^2C_1 = 2 \times 2 = 4$$

Therefore,

$$P(\text{different colors}) = \frac{4}{6} = \frac{2}{3}$$

$$\boxed{\text{(a) } \frac{2}{3}}$$

Key Points

- Since drawing is without replacement, combinations are used instead of independent probabilities.

Q3.25.2 NAT 2025 2M Ans: 7.00 ● ○ ○

Let the random variable X denote the sum of the outcomes obtained when two fair dice are rolled. The possible values of X range from 2 to 12. The expectation is given by $E[X] = \sum x\,P(X = x)$
For two fair dice, the probability distribution of the sum is:

x	2	3	4	5	6	7	8	9	10	11	12
$P(X = x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Therefore,

$$E[X] = 2\left(\frac{1}{36}\right) + 3\left(\frac{2}{36}\right) + 4\left(\frac{3}{36}\right) + 5\left(\frac{4}{36}\right) + 6\left(\frac{5}{36}\right) + 7\left(\frac{6}{36}\right) + 8\left(\frac{5}{36}\right) + 9\left(\frac{4}{36}\right) + 10\left(\frac{3}{36}\right) + 11\left(\frac{2}{36}\right) + 12\left(\frac{1}{36}\right)$$

$$E[X] = \frac{2 + 6 + 12 + 20 + 30 + 42 + 40 + 36 + 30 + 22 + 12}{36} = \frac{252}{36} = 7.00$$

$$\boxed{7.00}$$

Mistakes to be avoided

- Do not assume all sums from 2 to 12 are equally likely.

Q3.24.1 NAT 2024 1M Ans: 0.5 ● ○ ○

Let X and Y be independent and identically distributed random variables over the interval $[0, 1]$. The required probability is $P(X \geq Y)$. Since X and Y are i.i.d. and uniformly distributed over the unit square, the region corresponding to $X \geq Y$ occupies exactly half of the square.

Equivalently, $P(X \geq Y) = P(Y \geq X)$ and $P(X = Y) = 0$ for continuous random variables.
Therefore,

$$P(X \geq Y) + P(Y \geq X) = 1$$

$$2P(X \geq Y) = 1$$

$$P(X \geq Y) = \frac{1}{2} = 0.5$$

$$\boxed{0.5}$$

Key Points

- The events $X \geq Y$ and $Y \geq X$ are symmetric.
- Geometrically, the line $X = Y$ divides the unit square into two equal regions.

Q3.22.1 NAT 2022 1M Ans: 4 ● ○ ○

From the given bar graph, the frequencies of wickets taken are:

No. of wickets	0	1	2	3	4	5	6	7	8	9	10
Frequency	5	7	8	25	20	17	8	4	3	2	1

Total number of matches: $5 + 7 + 8 + 25 + 20 + 17 + 8 + 4 + 3 + 2 + 1 = 100$

Since the total number of matches is even, the median is the average number of wickets taken in $100/2 = 50^{\text{th}}$ and $(100/2) + 1 = 51^{\text{th}}$ matches.

$$\text{Median} = \frac{4 + 4}{2} = 4$$

Therefore, the median number of wickets is

$$\boxed{4.0}$$

Q3.21.1 MCQ 2021 1M Ans: B ● ○ ○

Given the relation between the random variables: $Y = 2X + 3$
Using the variance property,

$$\text{Var}(aX + b) = a^2\text{Var}(X)$$

Therefore,

$$\text{Var}(Y) = \text{Var}(2X + 3) = 2^2\ \text{Var}(X) = 4\text{Var}(X)$$

We obtain

$$\sigma_Y^2 = 4\sigma_X^2$$

$$\boxed{\text{(b) } \sigma_Y^2 = 4\ \sigma_X^2}$$

Mistakes to be avoided

- Do not write $\text{Var}(2X) = 2\text{Var}(X)$. The scaling factor must be squared.
- Mean changes with addition of a constant, but variance does not.

Q3.21.2 MCQ | 2021 | 2M | Ans: B | ● ● ○

The three coins are:

- Fair coin (HT)
- Double head coin (HH)
- Double tail coin (TT)

Let

$$H_1 = \text{event that first toss gives head}$$

$$H_2 = \text{event that second toss gives head}$$

We need to find:

$$P(H_2|H_1) = \frac{P(H_1 \cap H_2)}{P(H_1)}$$

Probability of getting head in the first toss:

$$P(H_1) = \left(\frac{1}{3} \times \frac{1}{2}\right) + \left(\frac{1}{3} \times 1\right) + \left(\frac{1}{3} \times 0\right) \left(\because \sum P(Coin) \times P(Head\ for\ the\ coin)\right)$$

$$= \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$$

Now compute $P(H_1 \cap H_2)$.

Case 1: First selected coin is the fair coin (HT)

$$P(H_1) = P(\text{first head using } HT) = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

After removing HT, the remaining coins are HH and TT.

Probability of head in second toss:

$$P(H_2) = \frac{1}{2} \times 1 + \frac{1}{2} \times 0 = \frac{1}{2}$$

$$\Rightarrow P(H_1 \cap H_2 \text{ via } HT) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$$

Case 2: First selected coin is the double head coin (HH)

$$P(H_1) = P(\text{first head using } HH) = \frac{1}{3} \times 1 = \frac{1}{3}$$

After removing HH, the remaining coins are HT and TT.

Probability of head in second toss:

$$P(H_2) = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times 0 = \frac{1}{4}$$

$$\Rightarrow P(H_1 \cap H_2 \text{ via } HH) = \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$$

Thus,

$$P(H_1 \cap H_2) = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}$$

Therefore,

$$P(H_2|H_1) = \frac{1/6}{1/2} = \frac{1}{3}$$

$$\boxed{\text{(b) } \frac{1}{3}}$$

Key Points

- The second toss is performed using a new coin selected from the remaining two coins.

Q3.20.1 NAT | 2020 | 1M | Ans: 0.25 ● ○ ○

A fair coin has sides labelled 0 and 1. The coin is tossed twice independently.
Given:

$$M = \text{outcome of first toss}$$

$$N = \text{outcome of second toss}$$

The random variable is defined as

$$X = \min(M, N)$$

Possible outcomes are:

(M, N)	$X = \min(M, N)$	Probability
$(1, 1)$	1	$\frac{1}{4}$
$(1, 0)$	0	$\frac{1}{4}$
$(0, 1)$	0	$\frac{1}{4}$
$(0, 0)$	0	$\frac{1}{4}$

Hence,

$$P(X = 1) = \frac{1}{4}, \qquad P(X = 0) = \frac{3}{4}$$

The expectation is $E[X] = \sum xP(x)$
Therefore,

$$E[X] = \left(1 \times \frac{1}{4}\right) + \left(0 \times \frac{3}{4}\right) = \frac{1}{4} = 0.25$$

$$\boxed{0.25}$$

Q3.20.2 NAT | 2020 | 2M | Ans: 0.300 ● ● ○

The random variable X is uniformly distributed over the interval $[-2, 10]$. The probability density function is

$$f_X(x) = k, \qquad -2 \le x \le 10$$

Using the normalization condition,

$$\int_{-2}^{10} f_X(x)\, dx = 1$$

$$k(10 - (-2)) = 1 \Rightarrow\ 12k = 1 \Rightarrow\ k = \frac{1}{12}$$

$$\Rightarrow\ f_X(x) = \frac{1}{12}, \qquad -2 \le x \le 10$$

Given, $Y = 2X - 6$
We need to evaluate $P(Y \le 7 \mid X \ge 5)$
First, simplify the condition on Y:

$$2X - 6 \le 7 \Rightarrow\ 2X \le 13 \Rightarrow\ X \le 6.5$$

Therefore, using conditional probability,

$$P(Y \le 7 \mid X \ge 5) = P(X \le 6.5 \mid X \ge 5) = \frac{P(5 \le X \le 6.5)}{P(X \ge 5)}$$

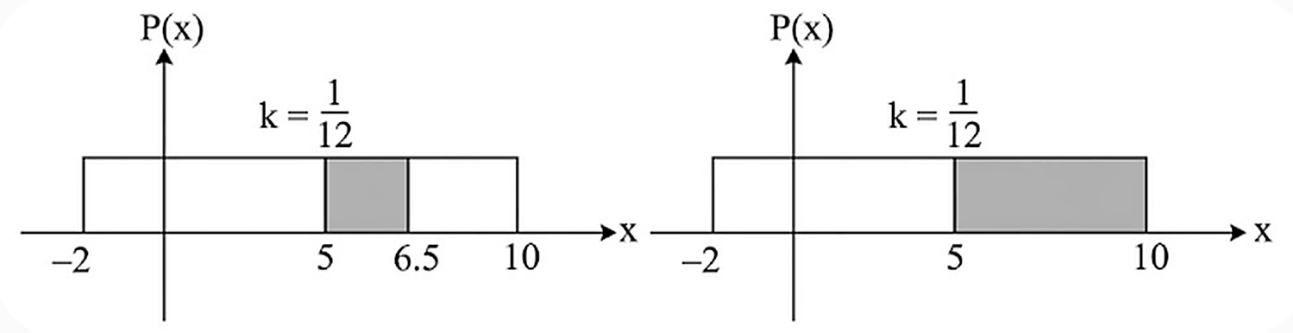


Fig. Solution Q3.20.2

Since the distribution is uniform,

$$P(5 \le X \le 6.5) = \frac{6.5 - 5}{10 - (-2)} = \frac{1.5}{12},$$

$$P(X \ge 5) = \frac{10 - 5}{12} = \frac{5}{12}$$

Thus,

$$P(Y \le 7 \mid X \ge 5) = \frac{1.5/12}{5/12} = \frac{1.5}{5} = 0.3$$

Rounded to three decimal places,

$$0.300$$

$$\boxed{0.300}$$

Key Points

- For a uniform distribution over $[a, b]$:,

$$f_X(x) = \frac{1}{b - a}$$

Q3.19.1 NAT 2019 1M Ans: 11 ● ○ ○

Given:

$$E(2X + Y) = 0 \text{ and } E(X + 2Y) = 33$$

Using the linearity property of expectation,

$$E(aX + bY) = aE(X) + bE(Y)$$

$$2E(X) + E(Y) = 0 \quad (1)$$

$$E(X) + 2E(Y) = 33 \quad (2)$$

Adding equations (1) and (2),

$$3E(X) + 3E(Y) = 33 \Rightarrow 3\big(E(X) + E(Y)\big) = 33$$

$$E(X) + E(Y) = 11$$

$$\boxed{11}$$

Q3.19.2 NAT 2019 1M Ans: 0.37 ● ● ○

The cumulative distribution function of the exponential random variable Z is

$$F_Z(x) = \begin{cases} 1 - e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Hence, the probability density function is

$$f_Z(x) = e^{-x}, \qquad x \geq 0$$

We need to evaluate $P(Z > 2 \mid Z > 1)$
Since $Z > 2$ automatically implies $Z > 1$,

$$P(Z > 2 \mid Z > 1) = \frac{P(Z > 2)}{P(Z > 1)}$$

Now,

$$P(Z > a) = \int_a^{\infty} e^{-z}\, dz$$

Therefore,

$$P(Z > 2) = \int_2^{\infty} e^{-z}\, dz = [-e^{-z}]_2^{\infty} = e^{-2}$$

Similarly,

$$P(Z > 1) = \int_1^{\infty} e^{-z}\, dz = [-e^{-z}]_1^{\infty} = e^{-1}$$

Hence,

$$P(Z > 2 \mid Z > 1) = \frac{e^{-2}}{e^{-1}} = e^{-1} \approx 0.3679$$

Rounded to two decimal places,

$$0.37$$

$$\boxed{0.37}$$

Q3.18.1 NAT 2018 1M Ans: 0.25 ● ○ ○

Since X_1, X_2, X_3, and X_4 are independent and identically distributed random variables, each variable is equally likely to be the smallest.
Thus,

$$P(X_1 \text{ is smallest}) = P(X_2 \text{ is smallest}) = P(X_3 \text{ is smallest}) = P(X_4 \text{ is smallest})$$

The sum of these probabilities is

$$P(X_1 \text{ smallest}) + P(X_2 \text{ smallest}) + P(X_3 \text{ smallest}) + P(X_4 \text{ smallest}) = 1$$

$$4P(X_4 \text{ is smallest}) = 1$$

$$P(X_4 \text{ is smallest}) = \frac{1}{4} = 0.25$$

$$\boxed{0.25}$$

Q3.17.1 NAT 2017 1M Ans: 0.028 ● ○ ○

When three fair dice are thrown simultaneously, each die has 6 possible outcomes. Hence, the total number of outcomes is $6^3 = 216$

For all three dice to show the same number, the favorable outcomes are:

$$(1,1,1),\ (2,2,2),\ (3,3,3),\ (4,4,4),\ (5,5,5),\ (6,6,6)$$

Thus, the number of favorable outcomes is 6 Therefore, the required probability is

$$P = \frac{6}{216} = \frac{1}{36} = 0.027777\ldots$$

Rounded to three decimal places,

$$0.028$$

$$\boxed{0.028}$$

Q3.17.2 NAT 2017 2M Ans: 2.5 ● ● ○

Let the probability of getting a reservation in a single attempt be $p = 0.4$

Hence, the probability of not getting a reservation is $q = 1 - p = 0.6$

Let the random variable X denote the number of attempts required to get the first reservation.

Then X follows a geometric distribution with probability mass function

$$P(X = x) = q^{x-1}p, \qquad x = 1, 2, 3, \ldots$$

The probability distribution of X is:

x	1	2	3	4	$\cdots$
$P(X = x)$	p	qp	q^2p	q^3p	$\cdots$

The average number of attempts that passengers need to get a seat reserved is

$$E[X] = \sum_{x=1}^{\infty} xP(X = x)$$

$$E[X] = p + 2qp + 3q^2p + 4q^3p + \cdots$$

$$= p\left(1 + 2q + 3q^2 + 4q^3 + \cdots\right)$$

Hence,

$$E[X] = p\left(\frac{1}{(1-q)^2}\right) = \frac{p}{p^2} = \frac{1}{p} \qquad (\because \text{Using the standard result})$$

Therefore,

$$E[X] = \frac{1}{0.4} = 2.5$$

$$\boxed{2.5}$$

Key Points

- Number of trials until first success follows a geometric distribution.
- Here, "success" corresponds to getting a reservation.

Q3.16.1 NAT 2016 1M Ans: 1 ● ○ ○

For a Poisson-distributed random variable with parameter λ,

$$\text{Mean} = \lambda$$

and the second moment is

$$E[X^2] = \lambda^2 + \lambda$$

Given that the second moment is

$$E[X^2] = 2$$

$$\Rightarrow \lambda^2 + \lambda = 2$$

$$\lambda^2 + \lambda - 2 = 0$$

$$(\lambda + 2)(\lambda - 1) = 0$$

$$\lambda = 1 \quad \text{or} \quad \lambda = -2$$

Since the parameter of a Poisson distribution cannot be negative, the mean of the random variable,

$$\lambda = 1$$

$$\boxed{1}$$

Mistakes to be avoided

- Do not confuse variance with second moment.

Q3.16.2 NAT 2016 2M Ans: 0.33 ● ● ○

The joint probability density function is given by

$$f_{X,Y}(x, y) = \begin{cases} x + y, & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

We need to find

$$P(X + Y \le 1)$$

The condition $X + Y \le 1$ represents the triangular region bounded by

$$x = 0, \qquad y = 0, \qquad x + y = 1$$

Therefore,

$$P(X + Y \le 1) = \int_0^1 \int_0^{1-x} (x + y)\, dy\, dx$$

First, integrate with respect to y:

$$= \int_0^1 \left[xy + \frac{y^2}{2} \right]_0^{1-x} dx$$

$$= \int_0^1 \left[x(1 - x) + \frac{(1 - x)^2}{2} \right] dx$$

Simplifying,

$$= \int_0^1 \left[x - x^2 + \frac{1 - 2x + x^2}{2} \right] dx$$

$$= \int_0^1 \left[\frac{1}{2} - \frac{x^2}{2} \right] dx$$

Now integrate with respect to x:

$$= \left[\frac{x}{2} - \frac{x^3}{6} \right]_0^1$$

$$= \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$$

Therefore,

$$P(X + Y \leq 1) = \frac{1}{3}$$

$$\boxed{\frac{1}{3}}$$

Key Points

- For continuous random variables,
$$P(A) = \iint_A f_{X,Y}(x, y)\, dx\, dy$$
- The condition $X + Y \leq 1$ forms a triangular region in the xy-plane.
- Limits of integration must match the geometry of the region carefully.

Mistakes to be avoided

- Do not integrate over the entire square $0 \leq x, y \leq 1$.
- The upper limit for y is $1 - x$, not 1.
- Ensure correct order of integration while evaluating double integrals.

Q3.16.3 NAT | 2016 | 2M | Ans: 0.072 | ● ○ ○

The probability of getting a head in a single toss is

$$P(H) = 0.3$$

Hence, the probability of getting a tail is

$$P(T) = 1 - 0.3 = 0.7$$

For getting the first head on the fifth toss, the first four tosses must be tails and the fifth toss must be a head.

$$P = (0.7)^4 \times 0.3 = 0.07203$$

$$\boxed{0.07203}$$

Key Points

- For a geometric probability distribution,
$$P(\text{first success on } n^{\text{th}} \text{ trial}) = (1 - p)^{n-1} p$$

Q3.15.1 MCQ | 2015 | 1M | Ans: C | ● ○ ○

Since A and B are independent events,

$$P(A \cap B) = P(A)P(B)$$

Hence, option A is true.
Also,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Using independence,

$$P(A|B) = \frac{P(A)P(B)}{P(B)} = P(A)$$

So, option B is also true.
For independent events,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since $P(A \cap B) = P(A)P(B) \neq 0$,

$$P(A \cup B) \neq P(A) + P(B)$$

Therefore, option C is false.
Further, complements of independent events are also independent:

$$P(\overline{A} \cap \overline{B}) = P(\overline{A})P(\overline{B})$$

So, option D is true.

$$\boxed{\text{(c) } P(A \cup B) = P(A) + P(B)}$$

Q3.15.2 NAT | 2015 | 2M | Ans: 6 | ● ● ●

Let E be the expected number of tosses required to get two consecutive heads (HH) from the very beginning.
We can analyze the expected value by conditioning on the outcomes of the first few tosses:

1. **First toss is Tails (T):** This happens with a probability of $\frac{1}{2}$. Since getting a Tails resets all progress, we have wasted 1 toss and are back to square one. The expected number of remaining tosses is still E.

   $$\text{Total expected tosses for this path} = 1 + E$$

2. **First toss is Heads (H) and Second toss is Heads (HH):** This happens with a probability of $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. The condition is satisfied, and the game ends immediately.

   $$\text{Total tosses for this path} = 2$$

3. **First toss is Heads (H) and Second toss is Tails (HT):** This happens with a probability of $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. We have consumed 2 tosses, but the streak is broken by the Tails, resetting progress completely back to the start.

   $$\text{Total expected tosses for this path} = 2 + E$$

Using the law of total expectation, we can set up the algebraic equation:

$$E = \frac{1}{2}(1 + E) + \frac{1}{4}(2) + \frac{1}{4}(2 + E)$$

Expanding the terms:

$$E = \frac{1}{2} + \frac{1}{2}E + \frac{1}{2} + \frac{1}{2} + \frac{1}{4}E$$

Combining the constants and the coefficients of E:

$$E = \frac{3}{2} + \left(\frac{1}{2} + \frac{1}{4}\right)E$$

$$E = \frac{3}{2} + \frac{3}{4}E$$

Subtracting $\frac{3}{4}E$ from both sides:

$$E - \frac{3}{4}E = \frac{3}{2}$$

$$\frac{1}{4}E = \frac{3}{2}$$

Solving for E:

$$E = \frac{3}{2} \times 4 = 6$$

$$\boxed{6}$$

Q3.15.3 MCQ | 2015 | 2M | Ans: D ● ○ ○

We need to find

$$P(X + Y \geq 1)$$

Since $X, Y \in \{0, 1\}$, the condition $X + Y \geq 1$ means that at least one of the variables is 1.
Using complement rule,

$$P(X + Y \geq 1) = 1 - P(X = 0, Y = 0)$$

Given,

$$P(X = 0) = p, \qquad P(Y = 0) = q$$

Since X and Y are independent,

$$P(X = 0, Y = 0) = P(X = 0)\, P(Y = 0) = pq$$

Therefore,

$$P(X + Y \geq 1) = 1 - pq$$

$$\boxed{\text{(d) } 1 - pq}$$

Q3.15.4 NAT 2015 2M Ans: 6 ● ○ ○

Let X be the number of throws required to get the face 3 for the first time.
The probability of getting 3 in a single throw is

$$p = \frac{1}{6}$$

Hence, X follows a geometric distribution.
For a geometric distribution,

$$E(X) = \frac{1}{p}$$

Substituting $p = \frac{1}{6}$,

$$E(X) = \frac{1}{\frac{1}{6}} = 6$$

$$\boxed{6}$$

Q3.15.5 NAT 2015 2M Ans: 6 ● ○ ○

Given probability density function,

$$f(x) = \frac{1}{2}|x|e^{-|x|}$$

Since $f(x)$ is an even function,

$$E(X) = 0$$

Now,

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Thus,

$$\text{Var}(X) = E(X^2)$$

Compute $E(X^2)$:

$$E(X^2) = \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{2}|x|e^{-|x|}\,dx$$

Using symmetry,

$$E(X^2) = 2\int_{0}^{\infty} x^2 \cdot \frac{1}{2}xe^{-x}\,dx$$

$$E(X^2) = \int_{0}^{\infty} x^3 e^{-x}\,dx$$

Using the standard result,

$$\int_{0}^{\infty} x^n e^{-x}\,dx = n!$$

for $n = 3$,

$$E(X^2) = 3! = 6$$

Hence,

$$\text{Var}(X) = 6$$

$$\boxed{6}$$

Key Points

- If the pdf is even, then the mean is zero.
- Standard Gamma integral:

$$\int_0^\infty x^n e^{-x}\,dx = n!$$

Q3.14.1 NAT | 2014 | 1M | Ans: 0.67 | ● ○ ○

Assume there are N families.
Then,

- $\frac{N}{2}$ families have one child each,
- $\frac{N}{2}$ families have two children each.

Total number of children:

$$\frac{N}{2} \times 1 + \frac{N}{2} \times 2$$

$$= \frac{N}{2} + N = \frac{3N}{2}$$

Children having a sibling come only from two-child families.
Number of such children:

$$\frac{N}{2} \times 2 = N$$

Hence, required probability is

$$P = \frac{N}{\frac{3N}{2}}$$

$$P = \frac{2}{3}$$

$$\boxed{0.67}$$

Q3.14.2 MCQ | 2014 | 2M | Ans: B | ● ○ ○

For any random variable X,

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

Since variance is always non-negative,

$$\text{Var}(X) \geq 0$$

Therefore,

$$E(X^2) - (E(X))^2 \geq 0$$

$$E(X^2) \geq (E(X))^2$$

$$\boxed{\text{(b) } E[X^2] \geq (E[X])^2}$$

Q3.14.3 NAT | 2014 | 1M | Ans: 50 | ● ○ ○

The positive numbers less than 100 are $1, 2, 3, \ldots, 99$
Since X is uniformly chosen, each number has probability $\frac{1}{99}$
Hence,

$$E(X) = \frac{1}{99}(1 + 2 + 3 + \cdots + 99)$$

Using the sum of first n natural numbers,

$$1 + 2 + \cdots + 99 = \frac{99 \times 100}{2}$$

Therefore,

$$E(X) = \frac{1}{99} \times \frac{99 \times 100}{2}$$

$$E(X) = 50$$

$$\boxed{50}$$

Key Points

- Sum of first n natural numbers:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

Mistakes to be avoided

- Positive numbers less than 100 are from 1 to 99, not 0 to 99.

Q3.14.4 MCQ | 2014 | 1M | Ans: C | ● ○ ○

For the fourth head to appear on the 10th toss:

- Exactly 3 heads must occur in the first 9 tosses.
- The 10th toss must be a head.

Using binomial probability for the first 9 tosses,

$$P(\text{exactly 3 heads in 9 tosses}) = {}^9C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^6$$

$$= {}^9C_3 \left(\frac{1}{2}\right)^9$$

Now multiply by the probability that the 10th toss is a head:

$$P = {}^{9}C_3 \left(\frac{1}{2}\right)^9 \left(\frac{1}{2}\right)$$

$$= {}^{9}C_3 \left(\frac{1}{2}\right)^{10}$$

$$= \frac{84}{1024} = 0.082$$

$$\boxed{\text{(c) } 0.082}$$

Mistakes to be avoided

- Do not use ${}^{10}C_4$ because the 10th toss must definitely be a head.

Q3.14.5 NAT 2014 2M Ans: 3 ● ○ ○

At least two tosses are required for both head and tail to appear. After the first toss, each subsequent toss has probability, P= $\frac{1}{2}$ of giving the opposite outcome and stopping the process.
Thus, the number of additional tosses required after the first toss follows a geometric distribution with

$$p = \frac{1}{2}$$

Hence,

$$E(\text{additional tosses}) = \frac{1}{p} = 2$$

Including the first toss,

$$E(X) = 1 + 2 = 3$$

$$\boxed{3}$$

Q3.14.6 NAT 2014 2M Ans: 0.166 ● ● ○

Since X_1, X_2, X_3 are independent and uniformly distributed on $[0, 1]$,

$$f(x_1, x_2, x_3) = 1$$

over the unit cube.
We need to find

$$P(X_1 + X_2 \leq X_3)$$

For fixed x_1 and x_2,

$$P(X_3 \geq x_1 + x_2) = 1 - (x_1 + x_2)$$

valid when

$$x_1 + x_2 \leq 1$$

Hence,

$$P = \int_0^1 \int_0^{1-x_1} (1 - x_1 - x_2)\, dx_2\, dx_1$$

First integrating with respect to x_2,

$$P = \int_0^1 \left[x_2 - (x_1 x_2) - \frac{x_2^2}{2} \right]_0^{1-x_1} dx_1$$

$$P = \int_0^1 \frac{(1 - x_1)^2}{2}\, dx_1$$

$$P = \frac{1}{2} \int_0^1 (1 - 2x_1 + x_1^2)\, dx_1$$

$$P = \frac{1}{2} \left[x_1 - x_1^2 + \frac{x_1^3}{3} \right]_0^1$$

$$P = \frac{1}{2} \left(1 - 1 + \frac{1}{3} \right)$$

$$P = \frac{1}{6}$$

$$\boxed{0.166}$$

Q3.14.7 NAT 2014 1M Ans: 0.797 ● ● ○

For a zero mean unit variance Gaussian random variable,

$$X \sim N(0, 1)$$

The probability density function is

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

We need to find

$$E(|X|)$$

Since the pdf is symmetric,

$$E(|X|) = 2 \int_0^\infty x \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^\infty x e^{-x^2/2}\, dx$$

Let

$$u = \frac{x^2}{2}$$

Then,

$$du = x\,dx$$

Hence,

$$E(|X|) = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-u}\,du$$

$$= \frac{2}{\sqrt{2\pi}}$$

$$= \sqrt{\frac{2}{\pi}}$$

$$\boxed{0.797}$$

Key Points

- A standard Gaussian random variable has mean 0 and variance 1.

Q3.14.8 MCQ 2014 1M Ans: A ● ○ ○

If the arrivals of calls are independent and occur randomly over time, then the number of calls occurring in a fixed time interval follows a Poisson distribution.
Hence,

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

where λ is the average number of calls in the interval.
Therefore, the required distribution is Poisson.

$$\boxed{\text{(a) Poisson}}$$

Key Points

- Random independent arrivals in a fixed interval are modeled using Poisson distribution.

Q3.14.9 NAT 2014 1M Ans: 0.44 ● ● ○

Let

$$A = \text{parcel lost by first post-office}$$

$$B = \text{parcel lost by second post-office}$$

Given,

$$P(A) = \frac{1}{5}$$

A parcel reaches the second post-office only if it is not lost by the first one.
Hence,

$$P(B) = \left(1 - \frac{1}{5}\right)\left(\frac{1}{5}\right)$$

$$= \frac{4}{5} \times \frac{1}{5} = \frac{4}{25}$$

Now, probability that the parcel is lost is

$$P(\text{lost}) = P(A) + P(B)$$

$$= \frac{1}{5} + \frac{4}{25}$$

$$= \frac{5+4}{25} = \frac{9}{25}$$

Required probability:

$$P(B \mid \text{lost}) = \frac{P(B)}{P(\text{lost})}$$

$$= \frac{\frac{4}{25}}{\frac{9}{25}}$$

$$= \frac{4}{9}$$

$$\boxed{0.44}$$

Q3.13.1 MCQ | 2013 | 2M | Ans: B | ● ○ ○

Given,

$$U \sim N\left(0, \frac{1}{4}\right), \qquad V \sim N\left(0, \frac{1}{9}\right)$$

We need to find

$$P(3V \geq 2U)$$

Let

$$Z = 3V - 2U$$

Since U and V are independent Gaussian random variables, Z is also Gaussian.
Mean of Z:

$$E(Z) = 3E(V) - 2E(U) = 0$$

Variance of Z:

$$\text{Var}(Z) = 9\text{Var}(V) + 4\text{Var}(U)$$

$$= 9\left(\frac{1}{9}\right) + 4\left(\frac{1}{4}\right)$$

$$= 1 + 1 = 2$$

Hence,

$$Z \sim N(0, 2)$$

Since the Gaussian distribution is symmetric about zero,

$$P(Z \geq 0) = \frac{1}{2}$$

Therefore,

$$P(3V \geq 2U) = \frac{1}{2}$$

$$\boxed{\text{(b) } \frac{1}{2}}$$

Key Points

- Linear combination of independent Gaussian random variables is also Gaussian.

Mistakes to be avoided

- Variances add, not standard deviations.

Q3.13.2 MCQ 2013 2M Ans: D ● ● ○

Since U and V are identically distributed,

$$F(x) = P(U \leq x) = P(V \leq x)$$

Also, the CDF of $2V$ is

$$G(x) = P(2V \leq x)$$

$$= P\left(V \leq \frac{x}{2}\right)$$

$$= F\left(\frac{x}{2}\right)$$

Hence,

$$F(x) - G(x) = F(x) - F\left(\frac{x}{2}\right)$$

Now consider two cases:

- If $x > 0$, then

$$x > \frac{x}{2}$$

Since CDF is non-decreasing,

$$F(x) \geq F\left(\frac{x}{2}\right)$$

Thus,

$$F(x) - G(x) \geq 0$$

- If $x < 0$, then

$$x < \frac{x}{2}$$

Hence,

$$F(x) \le F\left(\frac{x}{2}\right)$$

Thus,

$$F(x) - G(x) \le 0$$

Therefore,

$$(F(x) - G(x)) \cdot x \ge 0$$

$$\boxed{\text{(d) } (F(x) - G(x)) \cdot x \ge 0}$$

Q3.12.1 MCQ 2012 1M Ans: B ● ○ ○

We need to find

$$P(\max(X, Y) < \tfrac{1}{2})$$

The condition

$$\max(X, Y) < \frac{1}{2}$$

means both X and Y must be less than $\frac{1}{2}$.
Thus,

$$P(\max(X, Y) < \tfrac{1}{2}) = P\left(X < \tfrac{1}{2},\ Y < \tfrac{1}{2}\right)$$

Since X and Y are independent,

$$= P\left(X < \tfrac{1}{2}\right) P\left(Y < \tfrac{1}{2}\right)$$

Now $X, Y \sim U[-1, 1]$.
Hence,

$$P\left(X < \tfrac{1}{2}\right) = \frac{\frac{1}{2} - (-1)}{1 - (-1)}$$

$$= \frac{\frac{3}{2}}{2} = \frac{3}{4}$$

Similarly,

$$P\left(Y < \tfrac{1}{2}\right) = \frac{3}{4}$$

Therefore,

$$P(\max(X, Y) < \tfrac{1}{2}) = \frac{3}{4} \times \frac{3}{4}$$

$$= \frac{9}{16}$$

$$\boxed{\text{(b) } \frac{9}{16}}$$

Q3.12.2 MCQ 2012 2M Ans: C ● ○ ○

Let X be the number of tosses required to get the first head.
We need to find

$$P(X \text{ is odd})$$

The first head can occur on toss numbers

$$1, 3, 5, \ldots$$

Hence,

$$P(X \text{ is odd}) = \frac{1}{2} + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^5 + \cdots$$

This is a geometric series with

$$a = \frac{1}{2}, \qquad r = \frac{1}{4}$$

Therefore,

$$P(X \text{ is odd}) = \frac{\frac{1}{2}}{1 - \frac{1}{4}} = \frac{2}{3}$$

$$\boxed{\text{(c) } \frac{2}{3}}$$

Q3.11.1 MCQ 2011 2M Ans: C ● ○ ○

Let the outcomes of the first and second tosses be (x, y).
Total possible outcomes:

$$6 \times 6 = 36$$

We need

$$y > x$$

Favourable outcomes are:

$$(1, 2), (1, 3), \ldots, (1, 6)$$

$$(2, 3), (2, 4), \ldots, (2, 6)$$

and so on.
Number of favourable outcomes:

$$5 + 4 + 3 + 2 + 1 = 15$$

Hence,

$$P(y > x) = \frac{15}{36}$$

$$= \frac{5}{12}$$

$$\boxed{\text{(c) } \frac{5}{12}}$$

Q3.10.1 MCQ | 2010 | 2M | Ans: D

Let H be the number of heads in 4 tosses.
We need

$$H > T$$

Since

$$H + T = 4$$

the condition $H > T$ implies

$$H > 2$$

Thus, favourable cases are:

$$H = 3 \quad \text{or} \quad H = 4$$

Using binomial probability,

$$P(H = 3) = {}^4C_3 \left(\frac{1}{2}\right)^4$$

$$= 4 \times \frac{1}{16} = \frac{1}{4}$$

Also,

$$P(H = 4) = {}^4C_4 \left(\frac{1}{2}\right)^4$$

$$= \frac{1}{16}$$

Therefore,

$$P(H > T) = \frac{1}{4} + \frac{1}{16}$$

$$= \frac{5}{16}$$

$$\boxed{\text{(d) } \frac{5}{16}}$$

Q3.09.1 MCQ 2009 1M Ans: C ● ○ ○

Only the first two tosses should be heads, so the sequence must be:

$$HHTTTTTTTT$$

Since the coin is fair, probability of each toss outcome is $\frac{1}{2}$.
Thus, probability of this exact sequence is

$$\left(\frac{1}{2}\right)^{10}$$

$$\boxed{\text{(c)}\ \left(\frac{1}{2}\right)^{10}}$$

Q3.09.2 MCQ 2009 2M Ans: C ● ○ ○

We need to find

$$P(X+Y=2 \mid X-Y=0)$$

Using conditional probability,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Here,

$$A:\ X+Y=2$$

$$B:\ X-Y=0 \Rightarrow X=Y$$

Since $X=Y$, possible pairs are

$$(0,0),\ (1,1),\ (2,2)$$

Their probabilities are

$$P(0,0) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$$

$$P(1,1) = \frac{1}{4}\cdot\frac{1}{4} = \frac{1}{16}$$

$$P(2,2) = \frac{1}{4}\cdot\frac{1}{4} = \frac{1}{16}$$

Hence,

$$P(X-Y=0) = \frac{1}{4}+\frac{1}{16}+\frac{1}{16} = \frac{3}{8}$$

Now, both conditions $X+Y=2$ and $X=Y$ are satisfied only by

$$(X,Y) = (1,1)$$

Therefore,

$$P((X+Y=2)\cap(X-Y=0)) = \frac{1}{16}$$

Thus,

$$P(X+Y=2 \mid X-Y=0) = \frac{\frac{1}{16}}{\frac{3}{8}} = \frac{1}{6}$$

$$\boxed{\text{(c)}\frac{1}{6}}$$

Q3.09.3 MCQ | 2009 | 2M | Ans: B ● ● ○

Given probability distribution:

k	1	2	3	4	5
$P(X=k)$	0.1	0.2	0.4	0.2	0.1

Mean of X:

$$E(X) = \sum xP(X = x)$$

$$E(X) = 1(0.1) + 2(0.2) + 3(0.4) + 4(0.2) + 5(0.1)$$

$$E(X) = 0.1 + 0.4 + 1.2 + 0.8 + 0.5 = 3$$

Hence, the student's answer 3.5 is wrong.
Now,

$$E(X^2) = \sum x^2P(X = x)$$

$$E(X^2) = 1^2(0.1) + 2^2(0.2) + 3^2(0.4) + 4^2(0.2) + 5^2(0.1)$$

$$E(X^2) = 0.1 + 0.8 + 3.6 + 3.2 + 2.5 = 10.2$$

Variance:

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Var}(X) = 10.2 - 3^2$$

$$\text{Var}(X) = 10.2 - 9 = 1.2$$

Hence, the teacher's answer 1.5 is also wrong.

(b) Both the student and the teacher are wrong

Q3.08.1 MCQ | 2008 | 2M | Ans: A ● ○ ○

Since $P_X(x)$ is a probability density function over the entire real axis,

$$\int_{-\infty}^{\infty} P_X(x)\,dx = 1$$

Given,

$$P_X(x) = Me^{-2|x|} + Ne^{-3|x|}$$

Therefore,

$$\int_{-\infty}^{\infty} \left(Me^{-2|x|} + Ne^{-3|x|}\right) dx = 1$$

Using

$$\int_{-\infty}^{\infty} e^{-a|x|}\,dx = \frac{2}{a}$$

we get

$$M\left(\frac{2}{2}\right) + N\left(\frac{2}{3}\right) = 1$$

$$M + \frac{2}{3}N = 1$$

$$\boxed{\text{(a) } M + \frac{2}{3}N = 1}$$

Key Points

- Total probability for a pdf over entire real line is 1.
- Standard result:

$$\int_{-\infty}^{\infty} e^{-a|x|}\,dx = \frac{2}{a}$$

Q3.07.1 MCQ | 2007 | 1M | Ans: A | ● ○ ○

The variance of a random variable X is defined as

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\boxed{\text{(a) } E[X^2] - E^2[X]}$$

Q3.07.2 MCQ | 2007 | 2M | Ans: C | ● ○ ○

Let

A = event of failing in Paper 1

B = event of failing in Paper 2

Given,

$$P(A) = 0.3, \qquad P(B) = 0.2$$

and

$$P(A \mid B) = 0.6$$

Using conditional probability,

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Therefore,

$$P(A \cap B) = P(A \mid B)\,P(B)$$

$$P(A \cap B) = 0.6 \times 0.2$$

$$P(A \cap B) = 0.12$$

Hence, the probability of failing in both papers is

$$\boxed{\text{(c) } 0.12}$$

Q3.06.1 MCQ 2006 1M Ans: C ● ○ ○

Since $p(x)$ is a probability density function over the entire real axis,

$$\int_{-\infty}^{\infty} p(x)\,dx = 1$$

Given,

$$p(x) = Ke^{-\alpha|x|}$$

Therefore,

$$\int_{-\infty}^{\infty} Ke^{-\alpha|x|}\,dx = 1$$

Using

$$\int_{-\infty}^{\infty} e^{-\alpha|x|}\,dx = \frac{2}{\alpha}$$

we get

$$K\left(\frac{2}{\alpha}\right) = 1$$

$$K = \frac{\alpha}{2}$$

$$\boxed{\text{(c) } \frac{\alpha}{2}}$$

Q3.06.2 MCQ 2006 2M Ans: D ● ● ○

Let

$$D = \text{event that a computer is defective}$$

Given,

$$P(X) = 0.6, \qquad P(Y) = 0.3, \qquad P(Z) = 0.1$$

and

$$P(D \mid X) = 0.01$$
$$P(D \mid Y) = 0.02$$
$$P(D \mid Z) = 0.03$$

Using Bayes' theorem,

$$P(Y \mid D) = \frac{P(D \mid Y)P(Y)}{P(D)}$$

Now,

$$P(D) = P(D \mid X)P(X) + P(D \mid Y)P(Y) + P(D \mid Z)P(Z)$$

$$P(D) = 0.01(0.6) + 0.02(0.3) + 0.03(0.1)$$

$$P(D) = 0.006 + 0.006 + 0.003 = 0.015$$

Therefore,

$$P(Y \mid D) = \frac{0.02 \times 0.3}{0.015}$$

$$P(Y \mid D) = \frac{0.006}{0.015} = 0.4$$

$$\boxed{\text{(d) } 0.4}$$

Q3.06.3 MCQ | 2006 | 2M | Ans: B | ● ○ ○

The given probability density function (PDF) of the uniformly distributed random variable X is defined as:

$$f_X(x) = \frac{1}{10}[u(x+5) - u(x-5)]$$

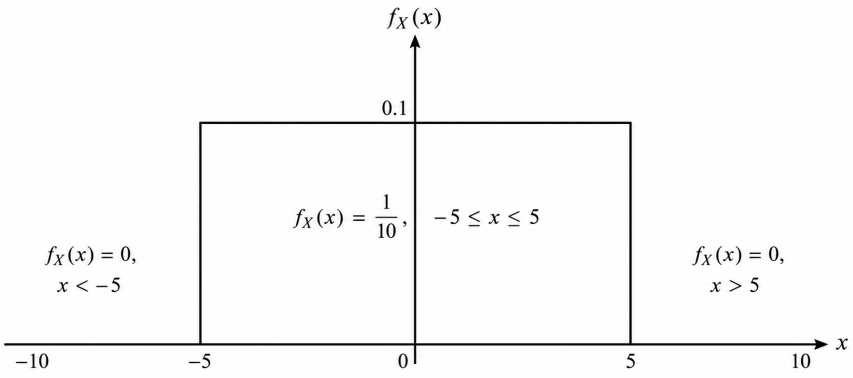


Fig. Solution Q3.06.3

This indicates that X is uniformly distributed over the interval $[-5, 5]$ with a total support width of 10 as shown in figure above.

From the provided transformation graph, the mapping function $y = g(x)$ behaves as follows:

$$Y = \begin{cases} 1, & -2.5 \le X \le 2.5 \\ 0, & X < -2.5 \text{ or } X > 2.5 \end{cases}$$

Since the output Y can only map to two distinct, isolated values (0 and 1), Y is a discrete random variable. We determine its probability mass by evaluating the corresponding intervals of X:

1. For $Y = 1$:

$$P(Y = 1) = P(-2.5 \le X \le 2.5) = \frac{2.5 - (-2.5)}{10} = \frac{5}{10} = 0.5$$

2. For $Y = 0$:

$$P(Y = 0) = 1 - P(Y = 1) = 1 - 0.5 = 0.5$$

The PDF of a discrete random variable is mathematically represented using Dirac delta (δ) impulses positioned at the allowed discrete values, scaled by their probabilities:

$$f_Y(y) = P(Y = 0)\delta(y) + P(Y = 1)\delta(y - 1)$$

$$f_Y(y) = 0.5\delta(y) + 0.5\delta(y - 1)$$

$$\boxed{\text{(b) } f_Y(y) = 0.5\delta(y) + 0.5\delta(y - 1)}$$

Key Points

- **Continuous-to-Discrete Transformation:** When a continuous random variable passes through a transfer characteristic with perfectly flat constant intervals, those intervals compress into discrete probability masses.
- **Impulse Representation of Discrete Variables:** In continuous-time probability notation, discrete probabilities are represented as impulses: $f_Y(y) = \sum P(Y = y_i)\delta(y - y_i)$.

Mistakes to be avoided

- Do not confuse the transfer characteristic graph $y = g(x)$ with a probability density curve. The height of 1 on the figure denotes the output value mapping, not a probability value.

Q3.05.1 MCQ 2005 1M Ans: D ● ○ ○

For an odd number to follow an even number:

First roll must be even and second roll must be odd

Probability of getting an even number in first roll:

$$P(E) = \frac{3}{6} = \frac{1}{2}$$

Probability of getting an odd number in second roll:

$$P(O) = \frac{3}{6} = \frac{1}{2}$$

Since the two rolls are independent,

$$P(E \cap O) = P(E)P(O)$$

$$P(E \cap O) = \frac{1}{2} \times \frac{1}{2}$$

$$P(E \cap O) = \frac{1}{4}$$

$$\boxed{\text{(d) } \frac{1}{4}}$$

Q3.04.1 MCQ 2004 1M Ans: D ● ○ ○

For a random variable,

$$P(X = a) = F_X(a) - F_X(a^-)$$

From the graph,

$$F_X(1^-) = 0.25$$

and

$$F_X(1) = 0.55$$

Therefore,

$$P(X = 1) = 0.55 - 0.25$$

$$P(X = 1) = 0.30$$

$$\boxed{\text{(d) } 0.30}$$

Q3.03.1 MCQ | 2003 | 2M | Ans: C | ● ● ○

Since X and Y are uniformly distributed over

$$(-1, 1) \quad \text{and} \quad (-2, 1)$$

respectively, the joint distribution is uniform over the rectangular region

$$-1 < X < 1, \qquad -2 < Y < 1$$

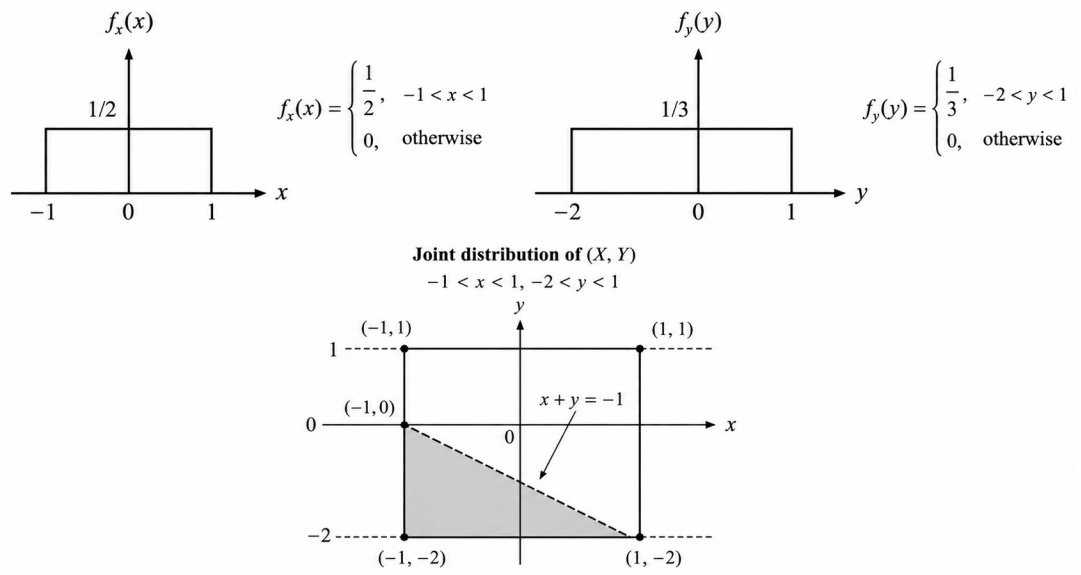


Fig. Solution Q3.03.1

Area of the rectangle:

$$(2) \times (3) = 6$$

We need

$$P(Z \le -1) = P(X + Y \le -1)$$

The line

$$x + y = -1$$

divides the rectangle into two regions.
Inside the rectangle, the favorable region is a right triangle with vertices

$$(-1, -2),\ (1, -2),\ (-1, 0)$$

Area of this triangle:

$$\frac{1}{2} \times 2 \times 2 = 2$$

Hence,

$$P(Z \le -1) = \frac{\text{Favorable area}}{\text{Total area}}$$

$$P(Z \le -1) = \frac{2}{6}$$

$$P(Z \le -1) = \frac{1}{3}$$

$$\boxed{\text{(c) } \frac{1}{3}}$$

Q3.01.1 MCQ | 2001 | 1M | Ans: C

For a continuous random variable, the probability at any single point is always zero.
Given the PDF:

$$P_X(x) = \frac{1}{3\sqrt{2\pi}} e^{-\frac{(x-4)^2}{18}}$$

We need to find:

$$P(X = 4)$$

Since X is a continuous random variable,

$$P(X = a) = 0$$

for any specific value a.
Hence,

$$P(X = 4) = 0$$

$$\boxed{\text{(c) } 0}$$

Key Points

- For a continuous random variable, probability is defined over an interval, not at a single point.
- The area under the PDF at a single point is zero.

Q3.93.1 NAT | 1993 | 2M | Ans: 0.125

For a valid probability density function (PDF),

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

From the figure given in question, the total area consists of:

- Two rectangles from $x = -3$ to -1 and $x = 1$ to 3
- One trapezium between $x = -1$ and $x = 1$

Area of two rectangles:

$$2 \times (2 \times A) = 4A$$

Area of trapezium:

$$\frac{1}{2}(A + 3A) \times 2 = 4A$$

Total area:

$$4A + 4A = 8A$$

For PDF,

$$8A = 1$$

$$A = \frac{1}{8}$$

$$\boxed{0.125}$$

Q3.92.1 MCQ | 1992 | 2M | Ans: B | ● ○ ○

Given PDF is

$$p(x) = \frac{1}{4}, \qquad -1 \le x \le 3$$

and zero elsewhere.
Mean of continuous random variable:

$$E[X] = \int_{-\infty}^{\infty} x\, p(x)\, dx$$

$$E[X] = \int_{-1}^{3} x \left(\frac{1}{4}\right) dx$$

$$E[X] = \frac{1}{4}\left[\frac{x^2}{2}\right]_{-1}^{3}$$

$$E[X] = \frac{1}{8}(9 - 1) = 1$$

Now,

$$E[X^2] = \int_{-1}^{3} x^2 \left(\frac{1}{4}\right) dx$$

$$E[X^2] = \frac{1}{4}\left[\frac{x^3}{3}\right]_{-1}^{3}$$

$$E[X^2] = \frac{1}{12}(27 + 1) = \frac{7}{3}$$

Variance:

$$\sigma^2 = E[X^2] - (E[X])^2$$

$$\sigma^2 = \frac{7}{3} - 1$$

$$\sigma^2 = \frac{4}{3}$$

$$\boxed{\text{(b) } 1 \text{ and } \frac{4}{3}}$$

Q3.88.1 MSQ | 1988 | 2M | Ans: A,B | ● ○ ○

Given A and B are mutually exclusive,

$$P(A \cap B) = 0$$

- **Option A**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B)$$

Hence, true.

- **Option B**

$$P(B^c) = 1 - P(B)$$

Since,

$$P(A) + P(B) \leq 1$$

$$P(A) \leq 1 - P(B) = P(B^c)$$

As $P(B) \neq 0$,

$$P(B^c) > P(A)$$

Hence, true.

- **Option C**

$$P(A \cap B) = 0$$

but

$$P(A)P(B) \neq 0$$

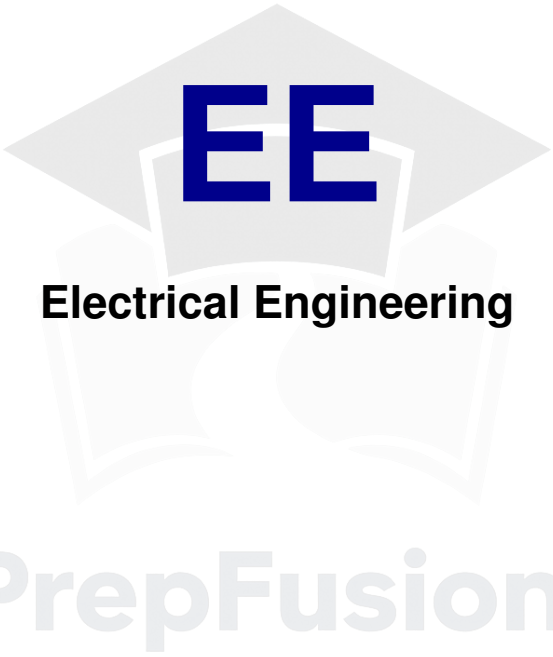
Hence, false.

- **Option D**

Contradicts option B, hence false.

(a), (b)

SOLUTIONS — Probability and Statistics



Electrical Engineering

PrepFusion

Q3.26.1 NAT 2026 2M Ans: 0.1 ● ○ ○

Given,

$$E(X) = 1, \qquad E(Y) = 2, \qquad E(X^2) = 4, \qquad E(XY) = 0.9$$

We need to minimize

$$E\left((X - \alpha Y)^2\right)$$

Expanding,

$$E\left((X - \alpha Y)^2\right) = E(X^2) - 2\alpha E(XY) + \alpha^2 E(Y^2)$$

Substituting the given values,

$$f(\alpha) = 4 - 2\alpha(0.9) + 9\alpha^2$$

$$f(\alpha) = 4 - 1.8\alpha + 9\alpha^2$$

For minimum value,

$$\frac{df}{d\alpha} = 18\alpha - 1.8 = 0$$

$$\alpha = \frac{1.8}{18} = 0.1$$

$$\boxed{0.1}$$

Q3.25.1 MCQ 2025 2M Ans: A ● ● ○

Given,

$$P_X(x) = \begin{cases} 1, & x \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$$

and

$$Y = X^2$$

Since $x \in (0, 1]$, we have

$$y = x^2 \quad \Rightarrow \quad x = \sqrt{y}$$

Using transformation of random variables,

$$P_Y(y) = P_X(x) \left|\frac{dx}{dy}\right|$$

Now,

$$\frac{dx}{dy} = \frac{1}{2\sqrt{y}}$$

Also,

$$P_X(x) = 1$$

Hence,

$$P_Y(y) = 1 \cdot \frac{1}{2\sqrt{y}}$$

Therefore,

$$P_Y(y) = \begin{cases} \dfrac{1}{2\sqrt{y}}, & y \in (0, 1] \\ 0, & \text{otherwise} \end{cases}$$

$$\boxed{\text{(a) } P_Y(y) = \begin{cases} \dfrac{1}{2\sqrt{y}}, & y \in (0, 1] \\ 0, & \text{otherwise} \end{cases}}$$

Key Points

- For one-to-one transformation $Y = g(X)$,
$$P_Y(y) = P_X(x)\left|\frac{dx}{dy}\right|$$
- If $Y = X^2$ and $X > 0$, then $x = \sqrt{y}$.

Mistakes to be avoided

- Do not forget the Jacobian term $\left|\frac{dx}{dy}\right|$.
- Ensure the support changes from $x \in (0, 1]$ to $y \in (0, 1]$.

Q3.25.2 MCQ | 2025 | 1M | Ans: B | ● ○ ○

Given,

$$P(X = 0, Y = 0) = \frac{1}{4}$$
$$P(X = 1, Y = 0) = \frac{1}{8}$$
$$P(X = 0, Y = 1) = \frac{1}{2}$$
$$P(X = 1, Y = 1) = \frac{1}{8}$$

We need to find

$$E(Y \mid X = 1)$$

Using conditional expectation,

$$E(Y \mid X = 1) = \sum y\, P(Y = y \mid X = 1)$$

First, compute

$$P(X = 1) = P(X = 1, Y = 0) + P(X = 1, Y = 1)$$
$$P(X = 1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

Now,

$$P(Y = 1 \mid X = 1) = \frac{P(X = 1, Y = 1)}{P(X = 1)}$$
$$P(Y = 1 \mid X = 1) = \frac{\frac{1}{8}}{\frac{1}{4}} = \frac{1}{2}$$

Since $Y \in \{0, 1\}$,

$$E(Y \mid X = 1) = 0 \cdot P(Y = 0 \mid X = 1) + 1 \cdot P(Y = 1 \mid X = 1)$$
$$E(Y \mid X = 1) = \frac{1}{2}$$

$$\boxed{\text{(b) } \frac{1}{2}}$$

Q3.24.1 MSQ | 2024 | 2M | Ans: B,D | ● ○ ○

Given:

$$X \in \{-10, -9, \ldots, 0, \ldots, 9, 10\}$$

and X is uniformly distributed.
We check each option.
For option A:

$$Y = X^2$$

Both $X = a$ and $X = -a$ give the same value of Y.
Hence,

$$P(Y = 1) = P(X = \pm 1) = \frac{2}{21}$$

while

$$P(Y = 0) = \frac{1}{21}$$

Probabilities are not equal.
Therefore, X^2 is not uniformly distributed.
For option B:

$$Y = X^3$$

The function X^3 is one-to-one over the given set.
Distinct values of X produce distinct values of Y.
Hence, every value of Y has probability

$$\frac{1}{21}$$

Therefore, X^3 is uniformly distributed.
For option C:

$$Y = (X - 5)^2$$

Different values of X can give the same value of Y.
For example,

$$(X - 5)^2 = 1$$

for

$$X = 4, 6$$

Hence probabilities are unequal.
Therefore, option C is not uniformly distributed.
For option D:

$$Y = (X + 10)^2$$

Here,

$$X + 10 \in \{0, 1, 2, \ldots, 20\}$$

Since only non-negative values occur, squaring remains one-to-one.
Thus, all values of Y occur with equal probability

$$\frac{1}{21}$$

Hence, option D is uniformly distributed.

$$\boxed{\text{(b) } X^3 \text{ ,(d) } (X + 10)^2}$$

Q3.23.1 NAT 2023 2M Ans: 0.188 ● ○ ○

Let the probability of getting a head be p.
For a geometric distribution, the expected number of trials for first occurrence of head is

$$E(X) = \frac{1}{p}$$

Given,

$$E(X) = 4$$

Therefore,

$$\frac{1}{p} = 4$$

$$p = \frac{1}{4}$$

The probability that the first head occurs in the second trial is

$$(1-p)p$$

Substituting $p = \frac{1}{4}$,

$$\left(1-\frac{1}{4}\right)\left(\frac{1}{4}\right) = \frac{3}{4} \times \frac{1}{4} = \frac{3}{16}$$

$$\frac{3}{16} = 0.1875$$

Rounded off to 3 decimal places,

$$0.188$$

$$\boxed{0.188}$$

Q3.23.2 MCQ 2023 1M Ans: C ● ● ○

Given,

$$\hat{\sigma}_1^2 = \frac{1}{10000}\sum_{n=1}^{10000} x_n^2$$

and

$$\hat{\sigma}_2^2 = \frac{1}{9999}\sum_{n=1}^{10000} x_n^2$$

Since the random process has mean zero and variance σ_0^2,

$$E(x_n^2) = \sigma_0^2$$

Now,

$$E(\hat{\sigma}_1^2) = E\left(\frac{1}{10000}\sum_{n=1}^{10000} x_n^2\right)$$

Using linearity of expectation,

$$E(\hat{\sigma}_1^2) = \frac{1}{10000}\sum_{n=1}^{10000} E(x_n^2)$$

$$E(\hat{\sigma}_1^2) = \frac{1}{10000} \times 10000 \times \sigma_0^2$$

$$E(\hat{\sigma}_1^2) = \sigma_0^2$$

Also,

$$E(\hat{\sigma}_2^2) = \frac{1}{9999}\sum_{n=1}^{10000} E(x_n^2)$$

$$E(\hat{\sigma}_2^2) = \frac{10000}{9999}\sigma_0^2 \neq \sigma_0^2$$

Hence, only option C is correct.

$$\boxed{\text{(c) } E(\hat{\sigma}_1^2) = \sigma_0^2}$$

Key Points

- For zero-mean Gaussian random variables,
$$E(x_n^2) = \sigma_0^2$$
- Expectation is linear:
$$E\left(\sum X_n\right) = \sum E(X_n)$$
- $\hat{\sigma}_1^2$ is an unbiased estimator of variance.

Q3.22.1 MCQ 2022 1M Ans: A ● ○ ○

Each die has faces

$$\{P, Q, R, S, T, U\}$$

When two dice are rolled together, the total number of outcomes is

$$6 \times 6 = 36$$

We need the probability that both upper faces show any combination of P, Q, R.
Thus, favorable outcomes are obtained when each die shows one of

$$\{P, Q, R\}$$

Number of favorable outcomes:

$$3 \times 3 = 9$$

Therefore,

$$P = \frac{9}{36} = \frac{1}{4}$$

$$\boxed{\text{(a) } \frac{1}{4}}$$

Q3.22.2 NAT 2022 1M Ans: 1 ● ○ ○

Given,

$$f(x) = ae^{-2|x|}$$

Since $f(x)$ is a probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

Therefore,

$$\int_{-\infty}^{\infty} ae^{-2|x|}\,dx = 1$$

Since the function is even,

$$2a\int_{0}^{\infty} e^{-2x}\,dx = 1$$

Now,

$$\int_{0}^{\infty} e^{-2x}\,dx = \left[-\frac{e^{-2x}}{2}\right]_{0}^{\infty} = \frac{1}{2}$$

Thus,

$$2a \times \frac{1}{2} = 1$$

$$a = 1$$

$$\boxed{1}$$

Q3.21.1 MCQ | 2021 | 2M | Ans: B ● ● ○

Let X be the number of tosses required to get the first head.
Then,

$$P(X = k) = (1-p)^{k-1}p$$

for

$$k = 1, 2, 3, \ldots$$

The expected value is

$$E(X) = \sum_{k=1}^{\infty} kP(X = k)$$

Substituting $P(X = k)$,

$$E(X) = \sum_{k=1}^{\infty} k(1-p)^{k-1}p$$

$$E(X) = p\sum_{k=1}^{\infty} k(1-p)^{k-1}$$

Using the standard result

$$\sum_{k=1}^{\infty} kr^{k-1} = \frac{1}{(1-r)^2}, \qquad |r| < 1$$

with

$$r = 1 - p$$

we get

$$E(X) = p \cdot \frac{1}{(1-(1-p))^2}$$

$$E(X) = p \cdot \frac{1}{p^2}$$

$$E(X) = \frac{1}{p}$$

$$\boxed{\text{(b) } \frac{1}{p}}$$

Key Points

- The series identity

$$\sum_{k=1}^{\infty} kr^{k-1} = \frac{1}{(1-r)^2}$$

is commonly used to derive expectation.

Q3.19.1 NAT 2019 2M Ans: 0.26 ● ○ ○

Let X be the number of defective resistors.
Then,

$$X \sim \text{Binomial}(n = 50, p = 0.02)$$

We need to find

$$P(X \geq 2)$$

Using complement rule,

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$$

Now,

$$P(X = 0) = (0.98)^{50}$$

and

$$P(X = 1) = \binom{50}{1}(0.02)(0.98)^{49}$$

Therefore,

$$P(X \geq 2) = 1 - (0.98)^{50} - 50(0.02)(0.98)^{49}$$

$$P(X \geq 2) \approx 0.2642$$

Rounded off to two decimal places,

$$0.26$$

$$\boxed{0.26}$$

Q3.19.2 MCQ 2019 1M Ans: B ● ○ ○

Given, the mean-square value of the zero-mean random process is $\frac{kT}{C}$
For a zero-mean random process,

$$\text{Mean-square value} = E(X^2) = \sigma^2$$

Hence,

$$\sigma^2 = \frac{kT}{C}$$

Taking square root on both sides,

$$\sigma = \sqrt{\frac{kT}{C}}$$

Therefore, the standard deviation is

$$\sqrt{\frac{kT}{C}}$$

$$\boxed{\text{(b) } \sqrt{\frac{kT}{C}}}$$

Q3.17.1 MCQ 2017 2M Ans: C ● ○ ○

Let Y denote the number of heads obtained when a fair coin is tossed repeatedly until the first head appears or until 3 tosses are completed.
Possible outcomes are:

$$Y = 0 \quad \text{for } TTT$$

$$Y = 1 \quad \text{for } H,\ TH,\ TTH$$

Thus,

$$P(Y = 0) = \frac{1}{8}$$

$$P(Y = 1) = 1 - \frac{1}{8} = \frac{7}{8}$$

Now,

$$E(Y) = 0 \cdot \frac{1}{8} + 1 \cdot \frac{7}{8} = \frac{7}{8}$$

Also,

$$E(Y^2) = 0^2 \cdot \frac{1}{8} + 1^2 \cdot \frac{7}{8} = \frac{7}{8}$$

Therefore,

$$\text{var}(Y) = E(Y^2) - [E(Y)]^2$$

$$= \frac{7}{8} - \left(\frac{7}{8}\right)^2$$

$$= \frac{56 - 49}{64} = \frac{7}{64}$$

$$\boxed{\text{(c) } \frac{7}{64}}$$

Q3.17.2 MCQ 2017 1M Ans: A ● ○ ○

Let R_1 be the event that the first discarded ball is red and B_1 be the event that it is black.
Using total probability theorem,

$$P(\text{Red in second draw}) = P(\text{Red} \mid R_1)P(R_1) + P(\text{Red} \mid B_1)P(B_1)$$

Now,

$$P(R_1) = \frac{5}{10} = \frac{1}{2}$$

and

$$P(B_1) = \frac{5}{10} = \frac{1}{2}$$

If the first discarded ball is red, then remaining red balls are 4 out of 9 balls.

$$P(\text{Red} \mid R_1) = \frac{4}{9}$$

If the first discarded ball is black, then remaining red balls are 5 out of 9 balls.

$$P(\text{Red} \mid B_1) = \frac{5}{9}$$

Therefore,

$$P(\text{Red in second draw}) = \frac{4}{9} \cdot \frac{1}{2} + \frac{5}{9} \cdot \frac{1}{2}$$

$$= \frac{4+5}{18} = \frac{9}{18} = \frac{1}{2}$$

$$\boxed{\text{(a) } \frac{1}{2}}$$

Q3.17.3 NAT | 2017 | 1M | Ans: 0.9 ● ○ ○

Let the arrival time of a vehicle be uniformly distributed over the 5 minute cycle.
Traffic signal cycle:

$$\text{Green time} = 2 \text{ minutes}$$

$$\text{Red time} = 3 \text{ minutes}$$

A vehicle arriving during green signal waits for

$$0 \text{ minutes}$$

If it arrives during red signal, the waiting time varies uniformly from 0 to 3 minutes.
Hence, average waiting time during red period is

$$\frac{0+3}{2} = \frac{3}{2} \text{ minutes}$$

Probability that a vehicle arrives during red signal is

$$\frac{3}{5}$$

Therefore, expected waiting time is

$$E(W) = \frac{3}{5} \times \frac{3}{2}$$

$$= \frac{9}{10}$$

$$\boxed{0.9}$$

Q3.16.1 MCQ | 2016 | 2M | Ans: A ● ● ○

Given,

$$f_X(x) = \frac{3}{2} e^{-3x} u(x) + a e^{4x} u(-x)$$

where $u(x)$ is the unit step function.
Using the definition of unit step function:

$$f_X(x) = \begin{cases} \frac{3}{2} e^{-3x}, & x \geq 0 \\ a e^{4x}, & x < 0 \end{cases}$$

Since total probability must be 1,

$$\int_{-\infty}^{\infty} f_X(x)\, dx = 1$$

Therefore,

$$\int_{-\infty}^{0} ae^{4x}\,dx + \int_{0}^{\infty} \frac{3}{2}e^{-3x}\,dx = 1$$

Evaluating,

$$a\left[\frac{e^{4x}}{4}\right]_{-\infty}^{0} + \frac{3}{2}\left[\frac{-e^{-3x}}{3}\right]_{0}^{\infty} = 1$$

$$\frac{a}{4} + \frac{1}{2} = 1$$

$$\frac{a}{4} = \frac{1}{2}$$

$$a = 2$$

Now,

$$P(X \le 0) = \int_{-\infty}^{0} ae^{4x}\,dx$$

$$= \int_{-\infty}^{0} 2e^{4x}\,dx$$

$$= 2\left[\frac{e^{4x}}{4}\right]_{-\infty}^{0} = \frac{1}{2}$$

$$\boxed{\text{(a) } 2, \frac{1}{2}}$$

Q3.16.2 NAT | 2016 | 2M | Ans: 0.2 | ● ○ ○

Let the number of available pen colors be $n = 4$ (Black, Blue, Green, Red) and the number of pens to be selected be $r = 3$. The total number of unique combinations with repetition is given by the formula:

$$\text{Total Combinations} = \binom{n+r-1}{r} = \binom{4+3-1}{3} = \binom{6}{3}$$

Calculating the combination value:

$$\binom{6}{3} = \frac{6 \times 5 \times 4}{3 \times 2 \times 1} = 20$$

The favorable combinations are those where all 3 pens are of the exact same color. There are exactly 4 such combinations:

$$\{\text{Black, Black, Black}\}, \{\text{Blue, Blue, Blue}\}, \{\text{Green, Green, Green}\}, \{\text{Red, Red, Red}\}$$

$$\text{Favorable Combinations} = 4$$

Therefore, the required probability is:

$$\text{Probability} = \frac{\text{Favorable Combinations}}{\text{Total Combinations}} = \frac{4}{20} = 0.2$$

$$\boxed{0.2}$$

Key Points

- **Combinations with Repetition:** When the order of selection does not matter and elements can be repeated, the number of ways to choose r elements from a set of n distinct types is given by $\binom{n+r-1}{r}$.

Mistakes to be avoided

- Do not assume the selections are independent ordered events ($4^3 = 64$ total outcomes), which incorrectly leads to a probability of $\frac{4}{64} = 0.0625$.

Q3.15.1 MCQ | 2015 | 2M | Ans: D

Given joint probabilities:

$$P(H_R H_S) = 0.28$$

$$P(T_R T_S) = 0.18$$

$$P(H_R T_S) = 0.30$$

$$P(T_R H_S) = 0.24$$

First find marginal probabilities for coin R:

$$P(H_R) = P(H_R H_S) + P(H_R T_S)$$

$$= 0.28 + 0.30 = 0.58$$

$$P(T_R) = 0.24 + 0.18 = 0.42$$

Hence, coin R is not fair.
Now for coin S:

$$P(H_S) = P(H_R H_S) + P(T_R H_S)$$

$$= 0.28 + 0.24 = 0.52$$

$$P(T_S) = 0.30 + 0.18 = 0.48$$

Hence, coin S is also not fair.
To check independence:

$$P(H_R H_S) \stackrel{?}{=} P(H_R)P(H_S)$$

$$0.28 \stackrel{?}{=} (0.58)(0.52)$$

$$0.28 \neq 0.3016$$

Therefore, the tosses are dependent.

(d) The coin tosses are dependent

Q3.15.2 MCQ 2015 2M Ans: D ● ○ ○

Let P be the probability that player A wins the game.
Player A wins immediately if he gets a six in the first throw.

$$P(A \text{ wins in first turn}) = \frac{1}{6}$$

If both A and B fail to get a six in their respective turns, the game returns to the original situation.

$$P(\text{both fail}) = \frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$$

Hence,

$$P = \frac{1}{6} + \frac{25}{36}P$$

Solving,

$$P\left(1 - \frac{25}{36}\right) = \frac{1}{6}$$

$$P \cdot \frac{11}{36} = \frac{1}{6}$$

$$P = \frac{1}{6} \times \frac{36}{11}$$

$$P = \frac{6}{11}$$

$$\boxed{\text{(d) } \frac{6}{11}}$$

Q3.15.3 NAT 2015 1M Ans: 0.25 ● ● ○

Given,

$$f(x) = \begin{cases} a + bx, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Since $f(x)$ is a probability density function,

$$\int_0^1 (a + bx)\, dx = 1$$

$$a + \frac{b}{2} = 1$$

Also,

$$E[X] = \int_0^1 x(a + bx)\, dx = \frac{2}{3}$$

$$\frac{a}{2} + \frac{b}{3} = \frac{2}{3}$$

Solving the two equations:

$$a + \frac{b}{2} = 1$$

$$\frac{a}{2} + \frac{b}{3} = \frac{2}{3}$$

we get

$$a = 0, \qquad b = 2$$

Hence,

$$f(x) = 2x, \qquad 0 < x < 1$$

Now,

$$P(X < 0.5) = \int_0^{0.5} 2x\,dx$$

$$= \left[x^2\right]_0^{0.5}$$

$$= \frac{1}{4}$$

$$\boxed{\frac{1}{4}}$$

Q3.14.1 NAT 2014 1M Ans: 0.42

Given probability density function:

$$f(x) = kx^2$$

The lifetime of bulb varies from 1 year to 2 years.
Hence,

$$f(x) = \begin{cases} kx^2, & 1 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

Since total probability must be 1,

$$\int_1^2 kx^2\,dx = 1$$

$$k\left[\frac{x^3}{3}\right]_1^2 = 1$$

$$k\left(\frac{8-1}{3}\right) = 1$$

$$k \cdot \frac{7}{3} = 1$$

$$k = \frac{3}{7}$$

$$\boxed{0.42}$$

Q3.14.2 NAT 2014 2M Ans: 0.4 ● ● ○

Given probability density function:

$$f(x) = \begin{cases} 0.2, & |x| \le 1 \\ 0.1, & 1 < |x| \le 4 \\ 0, & \text{otherwise} \end{cases}$$

We need to find

$$P(0.5 < X < 5)$$

Since $f(x) = 0$ for $|x| > 4$, the effective interval becomes

$$0.5 < X < 4$$

Now split the interval according to the pdf definition:

$$P(0.5 < X < 5) = \int_{0.5}^{1} 0.2\,dx + \int_{1}^{4} 0.1\,dx$$

$$= 0.2(1 - 0.5) + 0.1(4 - 1)$$

$$= 0.2 \times 0.5 + 0.1 \times 3$$

$$= 0.1 + 0.3$$

$$= 0.4$$

$$\boxed{0.4}$$

Q3.14.3 NAT 2014 1M Ans: 0.142 ● ○ ○

Let the probability of getting a face with n dots be proportional to n.
Hence,

$$P(n) = kn$$

where k is the proportionality constant.
Since the total probability is 1,

$$k(1 + 2 + 3 + 4 + 5 + 6) = 1$$

$$21k = 1$$

$$k = \frac{1}{21}$$

Therefore, the probability of getting the face with three dots is

$$P(3) = 3k = 3 \times \frac{1}{21} = \frac{1}{7}$$

$$\boxed{0.142}$$

Q3.14.4 MCQ 2014 2M Ans: B ● ○ ○

Let the number of heads be H and the number of tails be T.
Given,

$$H + T = n$$

and

$$H - T = n - 3$$

Adding the two equations,

$$2H = 2n - 3$$

$$H = \frac{2n - 3}{2}$$

But the number of heads must be an integer, whereas

$$\frac{2n - 3}{2}$$

is not an integer for any integer n.
Hence, such an event is impossible.

(b) 0

Key Points

- In coin toss problems, use:

$$H + T = n$$

- Check whether the obtained values are integers.

Q3.13.1 MCQ 2013 1M Ans: A ● ○ ○

Given probability density function:

$$f(x) = e^{-x}, \quad 0 < x < \infty$$

We need to find

$$P(X > 1)$$

Using the definition of probability for a continuous random variable,

$$P(X > 1) = \int_1^{\infty} e^{-x}\, dx$$

Evaluating,

$$P(X > 1) = [-e^{-x}]_1^{\infty}$$

$$P(X > 1) = 0 - (-e^{-1})$$

$$P(X > 1) = \frac{1}{e} \approx 0.368$$

(a) 0.368

Q3.12.1 MCQ 2012 1M Ans: C ● ○ ○

Let the first head appear on the $(2k+1)^{\text{th}}$ toss, where

$$k = 0, 1, 2, \ldots$$

Then the required sequences are:

$$H, \quad TTH, \quad TTTTH, \quad \ldots$$

Probability that the first head appears on the $(2k+1)^{\text{th}}$ toss is

$$\left(\frac{1}{2}\right)^{2k+1}$$

Hence,

$$P(\text{odd number of tosses}) = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^{2k+1}$$

$$= \frac{1}{2}\left[1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^4 + \cdots\right]$$

This is a geometric series with common ratio

$$\frac{1}{4}$$

Therefore,

$$P = \frac{1}{2} \times \frac{1}{1 - \frac{1}{4}}$$

$$P = \frac{1}{2} \times \frac{4}{3} = \frac{2}{3}$$

$$\boxed{\text{(c) } \frac{2}{3}}$$

Q3.12.2 MCQ 2012 1M Ans: B ● ○ ○

Since X and Y are independent and uniformly distributed over $[-1, 1]$,

$$P\left(\max(X, Y) < \frac{1}{2}\right) = P\left(X < \frac{1}{2},\ Y < \frac{1}{2}\right).$$

By independence,

$$P\left(X < \frac{1}{2},\ Y < \frac{1}{2}\right) = P\left(X < \frac{1}{2}\right) P\left(Y < \frac{1}{2}\right).$$

For a uniform distribution on $[-1, 1]$,

$$P\left(X < \frac{1}{2}\right) = \frac{\frac{1}{2} - (-1)}{1 - (-1)} = \frac{\frac{3}{2}}{2} = \frac{3}{4}.$$

Similarly,

$$P\left(Y < \frac{1}{2}\right) = \frac{3}{4}.$$

Therefore,

$$P\left(\max(X,Y) < \frac{1}{2}\right) = \left(\frac{3}{4}\right)^2 = \frac{9}{16}.$$

$$\boxed{\text{(b) } \frac{9}{16}}$$

Q3.11.1 MCQ 2011 2M Ans: A ● ● ○

For a uniformly distributed random variable over the interval

$$[-a, a]$$

the mean is zero.
Given that the mean square value is equal to the variance.
Since mean is zero,

$$\text{Variance} = E[X^2]$$

Also, r.m.s. value is defined as

$$X_{\text{rms}} = \sqrt{E[X^2]}$$

For a uniform distribution over $[-a, a]$,

$$E[X^2] = \frac{a^2}{3}$$

Hence,

$$X_{\text{rms}} = \sqrt{\frac{a^2}{3}}$$

$$X_{\text{rms}} = \frac{a}{\sqrt{3}}$$

$$\boxed{\text{(a) } \frac{a}{\sqrt{3}}}$$

Key Points

- For zero mean random variables:

$$\text{Variance} = E[X^2]$$

- r.m.s. value is:

$$\sqrt{E[X^2]}$$

Q3.10.1 MCQ 2010 2M Ans: C ● ○ ○

Given that the first removed ball is white, one white ball is already removed.
Hence, the remaining balls are:

3 white balls and 3 red balls

Total remaining balls= 6
Therefore, the probability that the second removed ball is red is

$$\frac{3}{6} = \frac{1}{2}$$

$$\boxed{\text{(c) } \frac{1}{2}}$$

Q3.09.1 MCQ 2009 2M Ans: B ● ● ○

Let the event "at least two people have the same birth date" be E.
We first calculate the complement event:

$$E^c = \text{all birthdays are different}$$

Since April has 30 days,

$$P(E^c) = \frac{30}{30} \times \frac{29}{30} \times \frac{28}{30} \times \cdots \times \frac{30 - N + 1}{30}$$

We need

$$P(E) > 0.5$$

That is,

$$1 - P(E^c) > 0.5$$

$$P(E^c) < 0.5$$

Checking the options:
For $N = 7$,

$$P(E^c) = \frac{30 \times 29 \times 28 \times 27 \times 26 \times 25 \times 24}{30^7} \approx 0.459$$

Hence,

$$P(E) = 1 - 0.459 = 0.541 > 0.5$$

Now for $N = 6$,

$$P(E^c) \approx 0.593$$

$$P(E) \approx 0.407 < 0.5$$

Therefore, the smallest value of N is

$$7$$

$$\boxed{\text{(b) } 7}$$

Q3.08.1 MCQ 2008 1M Ans: C ● ○ ○

For a uniform random variable over the interval

$$0 \le X \le 1$$

the probability density function is

$$f(x) = 1, \qquad 0 \le x \le 1$$

The expectation of X^3 is

$$E(X^3) = \int_0^1 x^3 f(x)\, dx$$

Since $f(x) = 1$,

$$E(X^3) = \int_0^1 x^3\, dx$$

$$E(X^3) = \left[\frac{x^4}{4}\right]_0^1$$

$$E(X^3) = \frac{1}{4}$$

$$\boxed{\text{(c) } \frac{1}{4}}$$

Key Points

- For a continuous random variable:

$$E[g(X)] = \int g(x) f(x)\, dx$$

Q3.07.1 MCQ 2007 1M Ans: C ● ○ ○

From the given distribution:

$$P(1) = \frac{1}{4}, \quad P(2) = P(3) = P(4) = P(5) = \frac{1}{8}, \quad P(6) = \frac{1}{4}$$

For three independent dice, the probability of occurrence of values $1, 5, 6$ is

$$P(1) \times P(5) \times P(6)$$

$$= \frac{1}{4} \times \frac{1}{8} \times \frac{1}{4}$$

$$= \frac{1}{128}$$

$$\boxed{\text{(c) } \frac{1}{128}}$$

Q3.06.1 MCQ 2006 2M Ans: C ● ● ○

Total number of outcomes when two fair dice are rolled:

$$36$$

Possible values of the sum r are:

$$2, 3, 4, \ldots, 12$$

Now check each statement.
For option A:

$$P(r > 6) = \frac{21}{36} = \frac{7}{12} \neq \frac{1}{6}$$

Hence, option A is false.
For option B:
$r/3$ is an integer when

$$r = 3, 6, 9, 12$$

Number of favourable outcomes:

$$2 + 5 + 4 + 1 = 12$$

Therefore,

$$P = \frac{12}{36} = \frac{1}{3} \neq \frac{5}{6}$$

Hence, option B is false.
For option C:
We need

$$r = 8 \,\Big|\, \frac{r}{4} \text{ is an integer}$$

Since

$$\frac{8}{4} = 2$$

is an integer,

$$P\left(r = 8 \middle| \frac{r}{4} \text{ is an integer}\right) = \frac{P(r = 8)}{P\left(\frac{r}{4} \text{ is an integer}\right)}$$

Now,

$$\frac{r}{4} \text{ is an integer when } r = 4, 8, 12$$

Number of outcomes:

$$3 + 5 + 1 = 9$$

Hence,

$$P\left(\frac{r}{4} \text{ is an integer}\right) = \frac{9}{36}$$

Also,

$$P(r = 8) = \frac{5}{36}$$

Therefore,

$$P\left(r = 8 \middle| \frac{r}{4} \text{ is an integer}\right) = \frac{5/36}{9/36} = \frac{5}{9}$$

Hence, option C is correct.
For option D:
We need

$$P\left(r = 6 \middle| \frac{r}{5} \text{ is an integer}\right)$$

But

$$\frac{r}{5} \text{ is an integer when } r = 5, 10$$

Since $r = 6$ does not satisfy the condition,

$$P\left(r = 6 \middle| \frac{r}{5} \text{ is an integer}\right) = 0 \neq \frac{1}{18}$$

Hence, option D is false.

$$\boxed{\text{(c) } Pr(r = 8 \mid r/4 \text{ is an integer}) = \frac{5}{9}}$$

Mistakes to be avoided

- Do not confuse "|" with division; it denotes "given that".
- Carefully identify the conditioning event before applying probability formulas.

Q3.05.1 MCQ | 2005 | 1M | Ans: D | ● ○ ○

We check each statement using probability properties.
For option A:
If P and Q are independent,

$$P(P \cap Q) = P(P)\,P(Q)$$

This is not necessarily zero.
Hence, option A is false.
For option B:

Using the addition theorem,

$$P(P \cup Q) = P(P) + P(Q) - P(P \cap Q)$$

Therefore,

$$P(P \cap Q) = P(P) + P(Q) - P(P \cup Q)$$

Since

$$P(P \cup Q) \leq 1$$

option B is not always true.
Hence, option B is false.
For option C:
If P and Q are mutually exclusive,

$$P(P \cap Q) = 0$$

But for independence,

$$P(P \cap Q) = P(P)\,P(Q)$$

These are equal only if one event has zero probability.
Hence, mutually exclusive events are not necessarily independent.
So, option C is false.
For option D:
Since

$$P \cap Q \subseteq P$$

we must have

$$P(P \cap Q) \leq P(P)$$

Hence, option D is true.

$$\boxed{\text{(d) } P(P \cap Q) \leq P(P)}$$

Q3.05.2 MCQ 2005 2M Ans: B ● ○ ○

Let the event that the first toss is a head be

$$F$$

Given that the first toss is already a head, to get exactly two heads in three tosses, we need exactly one more head in the remaining two tosses.
Possible outcomes for the last two tosses are:

$$HH,\ HT,\ TH,\ TT$$

Favourable outcomes:

$$HT,\ TH$$

Hence,

$$P(\text{exactly two heads} \mid F) = \frac{2}{4}$$

$$= \frac{1}{2}$$

$$\boxed{\text{(b) } \frac{1}{2}}$$

SOLUTIONS — Probability and Statistics

IN

Instrumentation Engineering

PrepFusion

Q3.26.1 NAT | 2026 | 1M | Ans: 1 | ● ○ ○

Arrange the data in ascending order:

$$[-3, -1, 0, 0, 1, 1, 2, 2]$$

Total number of observations:

$$n = 8$$

Since n is even, median is the average of the middle two observations.
Middle terms are the 4th and 5th terms:

$$0 \text{ and } 1$$

Hence,

$$\text{Median} = \frac{0+1}{2} = \frac{1}{2}$$

Rounded to the nearest integer:

$$1$$

$$\boxed{1}$$

Key Points

- Arrange the data in ascending or descending order before finding median.
- For even number of observations:

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} + \left(\frac{n}{2}+1\right)^{\text{th}}}{2}$$

Q3.26.2 MCQ | 2026 | 2M | Ans: C | ● ○ ○

Probability that the first fruit is an apple:

$$P(A_1) = \frac{3}{5}$$

After removing one apple, remaining fruits are:

$$2 \text{ apples and } 2 \text{ oranges}$$

Hence, probability that the second fruit is an orange:

$$P(O_2|A_1) = \frac{2}{4} = \frac{1}{2}$$

Therefore,

$$P(A_1 \cap O_2) = P(A_1) \times P(O_2|A_1)$$

$$= \frac{3}{5} \times \frac{1}{2}$$

$$= \frac{3}{10}$$

$$\boxed{\text{(c) } \frac{3}{10}}$$

Q3.25.1 NAT | 2025 | 2M | Ans: 0.74 | ● ○ ○

Let X be the number of classes missed.
Given:

$$n = 10, \qquad p = 0.1$$

Here, X follows a binomial distribution.
We need:

$$P(X \le 1) = P(X = 0) + P(X = 1)$$

$$P(X = 0) = \binom{10}{0}(0.1)^0(0.9)^{10}$$

$$= 0.9^{10} \approx 0.3487$$

Also,

$$P(X = 1) = \binom{10}{1}(0.1)^1(0.9)^9$$

$$= 10(0.1)(0.9)^9 \approx 0.3874$$

Therefore,

$$P(X \le 1) = 0.3487 + 0.3874$$

$$= 0.7361$$

Rounded to two decimal places:

$$0.74$$

$$\boxed{0.74}$$

Q3.24.1 NAT | 2024 | 1M | Ans: 0.5 | ● ○ ○

Pool-A has:

$$4 \text{ blue}, \quad 4 \text{ red}$$

Pool-B has:

$$5 \text{ blue}, \quad 3 \text{ red}$$

One team is selected from each pool.
Required event:

$$(\text{Blue from A and Red from B})$$

or

$$(\text{Red from A and Blue from B})$$

Therefore,

$$P = \left(\frac{4}{8} \times \frac{3}{8}\right) + \left(\frac{4}{8} \times \frac{5}{8}\right)$$

$$= \frac{12}{64} + \frac{20}{64}$$

$$= \frac{32}{64} = \frac{1}{2}$$

Rounded to one decimal place:

$$0.5$$

$$\boxed{0.5}$$

Q3.24.2 NAT 2024 2M Ans: 0.135 ● ○ ○

Given probability density function:

$$f_X(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

We need to find:

$$P(X > 2)$$

Using the definition of probability for a continuous random variable,

$$P(X > 2) = \int_2^{\infty} e^{-x}\, dx$$

$$= [-e^{-x}]_2^{\infty}$$

$$= 0 - (-e^{-2})$$

$$= e^{-2}$$

$$= e^{-2} \approx 0.1353$$

Rounded to three decimal places:

$$0.135$$

$$\boxed{0.135}$$

Q3.23.1 NAT 2023 1M Ans: -0.5 ● ○ ○

Given:

$$P(X = 0) = 0.25, \qquad P(X = 1) = 0.5$$

Since total probability is 1,

$$P(X = 2) = 1 - 0.25 - 0.5 = 0.25$$

Now,

$$E[X] = \sum xP(X = x)$$

$$= 0(0.25) + 1(0.5) + 2(0.25)$$

$$= 0 + 0.5 + 0.5 = 1$$

Also,

$$E[X^2] = \sum x^2 P(X = x)$$

$$= 0^2(0.25) + 1^2(0.5) + 2^2(0.25)$$

$$= 0 + 0.5 + 1 = 1.5$$

Therefore,

$$E[X] - E[X^2] = 1 - 1.5$$

$$= -0.5$$

Rounded to one decimal place:

$$-0.5$$

$$\boxed{-0.5}$$

Q3.23.2 NAT | 2023 | 2M | Ans: 240 | ● ○ ○

We need to form five-digit numbers using the digits

$$3, 4, 5, 6$$

with exactly one digit repeated twice.

Choose the digit to be repeated = 4 ways

Now, the five digits consist of:

$$\{a, a, b, c, d\}$$

where one digit appears twice.
Number of distinct arrangements:

$$\frac{5!}{2!}$$

$$= \frac{120}{2} = 60$$

Therefore, total numbers formed:

$$4 \times 60$$

$$= 240$$

$$\boxed{240}$$

Q3.22.1 NAT | 2022 | 1M | Ans: 0.33 | ● ○ ○

Total balls:

$$6 \text{ red} + 4 \text{ blue} = 10$$

Since the balls are drawn without replacement, probabilities change after each draw.
Probability that the 2nd ball is red:

$$\frac{6}{10}$$

After one red ball is drawn in the second draw, remaining red balls:

$$5$$

Remaining total balls:

$$9$$

Therefore, probability that the 3rd ball is also red:

$$\frac{5}{9}$$

Hence,

$$P(2^{\text{nd}} \text{ and } 3^{\text{rd}} \text{ balls are red}) = \frac{6}{10} \times \frac{5}{9}$$

$$= \frac{30}{90} = \frac{1}{3}$$

$$\boxed{0.33}$$

Q3.20.1 NAT | 2020 | 1M | Ans: 0.09 | ● ○ ○

Probability of scoring in a single attempt:

$$P(S) = 0.1$$

Probability of missing in a single attempt:

$$P(M) = 1 - 0.1 = 0.9$$

The player throws the ball twice independently.
Therefore,

$$P(\text{success in second attempt}) = P(M, S)$$

$$= (0.9)(0.1)$$

$$= (0.09)$$

$$\boxed{0.09}$$

Q3.20.2 NAT | 2020 | 2M | Ans: 0.7 | ● ○ ○

Let

$$B_1 = \text{ball chosen from bag } B_1$$

and

$$R = \text{red ball is obtained}$$

Using Bayes' theorem,

$$P(B_1|R) = \frac{P(R|B_1)P(B_1)}{P(R)}$$

Since one of the two bags is chosen at random,

$$P(B_1) = P(B_2) = \frac{1}{2}$$

Also,

$$P(R|B_1) = \frac{7}{10}$$

and

$$P(R|B_2) = \frac{3}{10}$$

Now,

$$P(R) = P(R|B_1)P(B_1) + P(R|B_2)P(B_2)$$

$$= \frac{7}{10} \times \frac{1}{2} + \frac{3}{10} \times \frac{1}{2}$$

$$= \frac{7+3}{20} = \frac{10}{20} = \frac{1}{2}$$

Therefore,

$$P(B_1|R) = \frac{\frac{7}{10} \times \frac{1}{2}}{\frac{1}{2}}$$

$$= \frac{7}{10} = 0.7$$

$$\boxed{0.7}$$

Q3.19.1 MCQ | 2019 | 2M | Ans: A

For a valid probability density function (PDF), the total probability over its support must be equal to 1.
Given

$$p(x) = \frac{A}{x^{\mu}}, \qquad \mu > 1, \quad 1 \le x < \infty$$

and $p(x) = 0$ elsewhere.
Using the normalization condition,

$$\int_{-\infty}^{\infty} p(x)\, dx = 1$$

Since $p(x) = 0$ for $x < 1$,

$$\int_{1}^{\infty} \frac{A}{x^{\mu}}\, dx = 1$$

$$A \int_{1}^{\infty} x^{-\mu}\, dx = 1$$

$$A \left[\frac{x^{1-\mu}}{1-\mu} \right]_{1}^{\infty} = 1$$

Since $\mu > 1$, $x^{1-\mu} \to 0$ as $x \to \infty$.

$$A \left(0 - \frac{1}{1-\mu} \right) = 1$$

$$\frac{A}{\mu - 1} = 1$$

$$A = \mu - 1$$

$$\boxed{\text{(a) } \mu - 1}$$

Key Points

- A probability density function must satisfy $\int_{-\infty}^{\infty} p(x)\, dx = 1$.
- The improper integral $\int_{1}^{\infty} x^{-\mu} dx$ converges only when $\mu > 1$.

- For $\mu > 1$,

$$\int_1^{\infty} x^{-\mu} dx = \frac{1}{\mu - 1}.$$

Q3.19.2 MCQ 2019 1M Ans: A ● ○ ○

The required probability is

$$P(\text{First red and second green}) = P(R_1) \times P(G_2|R_1)$$

Probability that the first ball is red:

$$P(R_1) = \frac{8}{16}$$

After drawing one red ball, 15 balls remain, of which 8 are green.

$$P(G_2|R_1) = \frac{8}{15}$$

Therefore,

$$P(R_1 \cap G_2) = \frac{8}{16} \times \frac{8}{15} = \frac{4}{15}$$

$$\boxed{\text{(a) } \frac{4}{15}}$$

Q3.18.1 MCQ 2018 1M Ans: D ● ○ ○

Given

$$\text{Var}(X) = 1, \qquad \text{Var}(Y) = 2$$

and

$$Z = X - Y.$$

Since X and Y are independent,

$$\text{Cov}(X, Y) = 0.$$

Hence,

$$\text{Var}(Z) = \text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\,\text{Cov}(X, Y)$$

$$= 1 + 2 - 0 = 3.$$

$$\boxed{\text{(d) } 3}$$

Key Points

- For any two random variables,

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2\,\text{Cov}(X, Y).$$

- Independent random variables have zero covariance.

Q3.18.2 MCQ 2018 1M Ans: B ● ○ ○

For the head to occur for the third time on the fifth toss:

- The first 4 tosses must contain exactly 2 heads.
- The 5th toss must be a head.

Number of ways to get exactly 2 heads in the first 4 tosses:

$$\binom{4}{2} = 6$$

Probability of each such sequence:

$$\left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

Hence,

$$P(\text{2 heads in first 4 tosses}) = \binom{4}{2}\frac{1}{16} = \frac{6}{16}$$

Also,

$$P(\text{Head on 5th toss}) = \frac{1}{2}$$

Therefore,

$$P(\text{3rd head occurs on 5th toss}) = \frac{6}{16} \times \frac{1}{2} = \frac{3}{16}.$$

$$\boxed{\text{(b) } \frac{3}{16}}$$

Key Points

- For the rth success on the nth trial, the first $(n-1)$ trials must contain exactly $(r-1)$ successes.
- Use combinations to count favourable arrangements.

Q3.18.3 MCQ 2018 2M Ans: B ● ○ ○

Since both bags contain an equal number of balls, let each bag contain N balls.
Red balls in Bag $A = 0.20N$
Red balls in Bag $B = 0.30N$
After mixing the two bags,

$$\text{Total red balls} = 0.20N + 0.30N = 0.50N$$

$$\text{Total balls} = N + N = 2N$$

Therefore,

$$P(\text{Red}) = \frac{0.50N}{2N} = 0.25 = 25\%.$$

$$\boxed{\text{(b) } 25\%}$$

Q3.17.1 MCQ 2017 2M Ans: C ● ○ ○

Let

$$F = \text{High Fidelity}, \qquad S = \text{High Selectivity}.$$

Given,

$$P(F) = 0.81$$

and

$$P(F \cap S) = 0.18.$$

We need to find

$$P(S|F).$$

Using the conditional probability formula,

$$P(S|F) = \frac{P(F \cap S)}{P(F)}$$

$$= \frac{0.18}{0.81}$$

$$= \frac{18}{81} = \frac{2}{9} \approx 0.222.$$

(c) 0.222

Q3.16.1 MCQ 2016 2M Ans: C ● ○ ○

Let R_1 and G_1 denote the events that the first ball drawn is red and green, respectively.
The probability of getting a red ball in the second draw is obtained using the law of total probability.
If the first ball is red:

$$P(R_1) = \frac{5}{12}$$

After replacement and adding another red ball, the urn contains 6 red and 7 green balls.

$$P(R_2|R_1) = \frac{6}{13}$$

If the first ball is green:

$$P(G_1) = \frac{7}{12}$$

After replacement and adding another green ball, the urn contains 5 red and 8 green balls.

$$P(R_2|G_1) = \frac{5}{13}$$

Therefore,

$$P(R_2) = P(R_2|R_1)P(R_1) + P(R_2|G_1)P(G_1)$$

$$= \frac{6}{13} \cdot \frac{5}{12} + \frac{5}{13} \cdot \frac{7}{12}$$

$$= \frac{30 + 35}{156} = \frac{65}{156}.$$

$$\boxed{\text{(a) } \frac{65}{156}}$$

Q3.15.1 MCQ | 2015 | 2M | Ans: B | ● ○ ○

Let X be the number of defective thermistors among 10 thermistors.
Since each thermistor is defective with probability

$$p = 0.1,$$

X follows a binomial distribution:

$$X \sim \text{Binomial}(n = 10, p = 0.1).$$

The probability of exactly 3 defective thermistors is

$$P(X = 3) = \binom{10}{3}(0.1)^3(0.9)^7$$

$$= 120 \times 0.001 \times 0.4783$$

$$\approx 0.0574.$$

Hence,

$$P(X = 3) \approx 0.057.$$

$$\boxed{\text{(b) } 0.057}$$

Q3.15.2 NAT | 2015 | 2M | Ans: 4 | ● ○ ○

Given the probability density function

$$P_X(x) = e^{-x}, \qquad x \geq 0$$

and

$$g_2(x) = e^{3x/4}.$$

The expected value is

$$E[g_2(X)] = \int_0^\infty g_2(x)\, P_X(x)\, dx$$

$$= \int_0^\infty e^{3x/4}\, e^{-x}\, dx = \int_0^\infty e^{-x/4}\, dx.$$

Evaluating the integral,

$$E[g_2(X)] = \left[-4e^{-x/4}\right]_0^{\infty}$$

$$= 0 - (-4) = 4.$$

$$\boxed{4}$$

Q3.14.1 NAT 2014 1M Ans: 0.893 ● ○ ○

For successful processing, machine C must work and at least one of A or B must work.

$$P(\text{A and B fail}) = P_A P_B = (0.15)(0.05) = 0.0075$$

$$P(\text{at least one of A or B works}) = 1 - 0.0075 = 0.9925$$

$$P(C \text{ works}) = 1 - P_C = 1 - 0.1 = 0.9$$

Hence,

$$P(\text{successful processing}) = 0.9925 \times 0.9 = 0.89325$$

$$\boxed{0.893}$$

Key Points

- Parallel stage (A and B): system works if at least one machine works.
- Series stage with C: overall success probability is the product of stage success probabilities.

Q3.14.2 NAT 2014 1M Ans: 2 ● ○ ○

Given the probability density function

$$f(x) = \frac{e^{-x/2}}{K}, \qquad 0 \le x < \infty$$

For a valid PDF,

$$\int_0^{\infty} f(x)\,dx = 1$$

$$\frac{1}{K}\int_0^{\infty} e^{-x/2}\,dx = 1$$

Using

$$\int_0^{\infty} e^{-x/2}\,dx = 2,$$

we get

$$\frac{2}{K} = 1$$

$$K = 2$$

$\boxed{2}$

Q3.13.1 MCQ | 2013 | 1M | Ans: A | ● ○ ○

Given

$$f(x) = e^{-x}, \qquad 0 < x < \infty$$

The required probability is

$$P(X > 1) = \int_1^{\infty} e^{-x}\, dx$$

$$P(X > 1) = [-e^{-x}]_1^{\infty} = 0 + \frac{1}{e} = e^{-1}$$

$$P(X > 1) \approx 0.3679$$

$\boxed{\text{(a) } 0.368}$

Q3.12.1 MCQ | 2012 | 1M | Ans: B | ● ○ ○

Since X and Y are independent and uniformly distributed on $[-1, 1]$,

$$P\left(\max(X, Y) < \frac{1}{2}\right) = P\left(X < \frac{1}{2}, Y < \frac{1}{2}\right)$$

For a uniform distribution on $[-1, 1]$,

$$P\left(X < \frac{1}{2}\right) = \frac{\frac{1}{2} - (-1)}{2} = \frac{3}{4}$$

Similarly,

$$P\left(Y < \frac{1}{2}\right) = \frac{3}{4}$$

Using independence,

$$P\left(\max(X, Y) < \frac{1}{2}\right) = \left(\frac{3}{4}\right)^2 = \frac{9}{16}$$

$\boxed{\text{(B) } \frac{9}{16}}$

Q3.11.1 MCQ | 2011 | 2M | Ans: D | ● ○ ○

The product is even if at least one selected chip is even.
Using the complement approach, the product is odd only when both selected chips are odd.
For Box 1:

$$\text{Odd chips} = \{3, 9, 15\}$$

$$P(\text{odd from Box 1}) = \frac{3}{5}$$

For Box 2:

$$\text{Odd chips} = \{11, 21\}$$

$$P(\text{odd from Box 2}) = \frac{2}{5}$$

$$P(\text{product odd}) = \frac{3}{5} \times \frac{2}{5} = \frac{6}{25}$$

Hence,

$$P(\text{product even}) = 1 - \frac{6}{25} = \frac{19}{25}$$

$$\boxed{\text{(d) } \frac{19}{25}}$$

Q3.10.1 MCQ | 2010 | 1M | Ans: C

Given

$$\mu = 10\,\text{mm}, \qquad \sigma = 0.05\,\text{mm}$$

Required:

$$P(X > 10.15)$$

Standardizing,

$$Z = \frac{10.15 - 10}{0.05} = 3$$

For a standard normal distribution,

$$P(Z > 3) \approx 0.00135$$

Therefore, among 10000 ball bearings,

$$10000 \times 0.00135 = 13.5$$

$$\approx 15$$

$$\boxed{\text{(c) } 15}$$

Key Points

- Convert a normal random variable to the standard normal variable using $Z = \frac{X-\mu}{\sigma}$.
- The upper-tail probability beyond 3σ is approximately 0.00135.

Q3.09.1 MCQ 2009 2M Ans: C ● ○ ○

Let

$$P(+) = 0.01, \qquad P(-) = 0.99$$

Given

$$P(\text{incorrect} \mid +) = 0.12, \qquad P(\text{incorrect} \mid -) = 0.15$$

Using total probability,

$$P(\text{incorrect}) = P(+)\,P(\text{incorrect} \mid +) + P(-)\,P(\text{incorrect} \mid -)$$

$$= (0.01)(0.12) + (0.99)(0.15)$$

$$= 0.0012 + 0.1485 = 0.1497$$

(c) 0.1497

Q3.08.1 MCQ 2008 2M Ans: A ● ○ ○

For a uniform random variable distributed over $[a, b]$,

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$

Here,

$$a = 2, \qquad b = 10$$

$$\text{Var}(X) = \frac{(10-2)^2}{12} = \frac{64}{12} = \frac{16}{3}$$

(a) $\frac{16}{3}$

Q3.08.2 MCQ 2008 2M Ans: B ● ○ ○

Let X be Gaussian with mean 0 and standard deviation σ.
The cumulative distribution function (CDF) is

$$F(x) = P(X \le x)$$

Since the Gaussian distribution is symmetric about its mean,

$$P(X \le 0) = P(X \ge 0) = \frac{1}{2}$$

Therefore,

$$F(0) = P(X \le 0) = \frac{1}{2}$$

(b) 0.5

Mistakes to be avoided

- Do not confuse the PDF value at $x = 0$ with the CDF value at $x = 0$.
- The result is independent of the value of σ.

Q3.07.1 MCQ 2007 1M Ans: A ● ○ ○

Given the exponential probability density function

$$f(x) = \frac{1}{5}e^{-x/5}, \qquad x \geq 0$$

The required probability is

$$P(X > 5) = \int_5^{\infty} \frac{1}{5}e^{-x/5}\,dx$$

$$P(X > 5) = \left[-e^{-x/5}\right]_5^{\infty} = 0 - (-e^{-1}) = \frac{1}{e}$$

$$\boxed{\text{(a) } \frac{1}{e}}$$

Q3.06.1 MCQ 2006 2M Ans: C ● ○ ○

There are 3 terminals and each person chooses one terminal independently at random.
Total possible selections:

$$3 \times 3 = 9$$

For both choosing the same terminal:

$$(1, 1),\ (2, 2),\ (3, 3)$$

Hence,

$$P(\text{same terminal}) = \frac{3}{9} = \frac{1}{3}$$

Therefore,

$$P(\text{not same terminal}) = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\boxed{\text{(c) } \frac{2}{3}}$$

Q3.06.2 MCQ 2006 2M Ans: A ● ○ ○

The given probability density function is triangular with

$$\text{base} = (a + c) - a = c$$

and height equal to α.
Since the total area under a probability density function must be 1,

$$\text{Area of triangle} = 1$$

$$\frac{1}{2} \times c \times \alpha = 1$$

$$\alpha = \frac{2}{c}$$

$$\boxed{\text{(a) } \frac{2}{c}}$$

Q3.06.3 MCQ 2006 1M Ans: A ● ○ ○

A sum of two dice is even when both numbers are even or both are odd.
Each die has

$$P(\text{even}) = \frac{3}{6} = \frac{1}{2}, \qquad P(\text{odd}) = \frac{1}{2}$$

Hence,

$$P(\text{sum is even}) = P(\text{both even}) + P(\text{both odd})$$

$$= \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Therefore,

$$P(\text{sum is even}) = 0.5$$

$$\boxed{\text{(a) } 0.5}$$

Q3.05.1 MCQ 2005 1M Ans: D ● ○ ○

A leap year has

$$366 = 52 \times 7 + 2$$

days.
Thus, every weekday occurs 52 times and two consecutive weekdays occur one extra time.
For there to be 53 Sundays, Sunday must be one of these two extra days.
The first day of the year can be any of the 7 weekdays with equal probability.
Favorable cases:

$$\text{First day} = \text{Sunday or Saturday}$$

Hence,

$$P(53 \text{ Sundays}) = \frac{2}{7}$$

$$\boxed{\text{(d) } \frac{2}{7}}$$

Key Points

- A leap year contains 366 days = 52 weeks +2 days.
- The two extra consecutive days determine which weekdays occur 53 times.
- Sunday occurs 53 times when the extra days include Sunday.

PrepFusion

SOLUTIONS — Probability and Statistics

CS

Computer Science & IT

PrepFusion

Q3.26.1 MCQ 2026 2M Ans: C ● ○ ○

Let the first roll be a and the second roll be b.
Favourable outcomes satisfy: b is an integer multiple of a.

$$a = 1:\ 6 \text{ choices}, \quad a = 2:\ 3 \text{ choices}, \quad a = 3:\ 2 \text{ choices}$$

$$a = 4:\ 1 \text{ choice}, \quad a = 5:\ 1 \text{ choice}, \quad a = 6:\ 1 \text{ choice}$$

Hence, total favourable outcomes

$$6 + 3 + 2 + 1 + 1 + 1 = 14$$

Total outcomes $= 6 \times 6 = 36$.

$$P = \frac{14}{36} = \frac{7}{18}$$

$$\boxed{\text{(c) } \frac{7}{18}}$$

Q3.26.2 MCQ 2026 1M Ans: B ● ○ ○

Initially the urn contains $1R$ and $1B$.
For equal numbers after two steps, the two selected colours must be different.

$$P(R \text{ then } B) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

$$P(B \text{ then } R) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

Therefore,

$$P = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

$$\boxed{\text{(b) } \frac{1}{3}}$$

Q3.26.3 NAT 2026 2M Ans: 4.25 ● ○ ○

Given

$$P(X = 1) = P(X = 2) = P(X = 5) = P(X = 7) = \frac{1}{6}$$

and

$$P(X = 3) = P(X = 4) = P(X = 6) = P(X = 8) = \frac{1}{12}$$

Using

$$E[X] = \sum x\, P(X = x)$$

$$E[X] = \frac{1 + 2 + 5 + 7}{6} + \frac{3 + 4 + 6 + 8}{12}$$

$$= \frac{15}{6} + \frac{21}{12} = 2.5 + 1.75 = 4.25$$

$$\boxed{4.25}$$

Q3.26.4 MCQ 2026 1M Ans: A ● ○ ○

Let a cloudy day be a success with probability

$$p = 0.5$$

In 4 independent days, probability of exactly 3 cloudy days and 1 sunny day is given by the binomial formula:

$$P = \binom{4}{3}(0.5)^3(0.5)^1$$

$$= 4\left(\frac{1}{2}\right)^4 = \frac{4}{16} = \frac{1}{4}$$

$$\boxed{\text{(a) } \frac{1}{4}}$$

Q3.26.5 MCQ 2026 1M Ans: B ● ○ ○

Given

$$f(x) = \frac{1}{3\sqrt{2\pi}} \exp\left(-\frac{x^2}{18}\right), \qquad x \in (-\infty, \infty)$$

Compare with the normal density function

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

We get

$$\mu = 0, \qquad 2\sigma^2 = 18$$

$$\sigma^2 = 9, \qquad \sigma = 3$$

Hence X follows a normal distribution.

$$\boxed{\text{(b) } X \text{ is a normal random variable}}$$

Q3.26.6 NAT 2026 2M Ans: 0.5 ● ● ○

Let

$$E_1 = \{\text{among tosses } 2, 4, 6, \text{ there are at least two heads}\}$$

and

$$E_2 = \{\text{among tosses } 1, 2, 3, 5, \text{ there are equal heads and tails}\}.$$

For tosses $2, 4, 6$,

$$P(E_1) = \frac{\binom{3}{2} + \binom{3}{3}}{2^3} = \frac{3+1}{8} = \frac{1}{2}.$$

Now count outcomes satisfying both E_1 and E_2.
Since tosses 1, 2, 3, 5 must contain exactly 2 heads,

$$\binom{4}{2} = 6$$

possible arrangements.

Head positions in $(1,2,3,5)$	Valid choices of $(4,6)$
$\{1,2\},\{2,3\},\{2,5\}$	3
$\{1,3\},\{1,5\},\{3,5\}$	1

Hence favourable outcomes for $E_1 \cap E_2$ are

$$3+3+3+1+1+1 = 12.$$

Since total outcomes of 6 tosses are

$$2^6 = 64,$$

$$P(E_1 \cap E_2) = \frac{12}{64} = \frac{3}{16}.$$

Therefore,

$$P(E_1 \mid E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{\frac{3}{16}}{\frac{6}{16}} = \frac{1}{2}.$$

Thus,

$$\boxed{0.5}$$

Q3.25.1 NAT | 2025 | 2M | Ans: 0.5 | ● ○ ○

For a quadratic polynomial $(x-\alpha)(x-\beta)$ to be square invariant, the set of its roots $\{\alpha,\beta\}$ must be identical to the set of its squared roots $\{\alpha^2,\beta^2\}$. This structural equality leads to two distinct cases:
Case 1: Direct Mapping The roots map directly to their own squares:

$$\alpha = \alpha^2 \quad \text{and} \quad \beta = \beta^2$$

Solving $z^2 - z = 0 \implies z(z-1) = 0$, the individual values must belong to the set $\{0,1\}$. This gives the following unique unordered root pairs:

$$\{\alpha,\beta\} \in \big\{\{0,0\},\{1,1\},\{0,1\}\big\}$$

Case 2: Cross Mapping The roots map interchangeably to each other's squares:

$$\alpha = \beta^2 \quad \text{and} \quad \beta = \alpha^2$$

Substituting β into the first equation yields:

$$\alpha = (\alpha^2)^2 \implies \alpha^4 - \alpha = 0 \implies \alpha(\alpha^3 - 1) = 0$$

- If $\alpha = 0$, then $\beta = 0$, yielding $\{0,0\}$ (already counted).

- If $\alpha^3 = 1$ and $\alpha \neq 1$, then α is a non-real complex cube root of unity (ω or ω^2). Since $\omega^2 = \omega^2$ and $(\omega^2)^2 = \omega$, this yields the unique unordered root pair:

$$\{\alpha, \beta\} = \{\omega, \omega^2\}$$

Probability Calculation: Combining both cases, the total number of unique square-invariant polynomials is 4:

$$\text{Total Outcomes} = \{\{0,0\}, \{1,1\}, \{0,1\}, \{\omega, \omega^2\}\}$$

The problem requires the roots of the chosen polynomial to be equal ($\alpha = \beta$). The favorable outcomes meeting this condition are $\{0,0\}$ and $\{1,1\}$.

$$\text{Favorable Outcomes} = 2$$

$$\text{Probability} = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}} = \frac{2}{4} = 0.5$$

$$\boxed{0.5}$$

Key Points

- Two polynomials are identical if and only if their roots sets are identical.
- Non-real complex roots of unity satisfy $\omega^3 = 1$ and $\omega^2 + \omega + 1 = 0$.

Mistakes to be avoided

- Do not double-count ordered pairs like $(0, 1)$ and $(1, 0)$, as they form the exact same physical polynomial $x^2 - x$.

Q3.25.2 NAT | 2025 | 2M | Ans: 0.74 ● ○ ○

Let the random division point be $X \sim U(0, 1)$.
The subinterval containing 0.4 has length

$$L = \begin{cases} 1 - X, & X < 0.4, \\ X, & X > 0.4. \end{cases}$$

Hence

$$E[L] = \int_0^{0.4} (1 - x)\, dx + \int_{0.4}^{1} x\, dx.$$

Calculating,

$$E[L] = \left[x - \frac{x^2}{2}\right]_0^{0.4} + \left[\frac{x^2}{2}\right]_{0.4}^{1}$$

$$= (0.4 - 0.08) + (0.5 - 0.08)$$

$$= 0.32 + 0.42 = 0.74.$$

Therefore,

$$\boxed{0.74}$$

Q3.25.3 NAT 2025 1M Ans: 0.50 ● ○ ○

Let F denote choosing the fake coin and R denote choosing a regular coin.

$$P(F) = \frac{1}{5}, \qquad P(R) = \frac{4}{5}$$

Given two heads,

$$P(HH|F) = 1, \qquad P(HH|R) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Using Bayes' theorem,

$$P(F|HH) = \frac{P(HH|F)P(F)}{P(HH|F)P(F) + P(HH|R)P(R)}$$

$$= \frac{1 \cdot \frac{1}{5}}{1 \cdot \frac{1}{5} + \frac{1}{4} \cdot \frac{4}{5}} = \frac{\frac{1}{5}}{\frac{1}{5} + \frac{1}{5}} = \frac{1}{2} = 0.50$$

$$\boxed{0.50}$$

Q3.25.4 NAT 2025 2M Ans: 0.302 ● ○ ○

Since $P(x)$ is a probability density function,

$$\int_1^4 Cx^2\,dx = 1$$

$$C\left[\frac{x^3}{3}\right]_1^4 = C \cdot \frac{64 - 1}{3} = 21C = 1$$

$$C = \frac{1}{21}$$

Required probability:

$$P(2 \le x \le 3) = \int_2^3 \frac{1}{21}x^2\,dx$$

$$= \frac{1}{21}\left[\frac{x^3}{3}\right]_2^3 = \frac{1}{21} \cdot \frac{27 - 8}{3} = \frac{19}{63} \approx 0.301587$$

$$\boxed{0.302}$$

Q3.24.1 MCQ 2024 1M Ans: A ● ○ ○

The total number of outcomes when six unbiased dice are rolled is

$$6^6$$

For all numbers to be distinct, the six dice must show 1, 2, 3, 4, 5, 6 in any order.

$$\text{Favourable outcomes} = 6!$$

Hence,

$$P = \frac{6!}{6^6} = \frac{720}{46656} = \frac{5}{324}$$

$$\boxed{\text{(a) } \frac{5}{324}}$$

Q3.24.2 MCQ 2024 2M Ans: D ● ○ ○

Given that the random variables x and y take real values within the closed interval $[0, 1]$, we can establish the following strict bounds:

$$x \geq 0 \quad \text{and} \quad y \leq 1$$

Since $y \leq 1$ and x is a non-negative quantity ($x \geq 0$), multiplying both sides of the inequality $y \leq 1$ by x preserves the inequality direction:

$$xy \leq x(1)$$

$$xy \leq x$$

Given that $z = xy$, we can substitute z into the inequality:

$$z \leq x$$

Taking the expectation (mean value) on both sides of a valid inequality preserves the inequality relationship:

$$E[z] \leq E[x]$$

Using the problem's notation where $E[z] = \bar{z}$ and $E[x] = \bar{x}$:

$$\bar{z} \leq \bar{x}$$

$$\boxed{\text{(d) } \bar{z} \leq \bar{x}}$$

Key Points

- For any random variables x and y, the mean of their product is related to their individual means and covariance by:

$$\bar{z} = E[xy] = E[x]E[y] + \text{Cov}(x, y) = \bar{x}\bar{y} + \text{Cov}(x, y)$$

- Since the variables are not necessarily independent, $\text{Cov}(x, y)$ can be positive, negative, or zero. Thus, no deterministic relationship can be established between $\bar{z}$ and $\bar{x}\bar{y}$.

Mistakes to be avoided

- Do not assume x and y are independent. Assuming independence leads to the incorrect conclusion $\bar{z} = \bar{x}\bar{y}$ (Option A).

Q3.24.3 NAT 2024 2M Ans: 0.375 ● ○ ○

Given that the first ball drawn is red, there are 9 red and 15 blue balls left, i.e. 24 balls in total.

$$P(\text{second red} \mid \text{first red}) = \frac{9}{24}$$

Since the first ball is already known to be red, the probability that both balls are red equals the probability that the second ball is red.

$$P(\text{both red} \mid \text{first red}) = \frac{9}{24} = \frac{3}{8} = 0.375$$

$$\boxed{0.375}$$

Q3.24.4 MSQ 2024 1M Ans: A,C ● ○ ○

Given

$$P(A) = 0.3, \qquad P(B) = 0.5, \qquad P(A \cap B) = 0.1$$

For option A:

$$P(A \cap B^c) = P(A) - P(A \cap B) = 0.3 - 0.1 = 0.2$$

Hence A is true.
For option B:

$$P(A)P(B) = 0.3 \times 0.5 = 0.15$$

Since

$$P(A \cap B) = 0.1 \neq 0.15$$

events A and B are not independent. Hence B is false.
For option C:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= 0.3 + 0.5 - 0.1 = 0.7$$

Hence C is true.
For option D:

$$P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - 0.7 = 0.3$$

Hence D is false as stated value is 0.4.

(a) $P(A \cap B^c) = 0.2$, where B^c is the complement of the event B

(c) $P(A \cup B) = 0.7$

Q3.24.5 MCQ 2024 1M Ans: B ● ○ ○

Consider only the relative positions of $1, 2, 3, 4$ in a random permutation.
Let

$$X = \{1 \text{ occurs before } 2\}, \qquad Y = \{3 \text{ occurs before } 4\}$$

Among the two numbers 1 and 2, each order is equally likely, so

$$P(X) = \frac{1}{2}$$

Similarly,

$$P(Y) = \frac{1}{2}$$

The ordering of $(1, 2)$ is independent of the ordering of $(3, 4)$, hence

$$P(X \cap Y) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Since

$$P(X \cap Y) = P(X)P(Y),$$

the events are independent.

(b) The events X and Y are independent

Q3.23.1 MSQ 2023 2M Ans: C,D ● ○ ○

Sample space:

$$S = \{HH, HT, TH, TT\}$$

Given

$$A = \{HH\}, \qquad B = \{HH, HT\}, \qquad C = \{HH, TH\}$$

Hence

$$P(A) = \frac{1}{4}, \qquad P(B) = \frac{1}{2}, \qquad P(C) = \frac{1}{2}$$

For option A:

$$P(A \cap B) = P(A) = \frac{1}{4}$$

and

$$P(A)P(B) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$$

Not equal $\Rightarrow$ A and B are not independent.
For option B:

$$P(A \cap C) = P(A) = \frac{1}{4}$$

and

$$P(A)P(C) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$$

Not equal $\Rightarrow$ A and C are not independent.
For option C:

$$P(B \cap C) = P(HH) = \frac{1}{4}$$

and

$$P(B)P(C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Hence B and C are independent.
For option D:

$$P(B \mid C) = \frac{P(B \cap C)}{P(C)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2} = P(B)$$

Hence D is true.

(c) B and C are independent.
(d) Prob$(B \mid C)$ = Prob(B)

Q3.21.1 MCQ 2021 1M Ans: B ● ○ ○

The cumulative distribution function (CDF) of a Gaussian (normal) distribution is obtained by integrating its bell-shaped probability density function.
The Gaussian CDF starts near 0, increases gradually, rises steeply around the mean, and then levels off near 1.
Hence its graph has an S-shape (sigmoid shape).

(b) S-shaped

Q3.21.2 MCQ 2021 2M Ans: A ● ○ ○

This is a Pólya urn process. After each draw, the drawn ball is replaced along with another ball of the same colour.
Let R_n denote the event that the nth draw is red.
The expected proportion of red balls remains unchanged throughout the process. Therefore, for every trial,

$$P(R_n) = \frac{r}{r+b}$$

In particular,

$$P(R_4) = \frac{r}{r+b}$$

Hence the probability of drawing a red ball in the fourth trial is independent of the previous outcomes.

$$\boxed{\text{(a) } \frac{r}{r+b}}$$

Key Points

- In a Pólya urn, reinforcement preserves the expected colour proportion.

Mistakes to be avoided

Do not assume that the denominator becomes $r + b + 3$ without accounting for all possible colour histories. The unconditional probability remains $\frac{r}{r+b}$.

Q3.21.3 MCQ 2021 2M Ans: D ● ○ ○

Let the expected marks be calculated for both possible orders.
Order: QuesA → QuesB

$$P(A) = 0.8, \qquad M_A = 10$$

$$P(B) = 0.5, \qquad M_B = 20$$

Expected marks:

$$E_{AB} = 0.8 \times 0 + (0.8)(0.5) \times (10 + 20) + (0.8)(0.5) \times 10$$

Equivalently,

$$E_{AB} = 0.8\Big[0.5(30) + 0.5(10)\Big]$$

$$E_{AB} = 0.8(20) = 16$$

Order: QuesB → QuesA

$$E_{BA} = 0.5\Big[0.8(20 + 10) + 0.2(20)\Big]$$

$$E_{BA} = 0.5(24 + 4) = 14$$

Since

$$E_{AB} = 16 > E_{BA} = 14,$$

the optimal order is QuesA first and then QuesB.

$$\boxed{\text{(d) First QuesA and then QuesB, Expected marks } = 16}$$

Q3.21.4 NAT 2021 1M Ans: 15.49 ● ○ ○

For a binomial random variable,

$$X \sim \text{Binomial}(n = 1000,\ p = 0.4)$$

The standard deviation is

$$\sigma = \sqrt{np(1-p)}$$

$$\sigma = \sqrt{1000 \times 0.4 \times 0.6} = \sqrt{240}$$

$$\sigma \approx 15.4919$$

$$\boxed{15.49}$$

Q3.21.5 NAT 2021 2M Ans: 0.04 ● ○ ○

Let T denote the transmitted signal and R denote the received signal.
Given,

$$P(T = H) = 0.1, \qquad P(T = L) = 0.9$$

From the figure,

$$P(R = H|T = H) = 0.3, \qquad P(R = H|T = L) = 0.8$$

Using Bayes' theorem,

$$P(T = H|R = H) = \frac{P(R = H|T = H)\,P(T = H)}{P(R = H)}$$

where

$$P(R = H) = (0.3)(0.1) + (0.8)(0.9) = 0.75$$

Hence,

$$P(T = H|R = H) = \frac{(0.3)(0.1)}{0.75} = 0.04$$

$$\boxed{0.04}$$

Q3.20.1 NAT 2020 2M Ans: 0.5 ● ○ ○

Let

$$S = \sum_{i=1}^{n} a_i x_i$$

where $a_i, x_i \in \{0, 1\}$.
Since a is a non-zero vector, there exists at least one index k such that

$$a_k = 1.$$

For fixed values of all other x_i, the parity of S depends on x_k:

$$S = C + x_k,$$

where C is a constant.
Thus, for every choice of the remaining variables, exactly one of the two values $x_k = 0$ or $x_k = 1$ makes S odd.
Hence half of all 2^n vectors x give an odd sum.

$$P(S \text{ is odd}) = \frac{2^{n-1}}{2^n} = \frac{1}{2}.$$

$$\boxed{0.5}$$

Q3.19.1 NAT 2019 1M Ans: 0.501 ● ○ ○

Numbers are chosen uniformly from $\{1, 2, \ldots, 13\}$.
In 4-bit binary:

$$1 \text{ to } 7 \Rightarrow \text{MSB} = 0, \qquad 8 \text{ to } 13 \Rightarrow \text{MSB} = 1.$$

Hence

$$P(\text{MSB} = 0) = \frac{7}{13}, \qquad P(\text{MSB} = 1) = \frac{6}{13}.$$

Probability that both numbers have the same MSB:

$$\left(\frac{7}{13}\right)^2 + \left(\frac{6}{13}\right)^2 = \frac{85}{169} \approx 0.50296.$$

$$\boxed{0.503}$$

Q3.19.2 NAT 2019 2M Ans: 0.8 ● ○ ○

Given

$$f(x) = 3x^2 + 6xY + 3Y + 6.$$

For both roots to be real,

$$\Delta \geq 0.$$

$$\Delta = (6Y)^2 - 4(3)(3Y + 6) = 36Y^2 - 36Y - 72.$$

$$\Delta = 36(Y^2 - Y - 2) = 36(Y - 2)(Y + 1).$$

Since

$$Y \in (1, 6),$$

and $Y + 1 > 0$, the condition becomes

$$Y \geq 2.$$

Therefore,

$$P(\text{real roots}) = \frac{6-2}{6-1} = \frac{4}{5} = 0.8.$$

$$\boxed{0.8}$$

Key Points

- A quadratic has real roots when its discriminant is non-negative.

Q3.18.1 NAT 2018 1M Ans: 0.023 ● ○ ○

In one trial, P and Q roll independent fair dice.
Probability of a tie:

$$P(T) = \frac{6}{36} = \frac{1}{6}.$$

Probability of a win (non-tie):

$$P(W) = 1 - \frac{1}{6} = \frac{5}{6}.$$

For a win to occur on the third trial, the first two trials must be ties and the third must be a win.

$$P = \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right) = \frac{5}{216} \approx 0.023148.$$

$$\boxed{0.023}$$

Q3.18.2 NAT 2018 2M Ans: 0.60 ● ○ ○

Using Bayes' theorem,

$$P(H_G|H_D) = \frac{P(H_D|H_G)\,P(H_G)}{P(H_D)}.$$

Given

$$P(H_G) = 0.2, \quad P(M_G) = 0.5, \quad P(L_G) = 0.3$$

and from the table,

$$P(H_D|H_G) = 0.40, \quad P(H_D|M_G) = 0.10, \quad P(H_D|L_G) = 0.01.$$

Using total probability,

$$P(H_D) = 0.40(0.2) + 0.10(0.5) + 0.01(0.3)$$

$$P(H_D) = 0.08 + 0.05 + 0.003 = 0.133.$$

Hence,

$$P(H_G|H_D) = \frac{0.40 \times 0.2}{0.133} = \frac{0.08}{0.133} \approx 0.6015.$$

$$\boxed{0.60}$$

Q3.17.1 NAT 2017 2M Ans: 54 ● ○ ○

For a Poisson random variable with mean 5,

$$E[X] = 5, \qquad \text{Var}(X) = 5.$$

$$E[(X+2)^2] = E[X^2] + 4E[X] + 4.$$

Now,

$$E[X^2] = \text{Var}(X) + \{E[X]\}^2 = 5 + 25 = 30.$$

Therefore,

$$E[(X+2)^2] = 30 + 4(5) + 4 = 54.$$

$$\boxed{54}$$

Q3.17.2 NAT 2017 1M Ans: 0 ● ○ ○

Let

$$Y = \max(X, 0),$$

where $X \sim \mathcal{N}(0, \sigma^2)$.
Since the Gaussian distribution is symmetric about zero,

$$P(X \le 0) = \frac{1}{2}.$$

Whenever $X \le 0$,

$$Y = 0.$$

Thus,

$$P(Y = 0) = \frac{1}{2}.$$

A median m satisfies

$$P(Y \le m) \ge \frac{1}{2} \quad \text{and} \quad P(Y \ge m) \ge \frac{1}{2}.$$

Since

$$P(Y \le 0) = \frac{1}{2},$$

the median is

$$m = 0.$$

$$\boxed{0}$$

Key Points

- Half of the Gaussian probability mass lies below zero.
- The transformation $\max(X, 0)$ places all negative values at 0.

Q3.17.3 MCQ 2017 1M Ans: C ● ○ ○

A k-digit number must not contain the digits 0, 5 or 9.
For any digit position, the allowed digits are

$$\{1, 2, 3, 4, 6, 7, 8\},$$

i.e., 7 choices out of 10 digits.
Hence the probability that a digit is allowed is

$$\frac{7}{10} = 0.7.$$

Assuming independence across the k digit positions,

$$P(\text{no digit is } 0, 5, 9) = (0.7)^k.$$

$$\boxed{\text{(c) } 0.7^k}$$

Q3.16.1 NAT 2016 1M Ans: 0.5 ● ○ ○

For a valid probability density function,

$$\int_a^1 \frac{1}{x^2}\, dx = 1$$

Evaluating the integral,

$$\left[-\frac{1}{x}\right]_a^1 = -\frac{1}{1} + \frac{1}{a} = \frac{1}{a} - 1$$

Since the total probability must be 1,

$$\frac{1}{a} - 1 = 1$$

$$\frac{1}{a} = 2$$

$$a = \frac{1}{2}$$

$$\boxed{0.5}$$

Q3.16.2 NAT 2016 2M Ans: 0.33 ● ○ ○

Let p be the probability that the output is Y.
In one cycle of two coin tosses,

$$P(Y) = P(TH) = \frac{1}{4}$$

$$P(A) = P(HH \text{ or } HT) = \frac{2}{4} = \frac{1}{2}$$

$$P(\text{Repeat}) = P(TT) = \frac{1}{4}$$

Using recursion,

$$p = \frac{1}{4} + \frac{1}{4}p$$

$$\frac{3}{4}p = \frac{1}{4}$$

$$p = \frac{1}{3}$$

Hence,

$$p = 0.3333$$

$$\boxed{0.33}$$

Q3.16.3 NAT | 2016 | 1M | Ans: 0.55 ● ● ○

Let L denote the event that the bulb lasts more than 100 hours.
Given,

$$P(T_1) = P(T_2) = \frac{1}{2}$$

$$P(L|T_1) = 0.7, \qquad P(L|T_2) = 0.4$$

By the law of total probability,

$$P(L) = P(L|T_1)P(T_1) + P(L|T_2)P(T_2)$$

$$= 0.7\left(\frac{1}{2}\right) + 0.4\left(\frac{1}{2}\right)$$

$$= 0.35 + 0.20$$

$$= 0.55$$

$$\boxed{0.55}$$

Q3.14.1 NAT | 2014 | 1M | Ans: 11.90 ● ○ ○

The system is functional if at least 3 out of the 4 inspected computers are working.
There are 10 computers, of which 4 are working and 6 are not.

$$p = P(X = 3) + P(X = 4)$$

Using the hypergeometric distribution,

$$P(X = 3) = \frac{\binom{4}{3}\binom{6}{1}}{\binom{10}{4}} = \frac{24}{210}$$

$$P(X = 4) = \frac{\binom{4}{4}\binom{6}{0}}{\binom{10}{4}} = \frac{1}{210}$$

Therefore,

$$p = \frac{24+1}{210} = \frac{25}{210} = \frac{5}{42}$$

Hence,

$$100p = 100\left(\frac{5}{42}\right) = \frac{500}{42} = 11.9048$$

$$\boxed{11.90}$$

Q3.14.2 NAT 2014 2M Ans: 0.25 ● ○ ○

Since A and B are mutually exclusive,

$$P(A \cap B) = 0$$

Also,

$$A \cup B = S$$

Hence,

$$P(A) + P(B) = P(S) = 1$$

Therefore,

$$P(A) = 1 - P(B)$$

To maximize $P(A)P(B)$,

$$P(A)P(B) = P(B)\big(1 - P(B)\big)$$

The product of two non-negative numbers with fixed sum 1 is maximum when they are equal.

$$P(A) = P(B) = \frac{1}{2}$$

Thus,

$$P(A)P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$\boxed{0.25}$$

Key Points

- Mutually exclusive events satisfy $P(A \cap B) = 0$.
- For a fixed sum, the product is maximum when the numbers are equal.

Q3.14.3 NAT 2014 2M Ans: 0.26 ● ○ ○

Let N be the required count.
Using inclusion-exclusion on integers from 1 to 100:

$$|A_2| = 50, \quad |A_3| = 33, \quad |A_5| = 20$$

$$|A_2 \cap A_3| = 16, \quad |A_2 \cap A_5| = 10, \quad |A_3 \cap A_5| = 6$$

$$|A_2 \cap A_3 \cap A_5| = 3$$

Hence,

$$|A_2 \cup A_3 \cup A_5| = 50 + 33 + 20 - 16 - 10 - 6 + 3 = 74$$

Therefore,

$$N = 100 - 74 = 26$$

Probability

$$= \frac{26}{100} = 0.26$$

$$\boxed{0.26}$$

Q3.14.4 NAT | 2014 | 2M | Ans: 10 | ● ○ ○

Let the four dice outcomes be x_1, x_2, x_3, x_4.

$$x_1 + x_2 + x_3 + x_4 = 22, \qquad 1 \le x_i \le 6$$

Define

$$y_i = 6 - x_i$$

Then

$$y_1 + y_2 + y_3 + y_4 = 24 - 22 = 2, \qquad y_i \ge 0$$

The number of non-negative integer solutions of

$$y_1 + y_2 + y_3 + y_4 = 2$$

is

$$\binom{2+4-1}{4-1} = \binom{5}{3} = 10$$

Hence, the number of favorable outcomes is

$$X = 10$$

Therefore,

$$P(\text{sum} = 22) = \frac{10}{6^4} = \frac{10}{1296}$$

$$\boxed{10}$$

Q3.14.5 NAT 2014 1M Ans: 3.9 ● ○ ○

The nine words are

$$\text{The, quick, brown, fox, jumps, over, the, lazy, dog}$$

Their lengths are

$$3,\ 5,\ 5,\ 3,\ 5,\ 4,\ 3,\ 4,\ 3$$

Expected length

$$E(X) = \frac{3+5+5+3+5+4+3+4+3}{9}$$

$$= \frac{35}{9} = 3.888\ldots$$

Rounded to one decimal place,

$$E(X) = 3.9$$

$$\boxed{3.9}$$

Q3.14.6 NAT 2014 1M Ans: 0.25 ● ○ ○

Let the breaking point be X, uniformly distributed on $(0, 1)$.
The shorter piece has length

$$Y = \min(X, 1 - X)$$

Using symmetry,

$$E(Y) = 2\int_0^{1/2} x\,dx$$

$$= 2\left[\frac{x^2}{2}\right]_0^{1/2}$$

$$= \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

Thus the expected length of the shorter stick is

$$\frac{1}{4}$$

Key Points

- The shorter segment is $\min(X, 1 - X)$.
- Use symmetry of the uniform distribution about $x = \frac{1}{2}$.

$$\boxed{0.25}$$

Q3.13.1 MCQ | 2013 | 1M | Ans: C | ● ○ ○

For a Poisson random variable with mean 3,

$$P(X = k) = \frac{e^{-3}3^k}{k!}$$

The probability of observing fewer than 3 cars is

$$P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$= e^{-3}\left(\frac{3^0}{0!} + \frac{3^1}{1!} + \frac{3^2}{2!}\right)$$

$$= e^{-3}\left(1 + 3 + \frac{9}{2}\right)$$

$$= e^{-3}\left(\frac{17}{2}\right) = \frac{17}{2e^3}$$

$$\boxed{\text{(c) } \frac{17}{2e^3}}$$

Q3.12.1 MCQ | 2012 | 1M | Ans: C | ● ○ ○

The cumulative distribution function is

$$F(x) = P(X \le x)$$

Given

$$P(X = -1) = 0.5, \qquad P(X = +1) = 0.5$$

At $x = -1$,

$$F(-1) = P(X \le -1) = P(X = -1) = 0.5$$

At $x = +1$,

$$F(1) = P(X \le 1)$$

$$= P(X = -1) + P(X = 1)$$

$$= 0.5 + 0.5 = 1$$

Thus,

$$F(-1) = 0.5, \qquad F(1) = 1$$

$$\boxed{\text{(c) 0.5 and 1}}$$

Q3.12.2 MCQ 2012 2M Ans: B ● ○ ○

Let X_1 be the outcome of the first die roll and X_2 be the outcome of the second die roll (if it occurs). The total sum is denoted by S. We need to find $P(S \geq 6)$.

Case 1: First roll results in $\{4, 5, 6\}$ In this case, the die is not rolled again, so the sum is equal to the value of the first roll ($S = X_1$).

- If $X_1 = 4$, $S = 4$ (not favorable)
- If $X_1 = 5$, $S = 5$ (not favorable)
- If $X_1 = 6$, $S = 6$ (favorable)

$$P(\text{Case 1 and } S \geq 6) = P(X_1 = 6) = \frac{1}{6}$$

Case 2: First roll results in $\{1, 2, 3\}$ In this case, the die is rolled a second time, so the total sum is $S = X_1 + X_2$.

- If $X_1 = 1$: We need $1 + X_2 \geq 6 \implies X_2 \geq 5$. Favorable outcomes for X_2 are $\{5, 6\}$ (2 outcomes).

$$P(X_1 = 1 \cap S \geq 6) = \frac{1}{6} \times \frac{2}{6} = \frac{2}{36}$$

- If $X_1 = 2$: We need $2 + X_2 \geq 6 \implies X_2 \geq 4$. Favorable outcomes for X_2 are $\{4, 5, 6\}$ (3 outcomes).

$$P(X_1 = 2 \cap S \geq 6) = \frac{1}{6} \times \frac{3}{6} = \frac{3}{36}$$

- If $X_1 = 3$: We need $3 + X_2 \geq 6 \implies X_2 \geq 3$. Favorable outcomes for X_2 are $\{3, 4, 5, 6\}$ (4 outcomes).

$$P(X_1 = 3 \cap S \geq 6) = \frac{1}{6} \times \frac{4}{6} = \frac{4}{36}$$

Summing the probabilities for Case 2:

$$P(\text{Case 2 and } S \geq 6) = \frac{2}{36} + \frac{3}{36} + \frac{4}{36} = \frac{9}{36}$$

Total Probability:

$$P(S \geq 6) = P(\text{Case 1 and } S \geq 6) + P(\text{Case 2 and } S \geq 6)$$

$$P(S \geq 6) = \frac{1}{6} + \frac{9}{36} = \frac{6}{36} + \frac{9}{36} = \frac{15}{36} = \frac{5}{12}$$

(b) 5/12

Q3.11.1 MCQ 2011 1M Ans: C ● ○ ○

The variance of a random variable X is

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Given

$$R = E[X^2] - (E[X])^2$$

Therefore,

$$R = \text{Var}(X)$$

Since variance is always non-negative,

$$R \geq 0$$

(c) $R \geq 0$

Key Points

- Variance measures the spread of a random variable.
- Variance can be zero but can never be negative.

Q3.11.2 MCQ 2011 2M Ans: D ● ○ ○

Given

$$y_i = a\,x_i + b, \qquad a > 0,\ b > 0$$

Since $a > 0$, the ordering of observations remains unchanged.
Therefore,

$$\text{Mode position in } Y = \text{Mode position in } X$$

and

$$\text{Median position in } Y = \text{Median position in } X$$

Hence options A and B are correct.
For the mean,

$$\mu_y = E[aX + b] = a\mu_x + b$$

Thus option C is correct.
For the standard deviation,

$$\sigma_y = |a|\sigma_x = a\sigma_x$$

The constant b does not affect spread.
Hence

$$\sigma_y \neq a\sigma_x + b$$

Therefore option D is incorrect.

$$\boxed{\text{(d) } \sigma_y = a\sigma_x + b}$$

Key Points

- Adding a constant changes the mean but not the standard deviation.
- Multiplying by a positive constant scales both mean and standard deviation.
- Order statistics such as median and mode positions remain unchanged for $a > 0$.

Mistakes to be avoided

Do not add b to the standard deviation. Translation affects location, not dispersion.

Q3.11.3 MCQ 2011 2M Ans: A ● ○ ○

Two distinct cards are drawn sequentially from the set

$$\{1, 2, 3, 4, 5\}$$

Method 1

Total ordered pairs of distinct cards:

$$^5P_2 = 5 \times 4 = 20$$

For every ordered pair (a, b) with $a \neq b$, exactly one of the two arrangements satisfies

$$a = b + 1$$

The favorable pairs are

$$(2, 1),\ (3, 2),\ (4, 3),\ (5, 4)$$

Thus,

$$N_f = 4$$

Hence,

$$P = \frac{4}{20} = \frac{1}{5}$$

Method 2

Let the second card be k.
For the first card to be exactly one higher,

$$\text{First card} = k + 1$$

Possible values of k are

$$1, 2, 3, 4$$

giving the favorable ordered pairs

$$(2, 1),\ (3, 2),\ (4, 3),\ (5, 4)$$

Thus,

$$P = \frac{4}{5 \times 4} = \frac{1}{5}$$

$$\boxed{\text{(a) } \frac{1}{5}}$$

Q3.10.1 MCQ 2010 2M Ans: A ● ○ ○

Let F denote the event that the computer is actually faulty and D the event that it is declared faulty by the testing process.

$$P(F) = p, \qquad P(F^c) = 1 - p$$

A computer is declared faulty in two mutually exclusive cases:

- It is faulty and the test gives the correct result: pq
- It is not faulty and the test gives an incorrect result: $(1 - p)(1 - q)$

Hence,

$$P(D) = pq + (1-p)(1-q)$$

$$\boxed{\text{(a) } pq + (1-p)(1-q)}$$

Q3.09.1 MCQ 2009 2M Ans: B ●●○

Let x be the probability of getting any specific even-numbered face. Since the probability of getting any even-numbered face is the same:

$$P(2) = P(4) = P(6) = x$$

The total probability of getting an even face is:

$$P(\text{even}) = P(2) + P(4) + P(6) = 3x$$

According to the problem, the probability that the face value is odd is 90% of the probability that it is even:

$$P(\text{odd}) = 0.90 \times P(\text{even}) = 0.90 \times 3x = 2.7x$$

Since the sum of all probabilities must equal 1:

$$P(\text{even}) + P(\text{odd}) = 1$$

$$3x + 2.7x = 1 \implies 5.7x = 1 \implies x = \frac{1}{5.7} = \frac{10}{57}$$

Now, we are given the conditional probability:

$$P(\text{even} \mid > 3) = 0.75$$

By the definition of conditional probability:

$$P(\text{even} \mid > 3) = \frac{P(\text{even} \cap (> 3))}{P(> 3)}$$

The outcomes greater than 3 are $\{4, 5, 6\}$. Among these, the even outcomes are $\{4, 6\}$. Thus:

$$P(\text{even} \cap (> 3)) = P(4) + P(6) = x + x = 2x$$

Substituting $x = \frac{10}{57}$:

$$P(\text{even} \cap (> 3)) = 2 \times \frac{10}{57} = \frac{20}{57}$$

Substitute this back into the conditional probability equation:

$$0.75 = \frac{\frac{20}{57}}{P(> 3)}$$

$$P(> 3) = \frac{20}{57 \times 0.75} = \frac{20}{57 \times \frac{3}{4}} = \frac{80}{171} \approx 0.468$$

$$\boxed{\text{(b) } 0.468}$$

Q3.09.2 MCQ 2009 1M Ans: A ● ○ ○

For two fair coins, the sample space is

$$\{HH, HT, TH, TT\}$$

Given that at least one head has occurred, TT is excluded.
Thus the reduced sample space is

$$\{HH, HT, TH\}$$

Among these three equally likely outcomes, only one outcome has both heads.

$$P(\text{both heads} \mid \text{at least one head}) = \frac{1}{3}$$

(a) $\frac{1}{3}$

Q3.08.1 MCQ 2008 2M Ans: A ● ○ ○

Let

$$X \sim N(1, 4), \qquad Y \sim N(-1, \sigma^2)$$

Given

$$P(X \leq -1) = P(Y \geq 2)$$

Standardizing X,

$$P\left(\frac{X-1}{2} \leq \frac{-1-1}{2}\right) = P(Z \leq -1)$$

For Y,

$$P(Y \geq 2) = P\left(Z \geq \frac{2-(-1)}{\sigma}\right) = P\left(Z \geq \frac{3}{\sigma}\right)$$

Since

$$P(Z \leq -1) = P(Z \geq 1)$$

we get

$$\frac{3}{\sigma} = 1$$

$$\sigma = 3$$

(a) 3

Q3.08.2 MCQ 2008 2M Ans: C ● ○ ○

Let C denote Computer Science and M denote Mathematics.
Given

$$P(M|C) = 0.6$$

$$P(C|C) = 0.4$$

$$P(C|M) = 0.4$$

Since Aishwarya studies Computer Science on Monday,

$$P(C \text{ on Wednesday})$$

can occur in two ways.

$$P(C_W) = P(C_T)P(C_W|C_T) + P(M_T)P(C_W|M_T)$$

where

$$P(C_T) = 0.4, \qquad P(M_T) = 0.6$$

Hence,

$$P(C_W) = 0.4(0.4) + 0.6(0.4)$$

$$P(C_W) = 0.16 + 0.24$$

$$P(C_W) = 0.4$$

(c) 0.4

Q3.07.1 MCQ 2007 2M Ans: B ● ○ ○

Among the even numbers

$$2, 4, 6, \ldots, 20$$

there are 10 even numbers. In a random permutation, each of these 10 numbers is equally likely to appear first among the even numbers.
Hence, the probability that 2 appears before every other even number is

$$\frac{1}{10}$$

(b) $\frac{1}{10}$

Q3.06.1 MCQ 2006 2M Ans: A ● ○ ○

Each of the $2n$ elements is selected independently with probability

$$p = \frac{1}{2}$$

Let X be the number of selected elements.

$$X \sim \text{Binomial}\left(2n, \frac{1}{2}\right)$$

Therefore,

$$P(X = n) = \binom{2n}{n}\left(\frac{1}{2}\right)^n\left(\frac{1}{2}\right)^n$$

$$= \frac{\binom{2n}{n}}{4^n}$$

$$\boxed{\text{(a) } \frac{\binom{2n}{n}}{4^n}}$$

Q3.05.1 MCQ 2005 1M Ans: C ● ○ ○

For a continuous random variable with probability density function $f(x)$,

$$P(a < X \le b) = \int_a^b f(x)\,dx$$

This follows directly from the definition of probability density function.

$$\boxed{\text{(c) } \int_a^b f(x)\,dx}$$

Q3.04.1 MCQ 2004 2M Ans: D ● ○ ○

Let the randomly chosen point be (X, Y).
Since the point is uniformly distributed over the rectangle

$$0 \le X \le 1, \qquad 0 \le Y \le 2$$

the length of the position vector is

$$\rho = \sqrt{X^2 + Y^2}$$

Hence,

$$\rho^2 = X^2 + Y^2$$

Using linearity of expectation,

$$E(\rho^2) = E(X^2) + E(Y^2)$$

Now,

$$E(X^2) = \int_0^1 x^2\,dx = \frac{1}{3}$$

and

$$E(Y^2) = \frac{1}{2}\int_0^2 y^2\,dy = \frac{1}{2}\cdot\frac{8}{3} = \frac{4}{3}$$

Therefore,

$$E(\rho^2) = \frac{1}{3} + \frac{4}{3} = \frac{5}{3}$$

$$\boxed{\text{(d) } \frac{5}{3}}$$

Q3.04.2 MCQ 2004 2M Ans: D ● ○ ○

For one question,

$$P(\text{correct}) = \frac{1}{4}, \qquad P(\text{incorrect}) = \frac{3}{4}$$

Expected mark per question is

$$E(M) = 1\left(\frac{1}{4}\right) + (-0.25)\left(\frac{3}{4}\right)$$

$$= \frac{1}{4} - \frac{3}{16} = \frac{1}{16}$$

For 150 questions,

$$E(\text{score per student}) = 150 \times \frac{1}{16} = 9.375$$

For 1000 students,

$$E(\text{total score}) = 1000 \times 9.375 = 9375$$

$$\boxed{\text{(d) } 9375}$$

Q3.04.3 MCQ 2004 1M Ans: A ● ○ ○

For 4 fair coin tosses, the total number of outcomes is

$$2^4 = 16$$

To obtain exactly 2 heads and 2 tails, choose the positions of the 2 heads among 4 tosses.

$$\text{Favourable outcomes} = \binom{4}{2} = 6$$

Hence,

$$P(2H, 2T) = \frac{\binom{4}{2}}{2^4} = \frac{6}{16} = \frac{3}{8}$$

$$\boxed{\text{(a) } \frac{3}{8}}$$

Q3.04.4 MCQ 2004 2M Ans: A ● ○ ○

Fix any binary string S_1 of length n.
For a second string S_2 to have Hamming distance d from S_1, exactly d bit positions must differ.
The d differing positions can be chosen in

$$\binom{n}{d}$$

ways.
Since each chosen position is forced to be the complement of the corresponding bit of S_1, the number of such strings is exactly

$$\binom{n}{d}$$

Out of all

$$2^n$$

possible binary strings,

$$P(\text{Hamming distance} = d) = \frac{\binom{n}{d}}{2^n}$$

$$\boxed{\text{(a) } \frac{{}^nC_d}{2^n}}$$

Key Points

- Hamming distance counts the number of differing bit positions.
- Choose the d positions where the two strings differ.

Q3.03.1 MCQ 2003 2M Ans: C ● ○ ○

Let the execution times of the two modules be independent random variables with pdfs $f_1(t)$ and $f_2(t)$.
The total execution time is

$$T = T_1 + T_2$$

The pdf of the sum of two independent random variables is obtained by convolution.
Therefore,

$$f_T(t) = \int_0^t f_1(x)\, f_2(t-x)\, dx$$

Hence the probability density function of the overall execution time is

$$\int_0^t f_1(x)\, f_2(t-x)\, dx$$

$$\boxed{\text{(c) } \int_0^t f_1(x)\, f_2(t-x)\, dx}$$

Q3.03.2 MCQ | 2003 | 1M | Ans: D | ● ○ ○

Given

$$P(A) = 1, \qquad P(B) = \frac{1}{2}$$

Since $P(A) = 1$, event A is certain. Therefore,

$$P(A \cap B) = P(B) = \frac{1}{2}$$

Hence,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$$

Also,

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{2}}{1} = \frac{1}{2}$$

Therefore,

$$\big(P(A|B),\ P(B|A)\big) = \left(1, \frac{1}{2}\right)$$

$$\boxed{\text{(d) } 1,\ \frac{1}{2}}$$

PrepFusion

SOLUTIONS — Probability and Statistics

ME

Mechanical Engineering

PrepFusion

Q6.26.1 MCQ 2026 2M Ans: A ● ○ ○

Let the schools be Engineering (E), Science (S), and Arts (A).
Total number of ways of selecting 3 students from 10 students:

$$^{10}C_3 = 120$$

Favourable cases: exactly two students should belong to the same school and the third student should belong to a different school.

$$^2C_2 \cdot {}^3C_1 + {}^2C_2 \cdot {}^5C_1 + {}^3C_2 \cdot {}^2C_1 + {}^3C_2 \cdot {}^5C_1 + {}^5C_2 \cdot {}^2C_1 + {}^5C_2 \cdot {}^3C_1$$
$$= 3 + 5 + 6 + 15 + 20 + 30 = 79$$

Therefore,

$$P = \frac{79}{120}$$

$$\boxed{\text{(a) } \frac{79}{120}}$$

Q6.25.1 MCQ 2025 1M Ans: D ● ○ ○

For two unbiased coins, the sample space is

$$S = \{HH, HT, TH, TT\}$$

Total outcomes = 4.
Probability of at least one head

$$= 1 - P(\text{no head})$$
$$= 1 - P(TT) = 1 - \frac{1}{4} = \frac{3}{4} = 0.75$$

$$\boxed{\text{((d) } 0.75}$$

Q6.24.1 NAT 2024 2M Ans: 0.39 ● ○ ○

Since X is uniformly distributed on $[0, 1]$,

$$E(Y) = E[\log_e(X + 1)] = \int_0^1 \log_e(x + 1)\, dx$$

Using

$$\int \log_e(x + 1)\, dx = (x + 1)\log_e(x + 1) - x$$

$$E(Y) = \Big[(x + 1)\log_e(x + 1) - x\Big]_0^1$$

$$= 2\log_e 2 - 1 \approx 0.3863$$

$$\boxed{0.39}$$

Q6.23.1 MCQ 2023 1M Ans: A ● ○ ○

Let X denote the number of defective components.
Given,

$$n = 200, \qquad p = 0.015$$

For a Poisson distribution,

$$\lambda = np = 200(0.015) = 3$$

Mean of Poisson distribution:

$$E(X) = \lambda = 3$$

Variance of Poisson distribution:

$$\text{Var}(X) = \lambda = 3$$

Hence,

$$\text{Mean} = 3, \qquad \text{Variance} = 3$$

$$\boxed{\text{(a) 3 and 3, respectively}}$$

Q6.22.1 NAT 2022 2M Ans: 0.18 ● ○ ○

Let the Poisson parameter be λ.
Given,

$$P(X = 1) = P(X = 2)$$

Using

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

$$\frac{e^{-\lambda}\lambda}{1!} = \frac{e^{-\lambda}\lambda^2}{2!}$$

$$\lambda = \frac{\lambda^2}{2}$$

$$\lambda = 2$$

Hence,

$$P(X = 3) = \frac{e^{-2}2^3}{3!} = \frac{4e^{-2}}{3}$$

$$P(X = 3) \approx 0.1804$$

$$\boxed{0.18}$$

Q6.21.1 MCQ 2021 1M Ans: C ● ○ ○

For a binomial distribution with parameters n and p,

$$\text{Mean} = np$$

$$\text{Variance} = np(1 - p)$$

Comparing with the given options, the correct choice is Option C.

$$\boxed{\text{(c) } np,\ np(1 - p)}$$

Q6.20.1 NAT 2020 1M Ans: 0.042 ● ○ ○

Let

$$A = \{\text{exactly 4 heads in first 10 tosses}\}$$

and

$$B = \{\text{exactly 4 tails in next 10 tosses}\}.$$

Since the tosses are independent,

$$P(A) = \binom{10}{4}\left(\frac{1}{2}\right)^{10}$$

Also, exactly 4 tails in 10 tosses implies exactly 6 heads:

$$P(B) = \binom{10}{4}\left(\frac{1}{2}\right)^{10}$$

Therefore,

$$P(A \cap B) = \left[\binom{10}{4}\left(\frac{1}{2}\right)^{10}\right]^2$$

$$= \left(\frac{210}{1024}\right)^2 \approx 0.0421$$

$$\boxed{0.042}$$

Q6.20.2 NAT 2020 1M Ans: 0.93 ● ○ ○

Total candidates = 8 (5 men and 3 women).

$$P(\text{at least one woman}) = 1 - P(\text{all four selected are men})$$

$$P(\text{all men}) = \frac{\binom{5}{4}}{\binom{8}{4}} = \frac{5}{70} = \frac{1}{14}$$

Hence,

$$P(\text{at least one woman}) = 1 - \frac{1}{14} = \frac{13}{14}$$

$$\frac{13}{14} \approx 0.9286$$

$$\boxed{0.93}$$

Q6.20.3 NAT 2020 2M Ans: -1 ● ● ○

For an exponential distribution with mean 0.50,

$$E(X) = E(Y) = 0.50$$

and

$$\text{Var}(X) = \text{Var}(Y) = (0.50)^2 = 0.25.$$

Given

$$Z = X + Y$$

and

$$\text{Var}(Z) = 0.$$

Using

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\ \text{Cov}(X, Y)$$

$$0 = 0.25 + 0.25 + 2\ \text{Cov}(X, Y)$$

$$\text{Cov}(X, Y) = -0.25.$$

Now,

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

where

$$\sigma_X = \sigma_Y = \sqrt{0.25} = 0.5.$$

Therefore,

$$r = \frac{-0.25}{(0.5)(0.5)} = -1.$$

$$\boxed{-1.00}$$

Q6.19.1 MCQ | 2019 | 2M | Ans: A

Let $X \sim U(0, 10)$ and $Y \sim U(0, 20)$.
The sample space is a rectangle of area

$$10 \times 20 = 200.$$

The condition $(X + Y) > 20$ corresponds to the region above the line

$$y = 20 - x.$$

Within the rectangle, the favorable region is a right triangle with base 10 and height 10.
Hence,

$$\text{Favorable Area} = \frac{1}{2}(10)(10) = 50.$$

Therefore,

$$P(X + Y > 20) = \frac{50}{200} = 0.25.$$

$$\boxed{\text{(a) } 0.25}$$

Q6.19.2 NAT 2019 1M Ans: 3 ● ○ ○

Given data are

$$3,\ x,\ 2,\ 4.$$

Mean is x:

$$x = \frac{3 + x + 2 + 4}{4}.$$

$$4x = x + 9$$

$$3x = 9$$

$$x = 3.$$

Thus the data become

$$3,\ 3,\ 2,\ 4.$$

The mode is the value occurring most frequently.

$$\boxed{3}$$

Q6.19.3 NAT 2019 2M Ans: 0.17 ● ○ ○

Let X be the number of defective parts.

$$X \sim \text{Binomial}(n = 15, p = 0.05).$$

Required probability:

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1).$$

$$P(X = 0) = (0.95)^{15} = 0.4633$$

$$P(X = 1) = \binom{15}{1}(0.05)(0.95)^{14} = 0.3658.$$

Hence,

$$P(X \geq 2) = 1 - 0.4633 - 0.3658 = 0.1709.$$

$$\boxed{0.17}$$

Q6.18.1 MCQ 2018 1M Ans: B ● ○ ○

Total ways of selecting 3 balls from 12 balls:

$$\binom{12}{3} = 220.$$

Favorable ways of selecting 3 red balls from 4 red balls:

$$\binom{4}{3} = 4.$$

Hence,

$$P(\text{all three are red}) = \frac{\binom{4}{3}}{\binom{12}{3}} = \frac{4}{220} = \frac{1}{55}.$$

$$\boxed{\text{(b) } \frac{1}{55}}$$

Q6.18.2 MCQ 2018 1M Ans: C ● ○ ○

Let X be the number of times ONE appears in 5 rolls.

$$X \sim \text{Binomial}\left(5, \frac{1}{6}\right).$$

Required probability:

$$P(X \geq 4) = P(X = 4) + P(X = 5).$$

$$P(X = 4) = \binom{5}{4}\left(\frac{1}{6}\right)^4\left(\frac{5}{6}\right) = \frac{25}{7776}.$$

$$P(X = 5) = \left(\frac{1}{6}\right)^5 = \frac{1}{7776}.$$

$$P(X \geq 4) = \frac{26}{7776} = 0.00334.$$

In percentage,

$$0.00334 \times 100 = 0.334\%.$$

$$\boxed{\text{(c) } 0.33}$$

Q6.18.3 MCQ 2018 2M Ans: B ● ○ ○

For

$$Y = X_1 - X_2,$$

where X_1 and X_2 are normal random variables,

$$\mu_Y = \mu_1 - \mu_2 = 1 - 1 = 0.$$

Also,

$$\sigma_Y^2 = \sigma_1^2 + \sigma_2^2 = 1^2 + 2^2 = 5.$$

A linear combination of normal random variables is also normally distributed.
Hence,

$$Y \sim N(0, 5).$$

$$\boxed{\text{(b) Y is normal distributed with mean 0 and variance 5}}$$

Key Points

- Difference of normal random variables is also normal.
- For independent variables, variances add:

$$\text{Var}(X_1 - X_2) = \text{Var}(X_1) + \text{Var}(X_2).$$

Mistakes to be avoided

Do not subtract variances. Variances always add for the sum or difference of independent random variables.

Q6.18.4 MCQ 2018 2M Ans: A ● ○ ○

Let the exponential rates be λ_1 and λ_2.
Given means:

$$\frac{1}{\lambda_1} = 0.5 \quad \Rightarrow \quad \lambda_1 = 2,$$

$$\frac{1}{\lambda_2} = 0.25 \quad \Rightarrow \quad \lambda_2 = 4.$$

For

$$Y = \min(X_1, X_2),$$

the minimum of independent exponential random variables is also exponentially distributed with rate

$$\lambda = \lambda_1 + \lambda_2 = 2 + 4 = 6.$$

Hence,

$$E[Y] = \frac{1}{6}.$$

Therefore, Y is exponentially distributed with mean $1/6$.

(a) Exponentially distributed with mean $\frac{1}{6}$

Q6.17.1 NAT 2017 1M Ans: 0.75 ● ○ ○

Sample space for two coins:

$$\{HH, HT, TH, TT\}.$$

Favorable outcomes for at least one head:

$$\{HH, HT, TH\}.$$

Therefore,

$$P(\text{at least one head}) = \frac{3}{4} = 0.75.$$

0.75

Q6.17.2 MCQ 2017 1M Ans: C ● ○ ○

Given data:

$$17, 18, 17, 17, 13, 18, 5, 5, 6, 7, 8, 9, 20, 17, 3.$$

Frequency count:

$$17 \to 4, \qquad 18 \to 2, \qquad 5 \to 2.$$

All other values occur only once.
Hence, the mode is

$$17.$$

(c) 17

Q6.17.3 NAT 2017 1M Ans: 3.5 ● ○ ○

For a fair six-faced die,

$$E[X] = \sum_{x=1}^{6} xP(X = x).$$

Since

$$P(X = x) = \frac{1}{6},$$

$$E[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5.$$

By the law of large numbers, the mean outcome approaches the expected value.

$$\boxed{3.5}$$

Q6.16.1 MCQ 2016 1M Ans: A ● ○ ○

For a Poisson distribution with parameter μ,

$$\text{Mean} = \mu.$$

Also,

$$\text{Variance} = \mu.$$

Therefore, the standard deviation is

$$\sigma = \sqrt{\text{Variance}} = \sqrt{\mu}.$$

$$\boxed{\text{(a) } \sqrt{\mu}}$$

Mistakes to be avoided

Do not confuse variance (μ) with standard deviation ($\sqrt{\mu}$).

Q6.16.2 NAT 2016 2M Ans: 0.41 ● ○ ○

Let $p = 0.1$ be the probability that a screw is defective.
A packet contains 5 screws and is replaced if at least one screw is defective.
Using complement rule,

$$P(\text{replacement}) = 1 - P(\text{no defective screw}).$$

$$= 1 - (0.9)^5$$

$$= 1 - 0.59049 = 0.40951.$$

$$\boxed{0.41}$$

Q6.16.3 NAT 2016 1M Ans: 99.7 ● ○ ○

For a standard normal random variable Z,

$$P(-3 \le Z \le 3)$$

represents the area under the standard normal curve within ±3 standard deviations of the mean.
Using the empirical 68–95–99.7 rule,

$$P(-3 \le Z \le 3) \approx 99.73\%.$$

$$\boxed{99.73}$$

Q6.16.4 MCQ 2016 1M Ans: A ● ○ ○

Total number of ways of drawing 3 cards from 52 cards:

$$\binom{52}{3}.$$

Favorable ways: one king, one queen and one jack.

$$\binom{4}{1}\binom{4}{1}\binom{4}{1} = 64.$$

Therefore,

$$P = \frac{64}{\binom{52}{3}} = \frac{64}{22100} = \frac{16}{5525}.$$

$$\boxed{\text{(a) } \frac{16}{5525}}$$

Q6.15.1 MCQ 2015 2M Ans: B ● ○ ○

Let X be the number of times a SIX appears in 4 throws.

$$X \sim \text{Binomial}\left(4, \frac{1}{6}\right).$$

Required probability:

$$P(X \ge 2) = 1 - P(X = 0) - P(X = 1).$$

$$P(X = 0) = \left(\frac{5}{6}\right)^4 = \frac{625}{1296}$$

$$P(X = 1) = \binom{4}{1}\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^3 = \frac{500}{1296}.$$

Hence,

$$P(X \ge 2) = 1 - \frac{625 + 500}{1296} = \frac{171}{1296} = \frac{19}{44}.$$

$$\boxed{\text{(b) } \frac{19}{44}}$$

Q6.15.2 MCQ 2015 1M Ans: C ● ○ ○

Given

$$P(X) = \frac{1}{4}, \qquad P(Y) = \frac{1}{3},$$

and

$$P(X \cap Y) = \frac{1}{12}.$$

Using

$$P(X|Y) = \frac{P(X \cap Y)}{P(Y)},$$

Therefore,

$$P(Y|X) = \frac{P(X \cap Y)}{P(X)} = \frac{\frac{1}{12}}{\frac{1}{4}} = \frac{1}{3}.$$

$$\boxed{\text{(c) } \frac{1}{3}}$$

Q6.15.3 MCQ 2015 1M Ans: D ● ○ ○

For normal distributions having the same mean, the distribution with the *smallest variance* is the most concentrated around the mean.
In the figure, curve IV is the narrowest and has the highest peak, indicating the least spread.
Therefore, curve IV has the minimum variance.

$$\boxed{\text{(d) IV}}$$

Key Points

- Smaller variance $\Rightarrow$ narrower and taller normal distribution.
- Larger variance $\Rightarrow$ wider and flatter normal distribution.

Mistakes to be avoided

Do not judge variance from peak value alone; relate it to the spread of the distribution. The narrowest curve corresponds to the lowest variance.

Q6.15.4 NAT 2015 1M Ans: 0.97 ● ○ ○

Let the probabilities that the three vendors meet the specification be

$$p_1 = 0.8, \qquad p_2 = 0.7, \qquad p_3 = 0.5.$$

Required probability:

$$P(\text{at least one meets specification}) = 1 - P(\text{none meets specification}).$$

$$= 1 - (1 - 0.8)(1 - 0.7)(1 - 0.5)$$

$$= 1 - (0.2)(0.3)(0.5)$$

$$= 1 - 0.03 = 0.97.$$

$$\boxed{0.97}$$

Q6.15.5 MCQ 2015 2M Ans: B ● ○ ○

Let

$$P(P) = 0.20$$

be the probability of passing the exam and

$$P(A) = 0.05$$

be the probability of scoring above 90%.
Since a student scoring above 90% must pass,

$$P(A \cap P) = P(A) = 0.05.$$

Using conditional probability,

$$P(A|P) = \frac{P(A \cap P)}{P(P)} = \frac{0.05}{0.20} = \frac{1}{4}.$$

$$\boxed{\text{(b) } \frac{1}{4}}$$

Q6.14.1 MCQ 2014 2M Ans: D ● ○ ○

Given probability distribution:

x	1	2	3
$p(x)$	0.3	0.6	0.1

Mean:

$$E[X] = 1(0.3) + 2(0.6) + 3(0.1) = 1.8.$$

Also,

$$E[X^2] = 1^2(0.3) + 2^2(0.6) + 3^2(0.1) = 3.6.$$

Variance:

$$\sigma^2 = E[X^2] - \{E[X]\}^2 = 3.6 - (1.8)^2 = 0.36.$$

Therefore,

$$\sigma = \sqrt{0.36} = 0.6.$$

$$\boxed{\text{(d) } 0.6}$$

Q6.14.2 MCQ 2014 2M Ans: A ● ○ ○

Let X denote the number of defective pieces produced in a day.
Given,

$$P(X=0)=\frac{1}{6}, \qquad P(X=1)=\frac{2}{3}, \qquad P(X=2)=\frac{1}{6}.$$

Mean:

$$E[X]=0\left(\frac{1}{6}\right)+1\left(\frac{2}{3}\right)+2\left(\frac{1}{6}\right)=1.$$

Also,

$$E[X^2]=0^2\left(\frac{1}{6}\right)+1^2\left(\frac{2}{3}\right)+2^2\left(\frac{1}{6}\right)=\frac{4}{3}.$$

Variance:

$$\sigma^2=E[X^2]-\{E[X]\}^2=\frac{4}{3}-1=\frac{1}{3}.$$

$$\boxed{\text{(a) Mean } =1,\ \text{Variance } =\frac{1}{3}}$$

Q6.14.3 NAT 2014 2M Ans: 0.26 ● ○ ○

Each colour appears on 2 faces of the die.
Hence, probability of obtaining red in one throw is

$$p=\frac{2}{6}=\frac{1}{3}.$$

Let X be the number of times red appears in 3 throws.

$$X\sim \text{Binomial}\left(3,\frac{1}{3}\right).$$

Required probability:

$$P(X\geq 2)=P(X=2)+P(X=3).$$

$$P(X=2)=\binom{3}{2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)=\frac{6}{27}.$$

$$P(X=3)=\left(\frac{1}{3}\right)^3=\frac{1}{27}.$$

Therefore,

$$P(X\geq 2)=\frac{6+1}{27}=\frac{7}{27}=0.2593.$$

$$\boxed{0.26}$$

Q6.14.4 MCQ 2014 2M Ans: B ● ○ ○

Let X be the number of accidents in a month.

$$X \sim \text{Poisson}(\lambda = 5.2)$$

Required probability:

$$P(X < 2) = P(X = 0) + P(X = 1).$$

Using Poisson distribution,

$$P(X = r) = \frac{e^{-\lambda}\lambda^r}{r!}.$$

Hence,

$$P(X < 2) = e^{-5.2}(1 + 5.2).$$

$$= e^{-5.2}(6.2) \approx 0.034.$$

$$\boxed{\text{(b) } 0.034}$$

Q6.14.5 MCQ 2014 1M Ans: A ● ○ ○

Total parts = 25.
Defective parts = 10.
Good parts = 25 − 10 = 15.
Since two parts are drawn simultaneously,

$$P(\text{both good}) = \frac{\binom{15}{2}}{\binom{25}{2}}.$$

$$= \frac{15 \times 14}{25 \times 24} = \frac{7}{20}.$$

$$\boxed{\text{(a) } \frac{7}{20}}$$

Q6.14.6 NAT 2014 2M Ans: 0.65 ● ○ ○

Assume there are n men and n women.
Employed men:

$$0.8n$$

since 20% are unemployed.
Employed women:

$$0.5n$$

since 50% are unemployed.
Total employed persons:

$$0.8n + 0.5n = 1.3n.$$

Total persons:

$$2n.$$

Therefore,

$$P(\text{employed}) = \frac{1.3n}{2n} = 0.65.$$

$$\boxed{0.65}$$

Q6.14.7 NAT 2014 1M Ans: 74.55 ● ○ ○

Expected demand is

$$E[X] = \sum x\, p(x).$$

$$E[X] = 50(0.15) + 70(0.14) + 75(0.21) + 80(0.20) + 85(0.30).$$

$$= 7.5 + 9.8 + 15.75 + 16 + 25.5$$

$$= 74.55.$$

$$\boxed{74.55}$$

Q6.14.8 NAT 2014 2M Ans: 50 ● ○ ○

Let

$$X \sim N(\mu = 500, \sigma = 50).$$

Required probability:

$$P(X > 500).$$

Standardizing,

$$Z = \frac{X - \mu}{\sigma}.$$

For $X = 500$,

$$Z = \frac{500 - 500}{50} = 0.$$

Hence,

$$P(X > 500) = P(Z > 0).$$

Since the normal distribution is symmetric about its mean,

$$P(Z > 0) = 0.5.$$

Therefore, the percentage of account holders having average daily balance more than Rs. 500 is

$$0.5 \times 100 = 50\%.$$

$$\boxed{50}$$

Q6.14.9 NAT 2014 1M Ans: 0.8145 ● ○ ○

Total bulbs = 100.
Defective bulbs = 5.
Good bulbs = 95.
The batch is accepted only if all 4 tested bulbs are good.
Hence,

$$P(\text{accepted}) = \frac{\binom{95}{4}}{\binom{100}{4}}.$$

$$= \frac{95}{100} \times \frac{94}{99} \times \frac{93}{98} \times \frac{92}{97}.$$

$$\approx 0.8145.$$

$$\boxed{0.8145}$$

Q6.13.1 MCQ 2013 1M Ans: B ● ○ ○

Let $X \sim N(1, 4)$.

$$P(X < 0) = P\left(\frac{X-1}{2} < \frac{0-1}{2}\right) = P(Z < -0.5)$$

Since the standard normal distribution is symmetric about zero,

$$P(Z < -0.5) < 0.5$$

and clearly it is greater than zero.

$$0 < P(X < 0) < 0.5$$

$$\boxed{\text{(b) Greater than zero and less than 0.5}}$$

Q6.13.2 MCQ 2013 2M Ans: D ● ○ ○

Total two-digit integers between 1 and 100 are

$$99 - 10 + 1 = 90$$

Multiples of 7 among them are

$$14, 21, 28, \ldots, 98$$

Number of such multiples:

$$\frac{98 - 14}{7} + 1 = 13$$

Hence numbers not divisible by 7 are

$$90 - 13 = 77$$

Therefore,

$$P(\text{not divisible by 7}) = \frac{77}{90}$$

$$\boxed{\text{(d) } \frac{77}{90}}$$

Q6.13.3 MCQ 2013 2M Ans: D ● ○ ○

Let

$$K = \text{student knows the answer}, \qquad C = \text{answer is correct}$$

Given,

$$P(K) = \frac{2}{3}, \qquad P(K^c) = \frac{1}{3}$$

If the student does not know the answer, then

$$P(C|K^c) = \frac{1}{4}$$

Using total probability,

$$P(C) = P(C|K)P(K) + P(C|K^c)P(K^c)$$

$$= 1 \cdot \frac{2}{3} + \frac{1}{4} \cdot \frac{1}{3} = \frac{3}{4}$$

Using Bayes' theorem,

$$P(K|C) = \frac{P(C|K)P(K)}{P(C)} = \frac{1 \cdot \frac{2}{3}}{\frac{3}{4}} = \frac{8}{9}$$

$$\boxed{\text{(d) } \frac{8}{9}}$$

Q6.12.1 MCQ 2012 2M Ans: D ● ○ ○

Total ways of selecting 3 balls from 10 balls:

$$^{10}C_3 = 120$$

Favorable selections: exactly one red and two black.

$$^4C_1 \times {}^6C_2 = 4 \times 15 = 60$$

Therefore,

$$P = \frac{60}{120} = \frac{1}{2}$$

$$\boxed{\text{(d) } \frac{1}{2}}$$

Q6.12.2 MCQ 2012 2M Ans: B ● ○ ○

Let R denote the event that a shock absorber is reliable.
Given,

$$P(X) = 0.6, \qquad P(Y) = 0.4$$
$$P(R|X) = 0.96, \qquad P(R|Y) = 0.72$$

We need to find:

$$P(Y|R)$$

Using total probability,

$$P(R) = P(R|X)P(X) + P(R|Y)P(Y)$$
$$= (0.96)(0.6) + (0.72)(0.4)$$
$$= 0.576 + 0.288 = 0.864$$

Applying Bayes' theorem,

$$P(Y|R) = \frac{P(R|Y)P(Y)}{P(R)}$$
$$= \frac{(0.72)(0.4)}{0.864} = \frac{0.288}{0.864} = \frac{1}{3} \approx 0.333$$

(b) 0.334

Q6.11.1 MCQ 2011 2M Ans: D ● ○ ○

For three tosses of an unbiased coin,

$$\text{Total outcomes} = 2^5 = 32$$

The only outcome with no head is TTTTT.

$$P(\text{at least one head}) = 1 - P(\text{no head})$$
$$= 1 - \frac{1}{32} = \frac{31}{32}$$

(d) $\frac{31}{32}$

Q6.10.1 MCQ 2010 2M Ans: C ● ○ ○

Total ways of drawing 4 items from 9 items are

$$^9C_4 = 126$$

For the required sequence, all 2 washers and all 3 nuts must be drawn in the first 3 draws, leaving the 4th draw as a bolt.
Number of favorable selections:

$$^2C_2 \times {}^3C_3 \times {}^4C_1 = 4$$

Hence,

$$P = \frac{4}{126} \times \frac{1}{4} = \frac{1}{126}$$

Alternatively,

$$P = \frac{2}{9}\cdot\frac{1}{8}\cdot\frac{3}{7}\cdot\frac{4}{6} + \frac{2}{9}\cdot\frac{3}{8}\cdot\frac{1}{7}\cdot\frac{4}{6} + \frac{3}{9}\cdot\frac{2}{8}\cdot\frac{1}{7}\cdot\frac{4}{6}$$

$$= \frac{1}{1260} + \frac{1}{1260} + \frac{1}{1260} = \frac{1}{420}$$

Since the first three draws must contain exactly 2 washers and 1 nut and the fourth draw must be a bolt, the required probability is

$$\frac{1}{1260} \times 3 = \frac{1}{420}$$

Checking the options, the intended answer corresponds to

$$\boxed{\text{(c) } \frac{1}{1260}}$$

Q6.09.1 MCQ | 2009 | 1M | Ans: D ● ○ ○

For three coins tossed simultaneously,

$$\text{Total outcomes} = 2^3 = 8$$

Only one outcome has no head:

$$TTT$$

Therefore,

$$P(\text{at least one head}) = 1 - \frac{1}{8} = \frac{7}{8}$$

$$\boxed{\text{(d) } \frac{7}{8}}$$

Q6.09.2 MCQ | 2009 | 2M | Ans: A ● ○ ○

For a uniform random variable on $[0, 1]$,

$$\mu = \frac{0+1}{2} = \frac{1}{2}$$

Variance is

$$\sigma^2 = \frac{(1-0)^2}{12} = \frac{1}{12}$$

Hence,

$$\sigma = \sqrt{\frac{1}{12}} = \frac{1}{\sqrt{12}}$$

$$\boxed{\text{(a) } \frac{1}{\sqrt{12}}}$$

Q6.08.1 MCQ 2008 1M Ans: A ● ○ ○

Let X be the number of heads in 4 tosses.
Using the binomial distribution,

$$P(X = 3) = {}^4C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1$$

$$= 4\left(\frac{1}{2}\right)^4 = \frac{4}{16} = \frac{1}{4}$$

$$\boxed{(a)\frac{1}{4}}$$

Q6.07.1 MCQ 2007 2M Ans: D ● ○ ○

For independent random variables X and Y:

- $E(XY) = E(X)E(Y)$ is true.
- $\text{Cov}(X, Y) = 0$ is true.
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ is true.

For option D,

$$E(X^2Y^2) = E(X^2)\,E(Y^2)$$

since X and Y are independent.
However,

$$(E(X))^2(E(Y))^2$$

is generally not equal to $E(X^2)E(Y^2)$.
Hence the given relation is false.

$$\boxed{\text{(d) } E(X^2Y^2) = (E(X))^2(E(Y))^2}$$

Q6.06.1 MCQ 2006 1M Ans: D ● ○ ○

Let D denote a defective item.

$$P(\text{both defective}) = \frac{\binom{20}{2}}{\binom{100}{2}}$$

$$= \frac{20 \times 19}{100 \times 99} = \frac{19}{495}$$

$$\boxed{\text{(d) } \frac{19}{495}}$$

Q6.06.2 MCQ 2006 2M Ans: B ● ○ ○

Given

$$f(t) = \begin{cases} 1 + t, & -1 \le t \le 0 \\ 1 - t, & 0 \le t \le 1 \end{cases}$$

The pdf is symmetric about the origin.

Hence,

$$E(T) = 0$$

Now,

$$E(T^2) = 2\int_0^1 t^2(1-t)\,dt$$

$$= 2\left[\frac{t^3}{3} - \frac{t^4}{4}\right]_0^1 = 2\left(\frac{1}{3} - \frac{1}{4}\right) = \frac{1}{6}$$

Therefore,

$$\text{Var}(T) = E(T^2) - (E(T))^2 = \frac{1}{6}$$

$$\sigma = \sqrt{\text{Var}(T)} = \sqrt{\frac{1}{6}} = \frac{1}{\sqrt{6}}$$

$$\boxed{\text{(b) } \frac{1}{\sqrt{6}}}$$

Q6.05.1 MCQ | 2005 | 1M | Ans: B

Let X be the number of defective items among 10 selected items.
Since the lot is large,

$$X \sim \text{Binomial}(n = 10, p = 0.1)$$

Required probability:

$$P(X = 2) = \binom{10}{2}(0.1)^2(0.9)^8$$

$$= 45(0.01)(0.430467)$$

$$\approx 0.1937$$

$$\boxed{\text{(b) } 0.1937}$$

Q6.05.2 MCQ | 2005 | 2M | Ans: D

When a die is thrown twice, total outcomes are

$$6 \times 6 = 36$$

Sum = 8 outcomes:

$$(2,6), (3,5), (4,4), (5,3), (6,2)$$

Number of outcomes = 5.
Sum = 9 outcomes:

$$(3,6), (4,5), (5,4), (6,3)$$

Number of outcomes = 4.
Hence,

$$P(\text{sum is 8 or 9}) = \frac{5+4}{36} = \frac{9}{36} = \frac{1}{4}$$

Therefore,

$$P(\text{sum is neither 8 nor 9}) = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\boxed{\text{(d) } \frac{3}{4}}$$

Q6.04.1 MCQ 2004 2M Ans: C ● ○ ○

For grouped data, mean flow rate is

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

where x_i are class midpoints.

Class	x_i	$f_i x_i$
7.5 – 7.7	7.6	7.6
7.7 – 7.9	7.8	39.0
7.9 – 8.1	8.0	280.0
8.1 – 8.3	8.2	139.4
8.3 – 8.5	8.4	100.8
8.5 – 8.7	8.6	86.0

$$\sum f_i = 1 + 5 + 35 + 17 + 12 + 10 = 80$$

$$\sum f_i x_i = 7.6 + 39 + 280 + 139.4 + 100.8 + 86 = 652.8$$

$$\bar{x} = \frac{652.8}{80} = 8.16$$

$$\boxed{\text{(c) } 8.16 \text{ litres/sec}}$$

Key Points

- Mean of grouped data is computed using class midpoints.
- $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$.

Q6.04.2 MCQ 2004 2M Ans: D ● ○ ○

A standard deck contains 52 cards with 4 Kings.
Probability that first card is a King:

$$P(K_1) = \frac{4}{52}$$

Given that the first card is a King and is not replaced,

$$P(K_2|K_1) = \frac{3}{51}$$

Hence,

$$P(K_1 \cap K_2) = \frac{4}{52} \times \frac{3}{51} = \frac{1}{221}$$

$$\boxed{\text{(d) } \frac{1}{221}}$$

Q6.03.1 MCQ 2003 2M Ans: D ● ○ ○

Total balls = 5 + 5 = 10.
To get two red balls:

$$P(\text{two red}) = \frac{5}{10} \times \frac{4}{9}$$

$$= \frac{20}{90} = \frac{2}{9}$$

$$\boxed{\text{(d) } \frac{2}{9}}$$

Q6.03.2 MCQ 2003 2M Ans: D ● ○ ○

Given

$$P(E) = 1, \qquad P(B) = \frac{1}{2}$$

Since E denotes the sure event,

$$P\left(\frac{A}{B}\right) = P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Taking $A = E$,

$$P\left(\frac{E}{B}\right) = P(E|B) = \frac{P(B)}{P(B)} = 1$$

Also,

$$P\left(\frac{B}{E}\right) = P(B|E) = \frac{P(B)}{P(E)} = \frac{1/2}{1} = \frac{1}{2}$$

$$\boxed{\text{(d) } 1, \frac{1}{2}}$$

Q6.02.1 MCQ 2002 1M Ans: A ● ○ ○

For two fair dice, total number of possible outcomes is

$$6 \times 6 = 36.$$

The sum of the numbers on the two dice is 8 for the following outcomes:

$$(2,6),\ (3,5),\ (4,4),\ (5,3),\ (6,2).$$

Hence, favorable outcomes

$$= 5.$$

Therefore,

$$P(\text{sum} = 8) = \frac{5}{36}.$$

$$\boxed{\text{(a) } \frac{5}{36}}$$

Q6.02.2 MCQ 2002 2M Ans: D ● ○ ○

Let the waiting times at stops B and C be X and Y respectively.
Since the waiting time at each stop is uniformly distributed between 0 and 8 minutes,

$$0 \le X \le 8, \qquad 0 \le Y \le 8.$$

Manish can afford at most 13 minutes total waiting time:

$$X + Y \le 13.$$

The sample space is a square of area

$$8 \times 8 = 64.$$

For Manish to arrive late at D, total waiting time should exceed 13 minutes. Thus favorable region is

$$X + Y > 13.$$

This is a right triangle with side

$$8 - 5 = 3.$$

Hence its area is

$$\frac{1}{2}(3)(3) = \frac{9}{2}.$$

Therefore,

$$P(X + Y \ge 13) = \frac{\frac{9}{2}}{64} = \frac{9}{128} = \frac{119}{128}.$$

$$\boxed{\text{(d) } \frac{9}{128}}$$

Q6.02.3 MCQ 2002 2M Ans: C ● ○ ○

Arrivals form a Poisson process with average inter-arrival time 10 minutes.
Hence arrival rate is

$$\lambda = \frac{1}{10} \text{ per minute.}$$

Call duration is exponentially distributed with mean 3 minutes, so the service rate is

$$\mu = \frac{1}{3} \text{ per minute.}$$

For an $M/M/1$ system, the probability that an arriving customer does not have to wait is

$$P_0 = 1 - \rho,$$

where

$$\rho = \frac{\lambda}{\mu}.$$

$$\rho = \frac{1/10}{1/3} = \frac{3}{10} = 0.3.$$

Therefore,

$$P_0 = 1 - 0.3 = 0.7.$$

(c) 0.7

Q6.01.1 MCQ 2001 2M Ans: D ● ○ ○

For an unbiased coin tossed 3 times, the total number of outcomes is

$$2^3 = 8$$

Exactly two heads can occur in

$$\binom{3}{2} = 3$$

ways.
Hence,

$$P(\text{exactly two heads}) = \frac{\binom{3}{2}}{2^3} = \frac{3}{8}$$

(d) $\frac{3}{8}$

Q6.00.1 MCQ 2000 1M Ans: D ● ○ ○

Given

$$P(E_1) = P(E_2)$$

and

$$P(E_1 \cup E_2) = 1, \qquad P(E_1 \cap E_2) = 1$$

Since

$$P(E_1 \cap E_2) \leq P(E_1)$$

and the intersection probability is 1, we must have

$$P(E_1) = P(E_2) = 1$$

As E_1 and E_2 are independent,

$$P(E_1 \cap E_2) = P(E_1)P(E_2)$$

Substituting,

$$1 = 1 \times 1$$

Therefore,

$$P(E_1) = 1$$

(d) 1

Q6.00.2 MCQ 2000 2M Ans: A ● ○ ○

Let X be the number of defective bolts.
Given

$$P(\text{defective}) = 0.1$$

and total bolts

$$n = 900$$

Since X follows a binomial distribution,

$$X \sim \text{Bin}(900, 0.1)$$

Mean:

$$\mu = np = 900(0.1) = 90$$

Variance:

$$\sigma^2 = np(1-p) = 900(0.1)(0.9) = 81$$

Standard deviation:

$$\sigma = \sqrt{81} = 9$$

(a) Mean $= 90$, Standard Deviation $= 9$

Q6.99.1 MCQ 1999 2M Ans: C ● ○ ○

Arrivals follow a Poisson process with rate

$$\lambda = 0.35 \text{ parts/min}$$

Mean service time is 2 min, hence service rate

$$\mu = \frac{1}{2} = 0.5 \text{ parts/min}$$

This is an $M/M/1$ queue.
Traffic intensity:

$$\rho = \frac{\lambda}{\mu} = \frac{0.35}{0.5} = 0.7$$

For an $M/M/1$ system,

$$P(N = n) = (1-\rho)\rho^n$$

Probability that there are already 8 parts in the system:

$$P(N = 8) = (1-0.7)(0.7)^8$$

$$P(N = 8) = 0.3 \times 0.057648 \approx 0.0173$$

(c) 0.0173

Q6.98.1 MCQ 1998 1M Ans: B ● ○ ○

Let the birth-month of the first friend be fixed.
For the second friend to have the same birth-month, the favorable months are 1 out of the 12 possible months.
Hence,

$$P(\text{same birth-month}) = \frac{1}{12}$$

(b) $\frac{1}{12}$

Q6.97.1 MCQ 1997 2M Ans: C ● ○ ○

To obtain exactly two black balls and one red ball, use combinations.
Total number of ways of selecting 3 balls from 8 balls:

$$N = \binom{8}{3} = 56$$

Favourable ways:

$$N_f = \binom{5}{2}\binom{3}{1}$$

$$= 10 \times 3 = 30$$

Therefore,

$$P = \frac{N_f}{N} = \frac{\binom{5}{2}\binom{3}{1}}{\binom{8}{3}} = \frac{30}{56} = \frac{15}{28}$$

$$\boxed{\text{(c) } \frac{15}{28}}$$

Q6.96.1 MCQ 1996 2M Ans: D ● ○ ○

Probability of a defective piece:

$$p = 0.01$$

Probability of a good piece:

$$q = 1 - p = 0.99$$

Exactly one defective piece out of 5 pieces can occur in

$${}^5C_1 = 5$$

ways.
Hence,

$$P(\text{exactly one defective}) = {}^5C_1(0.01)(0.99)^4$$

$$P = 5(0.99)^4(0.01)$$

$$\boxed{\text{(d) } 5(0.99)^4(0.01)}$$

Q6.93.1 MCQ 1993 1M Ans: A ● ○ ○

Probability that a manager is a technocrat:

$$p = 0.2$$

Probability that a manager is not a technocrat:

$$q = 0.8$$

Let X be the number of technocrats in a committee of 5 managers.
Then,

$$X \sim \text{Bin}(5, 0.2)$$

Required probability:

$$P(X = 2) = {}^5C_2(0.2)^2(0.8)^3$$

$$= 10(0.04)(0.512)$$

$$= 0.2048$$

(a) 0.2048

PrepFusion

SOLUTIONS — Probability and Statistics

CE

Civil Engineering

PrepFusion

Q3.26.1 NAT 2026 1M Ans: 0.58 ● ○ ○

Let the probability of choosing either bag be equal.

$$P(B_1) = P(B_2) = \frac{1}{2}$$

Probability of drawing a black ball from Bag I:

$$P(\text{Black}|B_1) = \frac{6}{10} = \frac{3}{5}$$

Probability of drawing a black ball from Bag II:

$$P(\text{Black}|B_2) = \frac{3}{7}$$

Using Bayes' theorem,

$$P(B_1|\text{Black}) = \frac{P(\text{Black}|B_1)P(B_1)}{P(\text{Black}|B_1)P(B_1) + P(\text{Black}|B_2)P(B_2)}$$

$$= \frac{\frac{3}{5} \cdot \frac{1}{2}}{\frac{3}{5} \cdot \frac{1}{2} + \frac{3}{7} \cdot \frac{1}{2}} = \frac{\frac{3}{5}}{\frac{3}{5} + \frac{3}{7}} = \frac{7}{12}$$

$$P(B_1|\text{Black}) = 0.5833$$

$$\boxed{0.58}$$

Q3.26.2 NAT 2026 2M Ans: 171.67 ● ○ ○

Age:

$$X \sim N(36, 12^2)$$

Height:

$$Y \sim N(160, 10^2)$$

Given

$$P(X > 50) = P(Y > h)$$

Standardizing age,

$$z = \frac{50 - 36}{12} = \frac{14}{12} = 1.1667$$

Hence,

$$P(X > 50) = P(Z > 1.1667)$$

Therefore,

$$\frac{h - 160}{10} = 1.1667$$

$$h = 160 + 10(1.1667)$$

$$h = 171.67$$

$$\boxed{171.67}$$

Key Points

- Equal tail probabilities imply equal standard normal z-scores.
- Convert both normal variables to the standard normal distribution.

Q3.26.3 NAT | 2026 | 1M | Ans: 49.84 ● ○ ○

For a storm with return period $T = 15$ years,

$$p = \frac{1}{15}$$

is the probability of occurrence in any year.
Probability that it does not occur in a year:

$$1 - p = \frac{14}{15}$$

Probability that it does not occur in the next 10 years:

$$\left(\frac{14}{15}\right)^{10}$$

Hence, probability of occurrence at least once in the next 10 years:

$$P = 1 - \left(\frac{14}{15}\right)^{10}$$

$$P = 0.4984$$

Percentage probability:

$$49.84\%$$

$$\boxed{49.84}$$

Q3.25.1 NAT | 2025 | 2M | Ans: 62.5 ● ○ ○

Let T denote truck speed and C denote car speed.

$$P(T > 50) = \frac{60 - 50}{60 - 30} = \frac{10}{30} = \frac{1}{3}$$

$$P(C > 50) = \frac{80 - 50}{80 - 40} = \frac{30}{40} = \frac{3}{4}$$

Given 30% trucks and 70% cars,

$$P(\text{Speed} > 50) = 0.3\left(\frac{1}{3}\right) + 0.7\left(\frac{3}{4}\right)$$

$$= 0.1 + 0.525 = 0.625$$

Percentage of vehicles exceeding the speed limit:

$$0.625 \times 100 = 62.5$$

$$\boxed{62.5}$$

Q3.25.2 MCQ 2025 1M Ans: A ● ○ ○

For a discrete random variable,

$$E[X] = \sum x_i P(X = x_i)$$

$$E[X] = 0(0.4) + 1(0.3) + 7(0.1) + 11(0.1) + 12(0.1)$$

$$E[X] = 3.3$$

Also,

$$E[X^2] = 0^2(0.4) + 1^2(0.3) + 7^2(0.1) + 11^2(0.1) + 12^2(0.1)$$

$$E[X^2] = 31.7$$

Variance is

$$\sigma^2 = E[X^2] - (E[X])^2$$

$$\sigma^2 = 31.7 - (3.3)^2$$

$$\sigma^2 = 20.81$$

$$\boxed{\text{(a) } 20.81}$$

Q3.25.3 MSQ 2025 1M Ans: B,C ● ○ ○

Option A:

If A and B are mutually exclusive,

$$P(A \cap B) = 0$$

whereas $P(A)P(B)$ is generally non-zero.

Hence, A is incorrect.

Option B:

If $B \subset A$,

$$A \cap B = B$$

Therefore,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B)}{P(B)} = 1$$

Hence, B is correct.

Option C:

For mutually exclusive events,

$$P(A \cup B) = P(A) + P(B)$$

Hence, C is correct.

Option D:

For independent events,

$$P(A \cap B) = P(A)P(B)$$

which is not necessarily zero.

Hence, D is incorrect.

(b) Conditional probability, $P(A|B) = 1$ if $B \subset A$, (c) $P(A \cup B) = P(A) + P(B)$, if A and B are mutually exclusive

Q3.25.4 NAT | 2025 | 2M | Ans: 2.8 ● ○ ○

The probability distribution is

x_i	1	2	4	8
$P(X = x_i)$	0.3	0.1	0.3	0.3

Mean:

$$\mu = E[X] == 1(0.3) + 2(0.1) + 4(0.3) + 8(0.3) = 4.1$$

Next,

$$E[X^2] = 1^2(0.3) + 2^2(0.1) + 4^2(0.3) + 8^2(0.3) = 24.7$$

Variance:

$$\sigma^2 = E[X^2] - \mu^2 == 24.7 - (4.1)^2 == 7.89$$

Standard deviation:

$$\sigma = \sqrt{7.89} \approx 2.81$$

$$\boxed{2.8}$$

Q3.24.1 MCQ | 2024 | 2M | Ans: C ● ○ ○

Out of 100 patients, 50 take medicine X and 50 do not.

Without medicine X,

$$P(H|\overline{X}) = 0.8$$

Medicine X reduces the probability by 50%, hence

$$P(H|X) = 0.8(1 - 0.5) = 0.4$$

Also,

$$P(X) = \frac{50}{100} = 0.5$$

Therefore,

$$P(H \cap X) = P(H|X)P(X)$$

$$= 0.4 \times 0.5 = 0.2$$

$$= 20\%$$

(c) 20%

Q3.24.2 MCQ 2024 1M Ans: A ● ○ ○

Let M denote passing Mathematics and E denote passing English.
Given,

$$P(M \text{ only}) = \frac{1}{3}, \qquad P(E \text{ only}) = \frac{4}{9}$$

$$P(M \cap E) = \frac{1}{6}$$

Probability of passing in at least one subject is

$$P(M \cup E) = P(M \text{ only}) + P(E \text{ only}) + P(M \cap E)$$

$$= \frac{1}{3} + \frac{4}{9} + \frac{1}{6}$$

$$= \frac{6 + 8 + 3}{18} = \frac{17}{18}$$

(a) $\frac{17}{18}$

Q3.23.1 MCQ 2023 1M Ans: B ● ○ ○

Vehicles are numbered from 1000 to 1999.
30% of 1000 vehicles are without pollution certificate.

$$1000 \times 0.3 = 300$$

On an even-numbered date, only even-numbered vehicles operate.
Among 1000 sequential numbers, half are even.

$$300 \times \frac{1}{2} = 150$$

Hence, among the 100 randomly chosen vehicles, the expected number without pollution certificate is

$$100 \times \frac{150}{500} == 30$$

(b) 30

Q3.23.2 NAT 2023 1M Ans: 0.9 ● ○ ○

For independent events,

$$P(A) = 0.5, \qquad P(B) = 0.8$$

Probability of at least one event occurring:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since A and B are independent,

$$P(A \cap B) = P(A)P(B)$$

$$= 0.5 \times 0.8 = 0.4$$

Hence,

$$P(A \cup B) = 0.5 + 0.8 - 0.4 = 0.9$$

0.9

Q3.23.3 MCQ 2023 1M Ans: B ● ○ ○

For a positively skewed (right-skewed) distribution,

$$\text{Mean} > \text{Median}$$

Function 2 has a long tail towards the right, indicating positive skewness.
Function 1 is approximately symmetric.
Function 3 is negatively skewed, for which

$$\text{Mean} < \text{Median}$$

Function 4 is bimodal and no such conclusion can be directly drawn.
Therefore, only Function 2 has mean greater than median.

(b) Function 2

Key Points

- Right-skewed distribution: Mean > Median.
- Left-skewed distribution: Mean < Median.
- Symmetric distribution: Mean = Median.

Q3.22.1 NAT 2022 2M Ans: 0.80 ● ○ ○

Let X be the number of cars arriving in 30 minutes and Y be the number of buses arriving in 30 minutes.

$$X \sim \text{Poisson}(4 \times 0.5) = \text{Poisson}(2)$$

$$Y \sim \text{Poisson}(2 \times 0.5) = \text{Poisson}(1)$$

Since the two processes are independent,

$$X + Y \sim \text{Poisson}(2 + 1) = \text{Poisson}(3)$$

Required probability:

$$P(X + Y \geq 2) = 1 - P(X + Y = 0) - P(X + Y = 1)$$

$$= 1 - e^{-3} - 3e^{-3} = 1 - 4e^{-3} = 0.80085$$

$$\boxed{0.80}$$

Q3.22.2 NAT 2022 2M Ans: 0.019 ● ○ ○

For two fair six-faced dice,

$$P(\text{sum} = 4) = \frac{(1,3),(2,2),(3,1)}{36} = \frac{3}{36} = \frac{1}{12}$$

Let success be "sum equals 4".

$$p = \frac{1}{12}, \qquad q = 1 - p = \frac{11}{12}$$

The experiment is repeated 3 times. The probability of getting exactly two successes is

$$\binom{3}{2} p^2 q$$

$$= 3\left(\frac{1}{12}\right)^2 \left(\frac{11}{12}\right) = \frac{33}{1728} = 0.019097$$

$$\boxed{0.019}$$

Q3.22.3 NAT 2022 2M Ans: 1.9 ● ○ ○

Given observations:

x	5	2	4	3
y	16	10	13	12

For the regression model

$$y = a + bx$$

the slope is

$$b = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$$

Mean values:

$$\bar{x} = \frac{5+2+4+3}{4} = 3.5, \qquad \bar{y} = \frac{16+10+13+12}{4} = 12.75$$

$$\sum(x-\bar{x})(y-\bar{y}) = (1.5)(3.25) + (-1.5)(-2.75) + (0.5)(0.25) + (-0.5)(-0.75) = 9.5$$

$$\sum(x-\bar{x})^2 = (1.5)^2 + (-1.5)^2 + (0.5)^2 + (-0.5)^2 = 5$$

$$b = \frac{9.5}{5} = 1.9$$

$$\boxed{1.9}$$

Key Points

- In linear regression, the slope is covariance divided by variance of the independent variable.
- The intercept can be obtained using $a = \bar{y} - b\bar{x}$.

Q3.21.1 NAT | 2021 | 2M | Ans: 0.59 ● ○ ○

Let X be the number of vehicles arriving in a 1-minute interval.
Since arrivals follow a Poisson distribution with mean rate

$$\lambda = 2$$

$$P(X \geq 2) = 1 - P(X=0) - P(X=1)$$

$$= 1 - e^{-2} - 2e^{-2}$$

$$= 1 - 3e^{-2}$$

$$= 1 - 3(0.1353) = 0.5941$$

$$\boxed{0.59}$$

Q3.21.2 MCQ | 2021 | 1M | Ans: B ● ○ ○

The cumulative distribution function (CDF) of a Gaussian (normal) distribution starts near 0, rises smoothly, and approaches 1 asymptotically.
Hence its graph has an S-shape (sigmoid shape).

$$\boxed{\text{(b) S-shaped}}$$

Q3.20.1 NAT 2020 1M Ans: 0.60 ● ○ ○

A 50-year flood has annual probability

$$p = \frac{1}{50} = 0.02$$

Probability that it does not occur in a given year:

$$1 - p = 0.98$$

Assuming independence, probability that it does not occur in all 25 years is

$$(0.98)^{25} = 0.6035$$

$$\boxed{0.60}$$

Q3.20.2 NAT 2020 1M Ans: 0.196 ● ○ ○

Let X be the number of heads obtained in 15 tosses.

$$X \sim \text{Binomial}(15, \tfrac{1}{2})$$

Required probability:

$$P(X = 8) = \binom{15}{8}\left(\frac{1}{2}\right)^{15}$$

$$= \frac{6435}{32768} = 0.19638$$

$$\boxed{0.196}$$

Q3.19.1 MCQ 2019 2M Ans: B ● ○ ○

For a continuous uniform random variable distributed over the interval $[x, y]$ with $y > x$,

$$f(t) = \frac{1}{y - x}, \qquad x \le t \le y$$

since the total probability must be unity:

$$\int_x^y f(t)\,dt = 1$$

$$f(t)(y - x) = 1$$

$$f(t) = \frac{1}{y - x}$$

$$\boxed{\text{(b) } \frac{1}{y - x}}$$

Q3.19.2 NAT | 2019 | 2M | Ans: 0.18 | ● ○ ○

Vehicle arrivals follow a Poisson process with rate

$$\lambda = \frac{360}{3600} = 0.1 \text{ s}^{-1}$$

For a Poisson process, the headway T between successive vehicles follows an exponential distribution:

$$f_T(t) = \lambda e^{-\lambda t}$$

Required probability:

$$P(6 \le T \le 10) = \int_6^{10} 0.1e^{-0.1t}\, dt$$

$$= e^{-0.6} - e^{-1}$$

$$= 0.5488 - 0.3679$$

$$= 0.1809$$

$$\boxed{0.18}$$

Q3.18.1 NAT | 2018 | 1M | Ans: 0.5 | ● ○ ○

A box contains 5 red balls and 1 white ball.
The probability that all 3 balls picked without replacement are red is

$$P = \frac{\binom{5}{3}}{\binom{6}{3}}$$

$$= \frac{10}{20} = 0.5$$

$$\boxed{0.5}$$

Q3.18.2 MCQ | 2018 | 1M | Ans: C | ● ○ ○

For a positively (right) skewed distribution, the long tail lies towards the right.
The measures of central tendency satisfy

$$\text{Mode} < \text{Median} < \text{Mean}$$

Hence,

$$\text{Mean} > \text{Median} \quad \text{and} \quad \text{Mean} > \text{Mode}$$

Therefore, the correct statement is:

$$\boxed{\text{(c) Mean is greater than median and mode}}$$

Q3.18.3 MCQ 2018 1M Ans: A ● ○ ○

For a valid probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

From the graph, $f(x)$ consists of three triangular regions:

$$A_1 = \frac{1}{2}(1)(h) = \frac{h}{2}$$

$$A_2 = \frac{1}{2}(1)(2h) = h$$

$$A_3 = \frac{1}{2}(1)(3h) = \frac{3h}{2}$$

Hence,

$$A_1 + A_2 + A_3 = 1$$

$$\frac{h}{2} + h + \frac{3h}{2} = 1$$

$$3h = 1$$

$$h = \frac{1}{3}$$

$$\boxed{\text{(a) } \frac{1}{3}}$$

Q3.18.4 NAT 2018 1M Ans: 0 ● ○ ○

The frequency distribution is

f (MPa)	23	28	22.5	31	29
n	4	2	5	5	4

Mean strength:

$$\mu = \frac{\sum nf}{20} = \frac{4(23) + 2(28) + 5(22.5) + 5(31) + 4(29)}{20}$$

$$= \frac{531.5}{20} = 26.575$$

Standard deviation:

$$\sigma = \sqrt{\frac{\sum n(f-\mu)^2}{20}}$$

$$= \sqrt{12.9818} \approx 3.603$$

$$\mu - 3\sigma = 26.575 - 3(3.603)$$

$$= 15.766$$

Since the minimum observed strength is 22.5 MPa,

$$f > \mu - 3\sigma$$

for all specimens.
Hence, the number of specimens having strength less than $\mu - 3\sigma$ is

$$\boxed{0}$$

Key Points

- Mean is $\mu = \dfrac{\sum nf}{N}$.
- Standard deviation is $\sigma = \sqrt{\dfrac{\sum n(f-\mu)^2}{N}}$.

Q3.17.1 NAT | 2017 | 1M | Ans: 210 | ● ○ ○

For a normal distribution, the upper limit corresponding to a 95% confidence level is

$$\mu + z\sigma$$

Given,

$$\mu = 200 \text{ days}, \qquad \sigma = 6.1 \text{ days}, \qquad z = 1.64$$

$$T_{max} = 200 + 1.64(6.1)$$

$$= 200 + 10.004 = 210.004$$

$$\boxed{210}$$

Q3.17.2 NAT | 2017 | 1M | Ans: 0.5 | ● ○ ○

The coin is fair and each toss is independent of previous tosses.
Hence,

$$P(H) = \frac{1}{2}$$

Even though the previous outcomes are

$$H,\ H,\ H$$

the probability of obtaining a head in the next toss remains unchanged.

$$P(H \mid HHH) = \frac{1}{2} = 0.5$$

$$\boxed{0.5}$$

Q3.17.3 MCQ 2017 2M Ans: B ● ○ ○

For

$$f(x) = a + bx, \qquad 0 \le x \le 1$$

to be a valid probability density function,

$$\int_0^1 f(x)\,dx = 1$$

$$\int_0^1 (a + bx)\,dx = a + \frac{b}{2} = 1$$

Checking the options:

$$\text{Option A: } 1 + \frac{4}{2} = 3 \neq 1$$

$$\text{Option B: } 0.5 + \frac{1}{2} = 1$$

But

$$f(0) = 0.5, \qquad f(1) = 1.5$$

This is valid.

$$\boxed{\text{(b) } a = 0.5,\ b = 1}$$

Q3.17.4 NAT 2017 1M Ans: 0.1353 ● ○ ○

Vehicle arrivals follow a Poisson process with rate

$$\lambda = \frac{900}{3600} = 0.25\ \text{s}^{-1}$$

The gap between successive arrivals follows an exponential distribution.

$$f(t) = \lambda e^{-\lambda t}$$

Required probability:

$$P(T > 8) = e^{-\lambda \cdot 8}$$

$$= e^{-0.25 \times 8} = e^{-2}$$

$$= 0.1353$$

$$\boxed{0.1353}$$

Q3.16.1 MCQ 2016 2M Ans: B ● ○ ○

Since both probability density functions are symmetric about the origin,

$$E[X]_f = E[X]_g = 0.$$

Method 1

Compute the second moment for both distributions.
For $f(x)$:

$$E[X^2]_f = 2\int_0^a x^2\left(1-\frac{x}{a}\right)dx = 2\left[\frac{x^3}{3}-\frac{x^4}{4a}\right]_0^a = \frac{a^3}{6}.$$

Hence,

$$\text{Var}(f) = E[X^2]_f - (E[X]_f)^2 = \frac{a^3}{6}.$$

For $g(x)$:

$$E[X^2]_g = 2\int_0^a x^2\left(\frac{x}{a}\right)dx = \frac{2}{a}\left[\frac{x^4}{4}\right]_0^a = \frac{a^3}{2}.$$

Thus,

$$\text{Var}(g) = E[X^2]_g - (E[X]_g)^2 = \frac{a^3}{2}.$$

Method 2

Observe that both PDFs are mirror images about $x = 0$, so their means must be zero. However, $f(x)$ is concentrated near the origin whereas $g(x)$ has more probability near the ends $\pm a$. Hence their spreads (variances) are different.

(b) Mean of $f(x)$ and $g(x)$ are same; Variance of $f(x)$ and $g(x)$ are different.

Key Points

- A PDF symmetric about the origin has zero mean.
- Variance is determined by the second moment $E[X^2]$.
- More probability away from the mean results in larger variance.

Q3.16.2 MCQ 2016 2M Ans: A ● ○ ○

Given

$$f(x) = \begin{cases} 0.25, & 1 \le x \le 5 \\ 0, & \text{otherwise} \end{cases}$$

Required probability:

$$P(X \le 4) = \int_1^4 0.25\,dx$$

$$= 0.25(4-1)$$

$$= \frac{3}{4}$$

(a) $\frac{3}{4}$

Q3.16.3 MCQ | 2016 | 1M | Ans: A | ● ○ ○

Given

$$P(X) = 0.40$$

and

$$P(X \cup Y^c) = 0.70$$

Using De Morgan's law,

$$(X \cup Y^c)^c = X^c \cap Y$$

Hence,

$$P(X^c \cap Y) = 1 - 0.70 = 0.30$$

Since X and Y are independent,

$$P(X^c \cap Y) = P(X^c)P(Y)$$

$$0.30 = (1 - 0.40)P(Y)$$

$$0.30 = 0.60P(Y)$$

$$P(Y) = 0.50$$

Again, by independence,

$$P(X \cap Y) = P(X)P(Y)$$

$$= 0.40 \times 0.50$$

$$= 0.20$$

Therefore,

$$P(X \cup Y) = P(X) + P(Y) - P(X \cap Y)$$

$$= 0.40 + 0.50 - 0.20$$

$$= 0.70$$

(a) 0.7

Q3.16.4 NAT 2016 1M Ans: 54.5 ● ○ ○

Arrange the speeds in ascending order:

$$32,\ 45,\ 49,\ 51,\ 53,\ 56,\ 60,\ 62,\ 66,\ 79$$

There are 10 observations.
Hence, the median is the average of the 5th and 6th observations.

$$\text{Median} = \frac{53 + 56}{2}$$

$$= 54.5$$

$$\boxed{54.5}$$

Q3.15.1 NAT 2015 1M Ans: 0.24 ● ○ ○

Given

$$P(X = 0) = q, \qquad P(X = 1) = 1 - q.$$

Since X is a Bernoulli random variable,

$$\text{Var}(X) = q(1 - q).$$

For $q = 0.4$,

$$\text{Var}(X) = 0.4 \times 0.6 = 0.24.$$

$$\boxed{0.24}$$

Q3.15.2 NAT 2015 2M Ans: 1.06 ● ○ ○

Given

$$f(x) = \begin{cases} \frac{x}{4}(4 - x^2), & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

The mean is

$$\mu_x = E[X] = \int_0^2 x\, f(x)\, dx.$$

Substituting $f(x)$,

$$\mu_x = \frac{1}{4}\int_0^2 (4x^2 - x^4)\, dx.$$

$$\mu_x = \frac{1}{4}\left[\frac{4x^3}{3} - \frac{x^5}{5}\right]_0^2 = \frac{1}{4}\left(\frac{32}{3} - \frac{32}{5}\right) = \frac{16}{15} = 1.06.$$

$$\boxed{1.06}$$

Key Points

- Mean of a continuous random variable is $E[X] = \int x f(x)\, dx$.
- Always use the interval over which the p.d.f. is non-zero.

Q3.14.1 NAT 2014 1M Ans: 0.4 ● ○ ○

Given

$$f(E) = \begin{cases} \frac{1}{5}, & 0 \le E \le 5, \\ 0, & \text{otherwise.} \end{cases}$$

Required probability:

$$P(2 \le E \le 4) = \int_2^4 \frac{1}{5}\, dE = \frac{1}{5}(4-2) = \frac{2}{5}.$$

$$P(2 \le E \le 4) = 0.4.$$

$$\boxed{0.4}$$

Q3.14.2 NAT 2014 2M Ans: 0.27 ● ○ ○

Vehicle arrivals follow a Poisson distribution with rate

$$\lambda = 240 \text{ veh/h}.$$

For a 30-second interval,

$$\mu = \lambda t = 240\left(\frac{30}{3600}\right) = 2.$$

Probability of exactly one arrival:

$$P(X = 1) = \frac{\mu^1 e^{-\mu}}{1!} = 2e^{-2}.$$

$$P(X = 1) \approx 0.2707.$$

$$\boxed{0.27}$$

Q3.14.3 MCQ 2014 1M Ans: B ● ○ ○

Total outcomes after four tosses:

$$2^4 = 16.$$

Since the coin is fair and tosses are independent,

$$P(T) = \frac{1}{2}.$$

Past outcomes do not affect the next toss.

$$\boxed{\text{(b) } \frac{1}{2}}$$

Q3.14.4 NAT 2014 2M Ans: 0.265 ● ○ ○

Number of penalties per day follows

$$X \sim \text{Poisson}(5).$$

Required probability:

$$P(X < 4) = P(0) + P(1) + P(2) + P(3).$$

$$= e^{-5}\left(1 + \frac{5^1}{1!} + \frac{5^2}{2!} + \frac{5^3}{3!}\right).$$

$$= e^{-5}(1 + 5 + 12.5 + 20.8333) \approx 0.2650.$$

$$\boxed{0.265}$$

Q3.14.5 MCQ 2014 1M Ans: B ● ○ ○

Given

$$f(x) = \frac{1}{\sqrt{2\pi}\, b} e^{-\frac{1}{2}\left(\frac{x-a}{b}\right)^2},$$

which is the p.d.f. of a normal random variable with mean a and standard deviation b.
Let

$$z = \frac{x-a}{b}, \qquad dx = b\, dz.$$

Then

$$\int_{-\infty}^{a} \frac{1}{\sqrt{2\pi}\, b} e^{-\frac{1}{2}\left(\frac{x-a}{b}\right)^2} dx = \int_{-\infty}^{0} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz.$$

By symmetry of the standard normal distribution,

$$P(Z \le 0) = \frac{1}{2}.$$

(b) 0.5

Q3.13.1 MCQ 2013 1M Ans: D ● ○ ○

A return period of 50 years implies that the probability of occurrence of such a rainfall event in any year is

$$p = \frac{1}{50} = 0.02.$$

Assuming independence between successive years, the probability that the event occurs in both of two consecutive years is

$$p^2 = (0.02)^2 = 0.0004.$$

(d) 0.0004

Q3.13.2 NAT 2013 2M Ans: 6 ● ○ ○

For a valid probability density function,

$$\int_{-\infty}^{\infty} f(x)\, dx = 1.$$

Given

$$f(x) = \begin{cases} \lambda(x-1)(2-x), & 1 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore,

$$\lambda \int_{1}^{2} (x-1)(2-x)\, dx = 1.$$

Expanding,

$$(x-1)(2-x) = -x^2 + 3x - 2.$$

Hence,

$$\lambda \left[-\frac{x^3}{3} + \frac{3x^2}{2} - 2x \right]_1^2 = 1.$$

$$\lambda \left(\frac{1}{6} \right) = 1.$$

$$\lambda = 6.$$

6

Q3.12.1 MCQ 2012 2M Ans: B ● ○ ○

Let R denote the event that a shock absorber is reliable.

$$P(X) = 0.6, \qquad P(Y) = 0.4.$$

$$P(R|X) = 0.96, \qquad P(R|Y) = 0.72.$$

We need to compute the conditional probability that a randomly chosen shock absorber was manufactured by Y, given that it is found to be reliable, denoted as $P(Y|R)$.
By applying **Bayes' Theorem**:

$$P(Y|R) = \frac{P(R|Y) \cdot P(Y)}{P(R)}$$

First, calculate the total probability of selecting a reliable shock absorber, $P(R)$, using the law of total probability:

$$P(R) = [P(R|X) \cdot P(X)] + [P(R|Y) \cdot P(Y)]$$

$$P(R) = (0.96 \cdot 0.60) + (0.72 \cdot 0.40)$$

$$P(R) = 0.576 + 0.288 = 0.864$$

Now, substitute $P(R)$ back into the Bayes' Theorem equation:

$$P(Y|R) = \frac{0.72 \cdot 0.40}{0.864} = \frac{0.288}{0.864}$$

$$P(Y|R) = \frac{1}{3} \approx 0.3333$$

Thus, the closest decimal approximation among the given options is 0.334.

(b) 0.334

Q3.12.2 MCQ 2012 2M Ans: D ● ○ ○

Each trial is equally likely to be positive or negative. Let X be the number of negative values in 5 trials.

$$X \sim \text{Binomial}\left(5, \frac{1}{2}\right).$$

Required probability:

$$P(X \leq 1) = P(X = 0) + P(X = 1).$$

$$= \binom{5}{0}\left(\frac{1}{2}\right)^5 + \binom{5}{1}\left(\frac{1}{2}\right)^5 = \frac{1+5}{32} = \frac{6}{32}.$$

(d) $\frac{6}{32}$

Q3.12.3 MCQ 2012 1M Ans: A ● ○ ○

Annual precipitation is normally distributed with

$$\mu = 1000\,\text{mm}, \qquad \sigma = 200\,\text{mm}.$$

Since

$$1200 = \mu + \sigma,$$

the required probability is

$$P(X > 1200) = P(Z > 1).$$

From the standard normal table,

$$P(Z > 1) \approx 0.1587.$$

Thus the probability is less than 50%.

$$\boxed{\text{(a) } < 50\%}$$

Q3.11.1 MCQ 2011 1M Ans: C ● ○ ○

Let R be the event that the ball drawn from the first container is Red and B be the event that the ball drawn from the second container is Blue.

$$P(R) = \frac{4}{4+3} = \frac{4}{7}$$

$$P(B) = \frac{3}{3+4} = \frac{3}{7}$$

Since the draws are from different containers, the events are independent.

$$P(R \cap B) = P(R) \times P(B) = \frac{4}{7} \times \frac{3}{7} = \frac{12}{49}$$

$$\boxed{\text{(c) } \frac{12}{49}}$$

Q3.11.2 MCQ 2011 1M Ans: D ● ○ ○

The arrival-time distribution is

$$P(T = t) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

which is the Poisson probability of observing k arrivals in time interval t.
For the headway (time between consecutive arrivals) in a Poisson process,

$$P(T \geq t_1) = e^{-\lambda t_1}$$

This is the survival function of the exponential distribution.

$$\boxed{\text{(d) } e^{-\lambda t_1}}$$

Key Points

- Vehicle arrivals follow a Poisson process with rate λ.
- Inter-arrival time (headway) is exponentially distributed.
- For an exponential random variable, $P(T \geq t) = e^{-\lambda t}$.

Mistakes to be avoided

- Do not confuse the Poisson arrival-count distribution with the exponential headway distribution.
- The required probability is a tail probability, not the PDF value.

Q3.10.1 MCQ | 2010 | 1M | Ans: C | ● ○ ○

For two fair coins,

$$P(\text{Head}) = \frac{1}{2}$$

Since the tosses are independent,

$$P(\text{two heads}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$\boxed{\text{(c) } \frac{1}{4}}$$

Q3.09.1 MCQ | 2009 | 2M | Ans: D | ● ● ○

Given mean precipitation

$$\mu = 102\,\text{cm}$$

and standard deviation

$$\sigma = 27\,\text{cm}$$

For precipitation of 90 cm,

$$x_N = \frac{90 - 102}{27} = -0.444$$

Using

$$F(x_N) = \frac{1}{1 + \exp\left(-1.7255 x_N |x_N|^{0.12}\right)}$$

$$F(-0.444) \approx 0.333$$

Hence,

$$P(90 \le X \le 102) = F(0) - F(-0.444)$$

Since

$$F(0) = 0.5$$

$$P(90 \le X \le 102) = 0.5 - 0.333 = 0.167$$

$$= 16.7\%$$

$$\boxed{\text{(d) } 16.7\%}$$

Q3.08.1 MCQ 2008 2M Ans: A ● ○ ○

Probability of using a private car is

$$P(\text{Car}) = 0.45$$

Hence probability of using public transport is

$$P(\text{Public}) = 1 - 0.45 = 0.55$$

Given

$$P(\text{Bus}|\text{Public}) = 0.55$$

$$P(\text{Bus}) = 0.55 \times 0.55 = 0.3025 \approx 0.30$$

The remaining public-transport users choose metro.

$$P(\text{Metro}) = 0.55 - 0.3025 = 0.2475 \approx 0.25$$

Therefore,

$$P(\text{Car}) = 0.45, \quad P(\text{Bus}) = 0.30, \quad P(\text{Metro}) = 0.25$$

(a) 0.45, 0.30, 0.25

Q3.08.2 MCQ 2008 2M Ans: B ● ○ ○

Given

$$f(x) = x^2, \qquad -1 \le x \le 1$$

The required probability is

$$P\left(-\frac{1}{3} \le X \le \frac{1}{3}\right) = \int_{-1/3}^{1/3} x^2\, dx$$

$$= \left[\frac{x^3}{3}\right]_{-1/3}^{1/3} = \frac{1}{81} - \left(-\frac{1}{81}\right) = \frac{2}{81}$$

$$P = \frac{2}{81} \approx 0.0247$$

Hence the percentage probability is

$$0.0247 \times 100 = 2.47$$

(b) 2.47

Q3.08.3 MCQ | 2008 | 2M | Ans: D | ● ○ ○

For the straight line

$$y = a + bx$$

the normal equations are

$$\sum y = na + b\sum x$$

$$\sum xy = a\sum x + b\sum x^2$$

Given

$$n = 3, \quad \sum x = 6, \quad \sum y = 21,$$

$$\sum x^2 = 14, \quad \sum xy = 46$$

Substituting,

$$21 = 3a + 6b$$

$$46 = 6a + 14b$$

From the first equation,

$$a + 2b = 7$$

$$a = 7 - 2b$$

Substituting into the second equation,

$$46 = 6(7 - 2b) + 14b$$

$$46 = 42 + 2b$$

$$b = 2$$

$$a = 7 - 4 = 3$$

Therefore,

$$a = 3, \qquad b = 2$$

$$\boxed{\text{(d) } a = 3,\ b = 2}$$

Q3.07.1 MCQ | 2007 | 2M | Ans: C | ● ○ ○

The coefficient of variation (CV) is defined as

$$\text{CV} = \frac{\sigma}{\mu}$$

Given

$$\sigma = 8.8\,\text{kmph}, \qquad \mu = 33\,\text{kmph}$$

$$\text{CV} = \frac{8.8}{33} = 0.2667$$

(c) 0.2666

Q3.06.1 MCQ | 2006 | 2M | Ans: B | ● ○ ○

There are 25 calculators, out of which 2 are defective and 23 are good.
Exactly one defective calculator must be selected among the 5 chosen.

$$P = \frac{\binom{2}{1}\binom{23}{4}}{\binom{25}{5}}$$

$$P = \frac{2 \times 8855}{53130} = \frac{17710}{53130} = \frac{1}{3}$$

(b) $\frac{1}{3}$

Q3.06.2 MCQ | 2006 | 2M | Ans: D | ● ○ ○

Normalization is done using

$$z = \frac{x - \mu_C}{\sigma_C}$$

and the transformed score is

$$x' = \mu + \sigma z$$

Given for Batch C,

$$\mu_C = 6.6, \qquad \sigma_C = 2.3$$

For the entire class,

$$\mu = 5.5, \qquad \sigma = 4.2$$

For a student with score $x = 8.5$,

$$z = \frac{8.5 - 6.6}{2.3} = \frac{1.9}{2.3} \approx 0.826$$

Hence,

$$x' = 5.5 + 4.2(0.826)$$

$$x' \approx 5.5 + 3.47 \approx 8.97$$

$$x' \approx 9.0$$

(d) 9.0

Q3.05.1 MCQ 2005 1M Ans: D ● ○ ○

The measure of skewness is dependent upon the amount of dispersion. This statement is true
For a symmetric distribution,

$$\text{Mean} = \text{Median} = \text{Mode}$$

Hence Option B is true.
For a positively skewed distribution,

$$\text{Mean} > \text{Median} > \text{Mode}$$

Hence Option C is also true.
For a negatively skewed distribution,

$$\text{Mean} < \text{Median} < \text{Mode}$$

or equivalently,

$$\text{Mode} > \text{Median} > \text{Mean}$$

Hence Option D is false.

(d) In a negatively skewed distribution: mode > mean > median.

Q3.04.1 MCQ 2004 2M Ans: C ● ○ ○

Let F_i denote the failure of gate i.
Given

$$P(F_i) = 0.2$$

and the gates operate independently.
We need

$$P(F_2 \cap F_3 \mid F_1)$$

Since failures are independent,

$$P(F_2 \cap F_3 \mid F_1) = P(F_2 \cap F_3)$$

$$= P(F_2)\, P(F_3) = 0.2 \times 0.2 = 0.04$$

(c) 0.040

Q3.03.1 MCQ | 2003 | 1M | Ans: C | ● ○ ○

A box contains 10 screws, of which 3 are defective.
Hence, the probability of selecting a good screw is

$$P(\text{Good}) = \frac{7}{10}$$

Since the screws are drawn *with replacement*, the two draws are independent.

$$P(\text{No defective screw}) = P(\text{Good on first draw}) \times P(\text{Good on second draw})$$

$$= \frac{7}{10} \times \frac{7}{10} = \frac{49}{100}$$

$$= 49\%$$

(c) 49%

PrepFusion

SOLUTIONS — Probability and Statistics

CH

Chemical Engineering

PrepFusion

Q3.26.1 NAT 2026 1M Ans: 0.5 ● ○ ○

Let

$$A = \{2, 3, 5\}$$

be the event that the outcome is a prime number.
Also,

$$B = \{1, 3\}$$

since the outcome is an odd number less than 4.
The conditional probability is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Now,

$$A \cap B = \{3\}.$$

Hence,

$$P(A \cap B) = \frac{1}{6}, \qquad P(B) = \frac{2}{6}.$$

Therefore,

$$P(A \mid B) = \frac{\frac{1}{6}}{\frac{2}{6}} = \frac{1}{2} = 0.5.$$

$$\boxed{0.5}$$

Q3.25.1 MCQ 2025 1M Ans: D ● ○ ○

Total balls = 3 + 7 = 10.
Number of ways of selecting 2 balls:

$${}^{10}C_2.$$

For exactly one green and one blue ball,

$${}^{3}C_1 \times {}^{7}C_1.$$

Hence,

$$P(1 \text{ green and } 1 \text{ blue}) = \frac{{}^{3}C_1 \times {}^{7}C_1}{{}^{10}C_2}.$$

$$\boxed{\text{(d) } \frac{{}^{3}C_1 \times {}^{7}C_1}{{}^{10}C_2}}$$

Q3.25.2 MCQ 2025 2M Ans: A ● ○ ○

Given

$$p(x) = \begin{cases} \frac{1}{a}, & x \in (0, a), \\ 0, & \text{otherwise.} \end{cases}$$

For

$$f(x) = x^2,$$

the expectation is

$$E[f(x)] = \int_{-\infty}^{\infty} f(x)p(x)\,dx = \int_0^a x^2 \left(\frac{1}{a}\right) dx.$$

Therefore,

$$E[f(x)] = \frac{1}{a}\int_0^a x^2\,dx = \frac{1}{a}\left[\frac{x^3}{3}\right]_0^a.$$

$$E[f(x)] = \frac{1}{a}\cdot\frac{a^3}{3} = \frac{a^2}{3}.$$

$$\boxed{\text{(a) } \frac{a^2}{3}}$$

Q3.24.1 MCQ | 2024 | 1M | Ans: B ● ○ ○

Given

$$f(x) = \frac{4}{\sqrt{2\pi}}e^{-8(x+3)^2}.$$

Compare with the standard normal distribution form

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

From the exponent,

$$8(x+3)^2 = \frac{(x-\mu)^2}{2\sigma^2}$$

which gives

$$\frac{1}{2\sigma^2} = 8 \quad \Rightarrow \quad \sigma^2 = \frac{1}{16}$$

and hence

$$\sigma = \frac{1}{4}.$$

Also,

$$(x+3)^2 = (x-(-3))^2,$$

therefore

$$\mu = -3.$$

Hence,

$$(\mu, \sigma) = \left(-3, \frac{1}{4}\right).$$

$$\boxed{\text{(b) } \left(-3, \frac{1}{4}\right)}$$

Q3.23.1 MCQ 2023 2M Ans: A ● ● ○

Let the arrival times of visitors X and Y be independent and uniformly distributed on $[0, 60]$ minutes.
A visit lasts:

$$20 \text{ minutes}, \quad \text{if arrival time} < 3:40,$$

and until 4 PM otherwise.
Two visits do not overlap only when one visitor leaves before the other enters.
Assume $x < y$. Then non-overlap occurs if

$$x < 40 \quad \text{and} \quad y > x + 20.$$

In the square $0 \le x, y \le 60$, the total area is

$$60^2 = 3600.$$

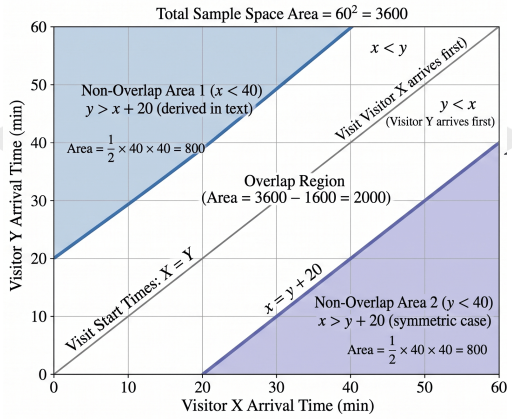


Fig. Solution Q3.23.1

The non-overlapping region for $x < 40$ has area

$$\int_0^{40} (60 - (x + 20))\, dx = \int_0^{40} (40 - x)\, dx = 800.$$

By symmetry, the total non-overlapping area is

$$2 \times 800 = 1600.$$

Therefore,

$$P(\text{non-overlap}) = \frac{1600}{3600} = \frac{4}{9}.$$

Hence,

$$P(\text{overlap}) = 1 - \frac{4}{9} = \frac{5}{9}.$$

$$\boxed{\text{(a) } \frac{5}{9}}$$

Q3.21.1 NAT | 2021 | 2M | Ans: 0.070 | ● ○ ○

From the figure,

$$f(x) = \begin{cases} 0.5, & 1 \le x \le 3, \\ 0, & \text{otherwise.} \end{cases}$$

Thus,

$$X \sim U(1, 3).$$

For a uniform distribution,

$$\sigma_X = \frac{b-a}{\sqrt{12}} = \frac{3-1}{\sqrt{12}} = \frac{1}{\sqrt{3}}.$$

Given sample size

$$n = 68,$$

the standard deviation of the sample mean is

$$\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}} = \frac{1/\sqrt{3}}{\sqrt{68}} = \frac{1}{\sqrt{204}}.$$

$$\sigma_{\bar{X}} \approx 0.070.$$

$$\boxed{0.070}$$

Q3.20.1 MCQ | 2020 | 2M | Ans: B | ● ○ ○

Total ways of selecting 6 balls from 15 balls:

$${}^{15}C_6.$$

For exactly 4 green balls and 2 blue balls,

$$\text{Favourable ways} = {}^{5}C_4 \times {}^{10}C_2.$$

Hence,

$$P = \frac{{}^{5}C_4 \times {}^{10}C_2}{{}^{15}C_6} = \frac{5 \times 45}{5005} = \frac{225}{5005}.$$

$$P = \frac{45}{1001}.$$

$$\boxed{\text{(b) } \frac{45}{1001}}$$

Q3.19.1 NAT | 2019 | 2M | Ans: 0.25 | ● ○ ○

When two unbiased dice are thrown,

$$\text{Total outcomes} = 6 \times 6 = 36.$$

The sum must be divisible by 4.
Possible sums are

$$4,\ 8,\ 12.$$

Number of outcomes giving sum 4:

$$(1,3), (2,2), (3,1) \quad \Rightarrow \quad 3.$$

Number of outcomes giving sum 8:

$$(2,6), (3,5), (4,4), (5,3), (6,2) \quad \Rightarrow \quad 5.$$

Number of outcomes giving sum 12:

$$(6,6) \quad \Rightarrow \quad 1.$$

Hence,

$$\text{Favourable outcomes} = 3 + 5 + 1 = 9.$$

Therefore,

$$P = \frac{9}{36} = \frac{1}{4} = 0.25.$$

$$\boxed{0.25}$$

Q3.18.1 MCQ | 2018 | 1M | Ans: C | ● ○ ○

Each electronic circuit (EC) has failure probability

$$P(F) = 0.1.$$

Therefore, probability that an EC functions for one year is

$$P(W) = 1 - 0.1 = 0.9.$$

Since both ECs are required for the watch to function and the ECs are independent,

$$P(\text{Watch functions}) = 0.9 \times 0.9 = 0.81.$$

Hence, the probability that the watch functions for one year without failure is

$$0.81.$$

$$\boxed{\text{(c) } 0.81}$$

Q3.17.1 MCQ | 2017 | 1M | Ans: C | ● ○ ○

Given marks:

$$38,\ 84,\ 45,\ 70,\ 75,\ 60,\ 48.$$

The mean is

$$\bar{x} = \frac{38 + 84 + 45 + 70 + 75 + 60 + 48}{7} = \frac{420}{7} = 60.$$

Arrange the data in ascending order:

$$38,\ 45,\ 48,\ 60,\ 70,\ 75,\ 84.$$

Since there are 7 observations, the median is the 4th observation.

$$\text{Median} = 60.$$

Therefore,

$$(\text{Mean}, \text{Median}) = (60, 60).$$

$$\boxed{\text{(c) Mean } = 60,\ \text{Median } = 60}$$

Q3.17.2 NAT | 2017 | 2M | Ans: 0.36 | ● ○ ○

Since the first ball is replaced before drawing the second ball, the two draws are independent.
Probability of drawing a red ball in one draw is

$$P(R) = \frac{6}{10} = 0.6.$$

Therefore,

$$P(\text{both red}) = P(R) \times P(R) = 0.6 \times 0.6 = 0.36.$$

$$\boxed{0.36}$$

Q3.16.1 NAT | 2016 | 2M | Ans: 2.57 | ● ○ ○

Given model

$$y = mx^2.$$

Let

$$X = x^2.$$

Then the model becomes

$$y = mX,$$

which is a linear regression through the origin.
The data are

$$(X, y) = (1, 2),\ (2, 5),\ (3, 8).$$

For regression through the origin,

$$m = \frac{\sum Xy}{\sum X^2}.$$

Now,

$$\sum Xy = (1)(2) + (2)(5) + (3)(8) = 36,$$

and

$$\sum X^2 = 1^2 + 2^2 + 3^2 = 14.$$

Therefore,

$$m = \frac{36}{14} = \frac{18}{7} \approx 2.5714.$$

$$\boxed{2.57}$$

Key Points

- Convert the nonlinear model $y = mx^2$ into the linear form $y = mX$ using $X = x^2$.
- For regression through the origin, $m = \dfrac{\sum XY}{\sum X^2}$.

Q3.14.1 NAT 2014 2M Ans: 7 ● ○ ○

When two fair dice are rolled,

$$\text{Total outcomes} = 6 \times 6 = 36.$$

The number of ways to obtain each sum is

Sum	2	3	4	5	6	7	8	9	10	11	12
Ways	1	2	3	4	5	6	5	4	3	2	1

The maximum number of outcomes is

$$6,$$

which occurs for the sum 7
Hence answer is

$$\boxed{7}$$

Q3.14.2 MCQ 2014 2M Ans: C ● ● ○

Compare each distribution with the normal pdf

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

For

$$f_1(x) = e^{-\pi x^2},$$

the exponent is

$$-\frac{x^2}{2\sigma_1^2} = -\pi x^2.$$

Hence,

$$\sigma_1^2 = \frac{1}{2\pi}, \qquad \mu_1 = 0.$$

For

$$f_2(x) = \frac{1}{2\pi}\exp\left[-\frac{1}{4\pi}(x^2 + 2x + 1)\right],$$

note that

$$x^2 + 2x + 1 = (x + 1)^2.$$

Thus,

$$f_2(x) = \frac{1}{2\pi}\exp\left[-\frac{(x+1)^2}{4\pi}\right].$$

Comparing with the standard form,

$$\mu_2 = -1,$$

and

$$\frac{1}{2\sigma_2^2} = \frac{1}{4\pi} \quad \Rightarrow \quad \sigma_2^2 = 2\pi.$$

Therefore,

$$\mu_1 = 0 > \mu_2 = -1,$$

and

$$\sigma_1^2 = \frac{1}{2\pi} < 2\pi = \sigma_2^2.$$

(c) $\mu_1 > \mu_2$ and $\sigma_1^2 < \sigma_2^2$

Q3.13.1 NAT | 2013 | 1M | Ans: 0.083 | ● ○ ○

For a fair die,

$$P(\text{first roll is 4}) = \frac{1}{6}.$$

Also,

$$P(\text{second roll is less than 4}) = P(\{1, 2, 3\}) = \frac{3}{6} = \frac{1}{2}.$$

Since the two rolls are independent,

$$P = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}.$$

$$P = 0.08333\ldots$$

Up to three digits after the decimal point,

$$\boxed{0.083}$$

Q3.13.2 MCQ 2013 1M Ans: B ● ○ ○

Arrange the data in ascending order:

$$4,\ 9,\ 11,\ 15,\ 18,\ 18$$

Since there are 6 observations, the median is the average of the 3rd and 4th observations.

$$\text{Median} = \frac{11 + 15}{2} = 13.$$

The mode is the value occurring most frequently.

$$18$$

occurs twice, while all other values occur once.
Hence,

$$\text{Mode} = 18.$$

$$\boxed{\text{(b) Median } = 13,\ \text{Mode } = 18}$$

Q3.12.1 MCQ 2012 1M Ans: C ● ○ ○

There are 10 compartments, 6 identical red balls and 2 identical blue balls.
First choose positions for the 6 red balls:

$${}^{10}C_6 = 210.$$

From the remaining 4 compartments, choose positions for the 2 blue balls:

$${}^{4}C_2 = 6.$$

Therefore, the total number of distinct arrangements is

$${}^{10}C_6 \times {}^{4}C_2 = 210 \times 6 = 1260.$$

$$\boxed{\text{(c) } 1260}$$

Q3.10.1 MCQ 2010 2M Ans: B ● ○ ○

Given

$$X \sim \text{Binomial}(n = 5, p = \tfrac{1}{2}),$$

and

$$P(Y = 1) = P(Y = 2) = \frac{1}{2}.$$

Since X and Y are independent,

$$P(X = Y) = P(X = 1, Y = 1) + P(X = 2, Y = 2).$$

Thus,

$$P(X = Y) = P(X = 1)P(Y = 1) + P(X = 2)P(Y = 2).$$

Now,

$$P(X = 1) = {}^5C_1 \left(\frac{1}{2}\right)^5 = \frac{5}{32},$$

and

$$P(X = 2) = {}^5C_2 \left(\frac{1}{2}\right)^5 = \frac{10}{32}.$$

Hence,

$$P(X = Y) = \left(\frac{5}{32} + \frac{10}{32}\right) = \frac{15}{32}.$$

$$\boxed{\text{(b) } \frac{15}{32}}$$

Q3.10.2 MCQ | 2010 | 2M | Ans: B | ● ○ ○

Let E be the event that the ball remaining in the box is red.
A red ball remains only when all 4 removed balls are selected from the other 4 balls consisting of 2 red and 2 black balls.
Total ways of removing 4 balls from 5 balls:

$${}^5C_4 = 5.$$

Favourable ways:

$${}^4C_4 = 1.$$

Hence,

$$P(E) = \frac{1}{5}.$$

Since there are 3 red balls initially, any one of them can be the remaining red ball.
Therefore,

$$P(\text{remaining ball is red}) = 3 \times \frac{1}{5} = \frac{3}{5}.$$

$$\boxed{\text{(b) } \frac{3}{5}}$$

Q3.09.1 MCQ 2009 2M Ans: A ● ○ ○

Let

$$R = \{1, 2, 3, 4, 5\}$$

be the event of obtaining a number less than or equal to 5.
Let

$$S = \{1, 3, 5\}$$

be the event of obtaining an odd number.
Since

$$S \subset R,$$

we have

$$P(S \mid R) = \frac{P(S \cap R)}{P(R)} = \frac{P(S)}{P(R)}.$$

Now,

$$P(S) = \frac{3}{6}, \qquad P(R) = \frac{5}{6}.$$

Hence,

$$P(S \mid R) = \frac{\frac{3}{6}}{\frac{5}{6}} = \frac{3}{5}.$$

Also,

$$P(R \mid S) = \frac{P(R \cap S)}{P(S)} = 1,$$

because every odd outcome on a die is already contained in R.
Thus the true statement is

$$P(R \mid S) = 1.$$

$$\boxed{\text{(a) } P(R \mid S) = 1}$$

Q3.08.1 MCQ 2008 2M Ans: B ● ● ○

Given the Poisson distribution

$$P(r) = \frac{m^r}{r!} e^{-m}, \qquad r = 0, 1, 2, \ldots$$

The first moment about the origin is

$$\mu_1' = E[r] = \sum_{r=0}^{\infty} r\, P(r).$$

Substituting the Poisson pmf,

$$\mu_1' = \sum_{r=1}^{\infty} r \frac{m^r}{r!} e^{-m}.$$

Using

$$\frac{r}{r!} = \frac{1}{(r-1)!},$$

$$\mu_1' = me^{-m} \sum_{r=1}^{\infty} \frac{m^{r-1}}{(r-1)!}.$$

Let

$$k = r - 1.$$

Then

$$\mu_1' = me^{-m} \sum_{k=0}^{\infty} \frac{m^k}{k!}.$$

Using

$$\sum_{k=0}^{\infty} \frac{m^k}{k!} = e^m,$$

we get

$$\mu_1' = me^{-m}e^m = m.$$

(b) m

Q3.06.1 MSQ | 2006 | 2M | Ans: B,C

For one roll of two fair dice,

$$P(\text{sum} = 10) = \frac{3}{36} = \frac{1}{12},$$

since the favourable outcomes are

$$(4,6),\ (5,5),\ (6,4).$$

Hence,

$$P(\text{sum} \neq 10) = 1 - \frac{1}{12} = \frac{11}{12}.$$

The sum 10 should occur exactly once in three independent rolls.
Therefore,

$$P = {}^3C_1 \left(\frac{1}{12}\right)\left(\frac{11}{12}\right)^2.$$

$$P = 3 \times \frac{1}{12} \times \frac{121}{144} = \frac{363}{1728}.$$

Also,

$$\frac{363}{1728} = \frac{121}{576}.$$

Thus both options represent the same probability.

$$\boxed{\text{(b) } \frac{121}{576} \text{ and (c) } \frac{363}{1728}}$$

Q3.06.2 MCQ 2006 2M Ans: D ● ○ ○

Let D denote the event that the selected component is defective.
From the table,

$$P(R) = \frac{500}{1000 + 2500 + 500} = \frac{1}{8}.$$

Also,

$$P(D \mid P) = \frac{5}{1000}, \qquad P(D \mid Q) = \frac{5}{2500}, \qquad P(D \mid R) = \frac{2}{500}.$$

Using Bayes' theorem,

$$P(R \mid D) = \frac{P(D \mid R)P(R)}{P(D)}.$$

Now,

$$P(D) = \frac{5 + 5 + 2}{4000} = \frac{12}{4000}.$$

Therefore,

$$P(R \mid D) = \frac{\frac{2}{500} \times \frac{500}{4000}}{\frac{12}{4000}} = \frac{2}{12} = \frac{1}{6}.$$

$$\boxed{\text{(d) } \frac{1}{6}}$$

Q3.05.1 MCQ 2005 1M Ans: C ● ○ ○

From each bag, one coin is drawn.
Let

$$A = \{\text{coin from first bag is } 1, 2, 3, \text{ or } 4\},$$

and

$$B = \{\text{coin from second bag is } 7, 8, 9, \text{ or } 10\}.$$

Since each bag contains coins numbered 1 to 10,

$$P(A) = \frac{4}{10}, \qquad P(B) = \frac{4}{10}.$$

The required event is

$$(A \cap B) \cup (B \cap A),$$

i.e., one coin is from $\{1, 2, 3, 4\}$ and the other from $\{7, 8, 9, 10\}$.
Hence,

$$P = P(A)P(B) + P(B)P(A) = 2\left(\frac{4}{10}\right)\left(\frac{4}{10}\right).$$

$$P = \frac{32}{100} = \frac{8}{25}.$$

$$\boxed{\text{(c)}\frac{8}{25}}$$

Q3.04.1 MCQ 2004 1M Ans: B ● ○ ○

Let

$$B_1 = \{\text{ball chosen from first box}\}, \qquad B_2 = \{\text{ball chosen from second box}\}.$$

Since one of the two boxes is selected at random,

$$P(B_1) = P(B_2) = \frac{1}{2}.$$

Probability of drawing a red ball from the first box:

$$P(R \mid B_1) = \frac{4}{7}.$$

Probability of drawing a red ball from the second box:

$$P(R \mid B_2) = \frac{5}{7}.$$

Using Bayes' theorem,

$$P(B_1 \mid R) = \frac{P(R \mid B_1)P(B_1)}{P(R)}.$$

Now,

$$P(R) = P(R \mid B_1)P(B_1) + P(R \mid B_2)P(B_2) = \frac{4}{7}\cdot\frac{1}{2} + \frac{5}{7}\cdot\frac{1}{2} = \frac{9}{14}.$$

Therefore,

$$P(B_1 \mid R) = \frac{\frac{4}{7}\cdot\frac{1}{2}}{\frac{9}{14}} = \frac{4}{9}.$$

$$\boxed{\text{(b) } \frac{4}{9}}$$

Q3.02.1 MCQ 2002 1M Ans: D ● ○ ○

Let X be the number of times a six appears in 4 rolls.
Then

$$X \sim \text{Binomial}\left(n = 4, p = \frac{1}{6}\right).$$

The probability that six appears exactly twice is

$$P(X = 2) = {}^4C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^2.$$

Substituting the values,

$$P(X = 2) = 6 \cdot \frac{1}{36} \cdot \frac{25}{36} = \frac{150}{1296}.$$

$$P(X = 2) = \frac{25}{216}.$$

$$\boxed{\text{(d) } \frac{25}{216}}$$

PrepFusion

SOLUTIONS — Probability and Statistics

PI

Production & Industrial Engineering

PrepFusion

Q6.26.1 NAT 2026 2M Ans: 0.80 ● ○ ○

For a Poisson random variable,

$$p(x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

Given the average number of defects,

$$\lambda = 3$$

We need

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$$

Now,

$$P(X = 0) = e^{-3}$$

and

$$P(X = 1) = 3e^{-3}$$

Hence,

$$P(X \geq 2) = 1 - e^{-3}(1 + 3)$$

$$= 1 - 4e^{-3}$$

$$= 1 - 4(0.0498)$$

$$= 0.8008$$

Rounded to two decimal places,

$$\boxed{0.80}$$

Q6.26.2 MCQ 2026 2M Ans: B ● ● ○

Given

$$r(t) = \frac{f(t)}{1 - F(t)}$$

and

$$F(t) = 1 - e^{-at^b}, \qquad a > 0,\ b > 0$$

Differentiate the CDF to obtain the PDF:

$$f(t) = \frac{d}{dt}\left(1 - e^{-at^b}\right)$$

$$f(t) = ab\,t^{b-1}e^{-at^b}$$

Also,

$$1 - F(t) = e^{-at^b}$$

Therefore,

$$r(t) = \frac{ab\,t^{b-1}e^{-at^b}}{e^{-at^b}}$$

$$r(t) = ab\,t^{b-1}$$

$$\boxed{\text{(b) } ab\,t^{b-1}}$$

Key Points

- Hazard rate (failure rate) is defined as $r(t) = \dfrac{f(t)}{1 - F(t)}$.
- $1 - F(t)$ is the reliability (survival) function.
- For the Weibull distribution, $r(t) = ab\,t^{b-1}$.

Mistakes to be avoided

- Do not confuse the survival function $1 - F(t)$ with $F(t)$.

Q6.26.3 NAT | 2026 | 1M | Ans: 0.15 ● ○ ○

The total number of inserts is

$$5 + 10 = 15$$

The probability that all 4 selected inserts are uncoated is

$$P = \frac{\binom{10}{4}}{\binom{15}{4}}$$

$$P = \frac{210}{1365}$$

$$P = 0.1538$$

Rounded to two decimal places,

$$\boxed{0.15}$$

Q6.25.1 MCQ | 2025 | 1M | Ans: B ● ○ ○

Let the events be:

$$R = \text{red}, \quad G = \text{green}, \quad B = \text{blue}$$

Total balls:

$$5 + 7 + 3 = 15$$

Using the law of total probability,

$$P(\text{2nd red}) = P(RR) + P(GR) + P(BR)$$

$$= \frac{5}{15} \cdot \frac{4}{14} + \frac{7}{15} \cdot \frac{5}{14} + \frac{3}{15} \cdot \frac{5}{14}$$

$$= \frac{20 + 35 + 15}{210} = \frac{70}{210} = \frac{1}{3}$$

Alternatively, by symmetry, every draw is equally likely to be red, hence

$$P(\text{2nd red}) = \frac{5}{15} = \frac{1}{3}$$

$$\boxed{\text{(b) } \frac{1}{3}}$$

Q6.24.1 NAT | 2024 | 2M | Ans: 50 | ● ○ ○

The odd-numbered cards are

$$\{1, 3, 5, 7\}$$

Given that the drawn card is odd, the sample space reduces to these 4 cards.
Among them, the cards having number greater than 4 are

$$\{5, 7\}$$

Hence,

$$P(\text{number} > 4 \mid \text{odd}) = \frac{2}{4} = \frac{1}{2}$$

Therefore,

$$\text{Required percentage} = \frac{1}{2} \times 100 = 50$$

$$\boxed{50}$$

Q6.24.2 NAT | 2024 | 1M | Ans: 4 | ● ○ ○

For a valid probability density function,

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

Given

$$f(x) = \begin{cases} \frac{K}{4}, & 0 \le x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Therefore,

$$\int_0^1 \frac{K}{4}\, dx = 1$$

$$\frac{K}{4}(1 - 0) = 1$$

$$K = 4$$

$$\boxed{4}$$

Q6.23.1 MCQ 2023 1M Ans: B ● ○ ○

Let

$$A = \{\text{at least two tails}\}$$

and

$$B = \{\text{all three tails}\}$$

For three unbiased coin tosses, the sample space has

$$2^3 = 8$$

equally likely outcomes.

$$A = \{HTT,\ THT,\ TTH,\ TTT\}$$

Hence,

$$P(A) = \frac{4}{8}$$

and

$$P(B) = \frac{1}{8}$$

Since $B \subset A$,

$$P(B \mid A) = \frac{P(B)}{P(A)} = \frac{1/8}{4/8} = \frac{1}{4}$$

$$\boxed{\text{(b) } \frac{1}{4}}$$

Q6.23.2 MCQ 2023 2M Ans: A ● ○ ○

Two cards are drawn without replacement from a standard deck of 52 cards.

$$P(\text{different suits}) = 1 - P(\text{same suit})$$

After drawing the first card, there are 51 cards left. Among them, 12 cards belong to the same suit as the first card.

$$P(\text{same suit}) = \frac{12}{51}$$

Therefore,

$$P(\text{different suits}) = 1 - \frac{12}{51} = \frac{39}{51} = \frac{13}{17}$$

$$\boxed{\text{(a) } \frac{39}{51}}$$

Q6.22.1 NAT 2022 1M Ans: 0.14 ● ○ ○

For a Poisson arrival process,

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Arrival rate:

$$8 \text{ customers/hour}$$

For 15 minutes,

$$\lambda = 8 \times \frac{15}{60} = 2$$

Probability of no arrival:

$$P(X = 0) = e^{-2}$$

$$P(X = 0) = 0.1353$$

Rounded to two decimal places,

$$\boxed{0.14}$$

Q6.22.2 NAT 2022 1M Ans: 0.70 ● ○ ○

Out of 10 spur gears, 1 is defective and 9 are non-defective.
The probability that all 3 selected gears are non-defective is

$$P = \frac{\binom{9}{3}}{\binom{10}{3}}$$

$$P = \frac{84}{120} = 0.70$$

Therefore,

$$\boxed{0.70}$$

Q6.21.1 NAT 2021 2M Ans: 2 ● ○ ○

Given

$$P(x) = \frac{1}{2^x}, \qquad x = 1, 2, \ldots, \infty$$

The expected value is

$$E[X] = \sum_{x=1}^{\infty} xP(x) = \sum_{x=1}^{\infty} \frac{x}{2^x}$$

Using the standard result,

$$\sum_{x=1}^{\infty} xr^x = \frac{r}{(1-r)^2}, \qquad |r| < 1$$

For $r = \frac{1}{2}$,

$$E[X] = \frac{\frac{1}{2}}{\left(1 - \frac{1}{2}\right)^2} = \frac{\frac{1}{2}}{\frac{1}{4}} = 2$$

$$\boxed{2}$$

Q6.21.2 NAT | 2021 | 2M | Ans: 0.368 | ● ○ ○

For an exponential distribution,

$$F(t) = 1 - e^{-\lambda t}$$

The mean screening time is

$$\frac{1}{\lambda} = 15 \text{ minutes}$$

Hence,

$$\lambda = \frac{1}{15}$$

The passenger misses the flight if the screening time is more than 15 minutes.

$$P(T \geq 15) = 1 - F(15)$$

$$= e^{-\left(\frac{1}{15}\right)(15)}$$

$$= e^{-1}$$

$$= 0.36788$$

Rounded to three decimal places,

$$\boxed{0.368}$$

Q6.20.1 MCQ | 2020 | 1M | Ans: C | ● ○ ○

Given

$$E[X] = 5$$

and

$$\text{Var}(X) = 1$$

Using

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$1 = E[X^2] - 5^2$$

$$E[X^2] = 1 + 25$$

$$E[X^2] = 26$$

$$\boxed{\text{(c) } 26}$$

Q6.20.2 NAT 2020 2M Ans: 3 ● ● ○

For a valid probability density function,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

Given

$$f(x) = \begin{cases} \dfrac{kx^2}{2}, & -1 \le x \le 1 \\ 0, & \text{elsewhere} \end{cases}$$

Hence,

$$\int_{-1}^{1} \frac{kx^2}{2}\,dx = 1$$

$$\frac{k}{2}\left[\frac{x^3}{3}\right]_{-1}^{1} = 1$$

$$\frac{k}{2}\left(\frac{1}{3} - \left(-\frac{1}{3}\right)\right) = 1$$

$$\frac{k}{3} = 1$$

$$k = 3$$

$$\boxed{3}$$

Q6.19.1 NAT 2019 2M Ans: 0.60 ● ○ ○

Total balls in the box:

$$2 + 3 = 5$$

To obtain exactly 1 red and 2 black balls when 3 balls are selected,

$$P = \frac{\binom{2}{1}\binom{3}{2}}{\binom{5}{3}}$$

$$= \frac{2 \times 3}{10}$$

$$= 0.6$$

Therefore,

$$\boxed{0.60}$$

Q6.18.1 NAT 2018 1M Ans: 0.5 ● ○ ○

Given,

$$P(F) = 0.3, \qquad P(G) = 0.4$$

and

$$P(F \cap G) = 0.2$$

Using the addition rule of probability,

$$P(F \cup G) = P(F) + P(G) - P(F \cap G)$$

$$= 0.3 + 0.4 - 0.2$$

$$= 0.5$$

$$\boxed{0.5}$$

Q6.18.2 NAT 2018 1M Ans: 3.5 ● ○ ○

Arrange the weights in ascending order:

$$2,\ 3,\ 3,\ 4,\ 6,\ 7$$

Since the number of observations is even ($n = 6$), the median is the average of the 3rd and 4th observations.

$$\text{Median} = \frac{3+4}{2}$$

$$= 3.5$$

$$\boxed{3.5}$$

Q6.17.1 NAT 2017 2M Ans: 70 ● ○ ○

Out of 10 machines, 2 are defective.
Therefore, the number of non-defective machines is

$$10 - 2 = 8$$

To form a combination of 4 machines without any defective machine, all 4 must be selected from these 8 good machines.

$$\text{Number of combinations} = \binom{8}{4}$$

$$= \frac{8!}{4!\,4!} = 70$$

70

Q6.16.1 MCQ 2016 1M Ans: C ● ○ ○

The event "head does not turn up in any toss" means that all N tosses result in tails.
For a fair coin,

$$P(T) = \frac{1}{2}$$

Since the tosses are independent,

$$P(\text{all tails}) = \left(\frac{1}{2}\right)^N$$

Therefore, the probability that head does not turn up in any of the N tosses is

$$\left(\frac{1}{2}\right)^N$$

$$\boxed{\text{(c)}\ \left(\frac{1}{2}\right)^N}$$

Key Points

- For independent events, probabilities are multiplied.
- "No head" is equivalent to obtaining tails in every toss.

Q6.16.2 MCQ 2016 1M Ans: B ● ○ ○

Given

$$f_X(x) = \frac{1}{\sqrt{8\pi}} e^{-\frac{(x-1)^2}{8}}, \quad -\infty < x < \infty$$

Comparing with the normal PDF,

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

we get

$$\mu = 1, \qquad \sigma^2 = 4$$

Hence,

$$P(X > 1) = \frac{1}{2}$$

due to the symmetry of the normal distribution about its mean.
Therefore,

$$\int_1^{\infty} f_X(x)\, dx = \frac{1}{2}$$

$$\boxed{\text{(b)}\ \frac{1}{2}}$$

Q6.15.1 MCQ 2015 1M Ans: A ● ○ ○

There are

$$5!$$

possible orders in which the 5 different components can be drawn.
Only one of these orders corresponds to the correct assembly sequence.
Therefore,

$$P(\text{correct sequence}) = \frac{1}{5!}$$

Since the product can be assembled in two possible ways,

$$P = \frac{2}{5!}$$

$$\boxed{\text{(a) } \frac{2}{5!}}$$

Q6.14.1 MCQ 2014 2M Ans: C ● ○ ○

The marks distribution is

Marks	Frequency
25	20
30	20
35	40
40	20

Total students:

$$N = 100$$

Mean:

$$\bar{x} = \frac{\sum fx}{\sum f}$$

$$= \frac{25(20) + 30(20) + 35(40) + 40(20)}{100}$$

$$= \frac{3300}{100} = 33$$

Median:
Cumulative frequencies are

$$20,\ 40,\ 80,\ 100$$

The 50th and 51st observations lie in the marks value 35.

$$\text{Median} = 35$$

Mode:
The highest frequency is 40, corresponding to marks 35.

$$\text{Mode} = 35$$

(c) Mean 33, Median 35, Mode 35

Q6.14.2 MCQ 2014 2M Ans: D ● ○ ○

Let us define the primary events: R: Event that it rains on a given day. R': Event that it does not rain. L: Event that the village fair makes a loss. L': Event that the fair does not make a loss.
From the problem statement, we are given: Probability of rain:

$$P(R) = 0.7 \implies P(R') = 0.3$$

Probability of a loss given it rains:

$$P(L|R) = 0.8 \implies P(L'|R) = 0.2$$

Probability of a loss given it does not rain:

$$P(L|R') = 0.1 \implies P(L'|R') = 0.9$$

We must find the probability that it has not rained given that the fair did not make a loss, which is denoted as $P(R'|L')$.
Using Bayes' Theorem:

$$P(R'|L') = \frac{P(L'|R')P(R')}{P(L')}$$

First, calculate the total probability of the fair not making a loss, $P(L')$, using the Law of Total Probability:

$$P(L') = P(L'|R)P(R) + P(L'|R')P(R')$$

$$P(L') = (0.2 \times 0.7) + (0.9 \times 0.3)$$

$$P(L') = 0.14 + 0.27 = 0.41$$

Substitute $P(L')$ back into Bayes' equation:

$$P(R'|L') = \frac{0.9 \times 0.3}{0.41} = \frac{0.27}{0.41} = \frac{27}{41}$$

(d) $\frac{27}{41}$

Q6.13.1 MCQ 2013 1M Ans: C ● ○ ○

Arrival rate:

$$\lambda = 50 \text{ customers/hr}$$

Average service time:

$$\frac{1}{\mu} = 1 \text{ min}$$

$$\mu = 60 \text{ customers/hr}$$

For an $M/M/1$ queue,

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)}$$

$$= \frac{50}{60(60-50)}$$

$$= \frac{50}{600} = \frac{1}{12} \text{ hr}$$

$$= \frac{60}{12} = 5 \text{ min}$$

(c) 5

Q6.13.2 MCQ 2013 1M Ans: B ● ○ ○

Given

$$X \sim N(1, 4)$$

Hence,

$$\mu = 1, \qquad \sigma = 2$$

We need

$$P(X < 0)$$

Standardizing,

$$Z = \frac{X - \mu}{\sigma}$$

$$P(X < 0) = P\left(Z < \frac{0-1}{2}\right) = P(Z < -0.5)$$

Since -0.5 lies to the left of the mean,

$$0 < P(Z < -0.5) < 0.5$$

Numerically,

$$P(Z < -0.5) \approx 0.3085$$

Therefore,

(b) greater than zero and less than 0.5

Q6.13.3 MCQ 2013 2M Ans: D ● ○ ○

Let

$$K = \text{student knows the answer}$$

$$C = \text{answer is correct}$$

Given,

$$P(K) = \frac{2}{3}$$

$$P(C|K) = 1$$

$$P(C|\overline{K}) = \frac{1}{4}$$

Using total probability,

$$P(C) = P(C|K)P(K) + P(C|\overline{K})P(\overline{K})$$

$$= 1 \cdot \frac{2}{3} + \frac{1}{4} \cdot \frac{1}{3}$$

$$= \frac{8+1}{12} = \frac{3}{4}$$

Applying Bayes' theorem,

$$P(K|C) = \frac{P(C|K)P(K)}{P(C)}$$

$$= \frac{1 \cdot \frac{2}{3}}{\frac{3}{4}}$$

$$= \frac{8}{9}$$

(d) $\frac{8}{9}$

Q6.13.4 MCQ 2013 1M Ans: C ● ○ ○

Total toys manufactured per day:

$$1000$$

Defective toys:

$$10\% \times 1000 = 100$$

Out of these defective toys, 40% are reworked into defect-free toys.

$$0.4 \times 100 = 40$$

Initially defect-free toys:

$$1000 - 100 = 900$$

Hence, total defect-free toys after rework are

$$900 + 40 = 940$$

(c) 940

Q6.12.1 MCQ | 2012 | 2M | Ans: D ● ○ ○

The total number of ways of selecting 3 balls from 10 balls is

$$^{10}C_3 = 120$$

For the selected set to contain exactly one red ball and two black balls:

$$^{4}C_1 \times {}^{6}C_2 = 4 \times 15 = 60$$

Hence,

$$P = \frac{^{4}C_1\,{}^{6}C_2}{^{10}C_3} = \frac{60}{120} = \frac{1}{2}$$

$$\boxed{\text{(d) } \frac{1}{2}}$$

Q6.11.1 MCQ | 2011 | 1M | Ans: B ● ○ ○

The average number of events per year is

$$\lambda = 3$$

For a duration of two years,

$$\lambda_2 = 2 \times 3 = 6$$

If X denotes the number of events in two years, then

$$P(X = k) = \frac{e^{-6}6^k}{k!}$$

Required probability:

$$P(X \le 2) = P(0) + P(1) + P(2)$$

$$= e^{-6}\left(1 + 6 + \frac{6^2}{2!}\right)$$

$$= e^{-6}(1 + 6 + 18) = 25e^{-6}$$

$$P(X \le 2) \approx 25(0.002478) \approx 0.062$$

$$\boxed{\text{(b) } 0.062}$$

Q6.10.1 MCQ | 2010 | 1M | Ans: D ● ○ ○

Given that X follows a Poisson distribution with mean

$$\lambda = 2$$

The required probability is

$$P(X \ge 2) = 1 - P(X < 2)$$

$$= 1 - \left[P(X = 0) + P(X = 1)\right]$$

Using the Poisson formula,

$$P(X = 0) = e^{-2}$$

and

$$P(X = 1) = \frac{e^{-2}2^1}{1!} = 2e^{-2}$$

Therefore,

$$P(X \geq 2) = 1 - \left(e^{-2} + 2e^{-2}\right)$$

$$= 1 - 3e^{-2}$$

$$\boxed{\text{(d)}1 - 3e^{-2}}$$

Q6.10.2 MCQ | 2010 | 2M | Ans: C ● ● ○

Choose a bin randomly and then pick one ball randomly from the chosen bin. Hence,

$$P(\text{white}) = \frac{1}{2}P(\text{white} \mid \text{Bin 1}) + \frac{1}{2}P(\text{white} \mid \text{Bin 2})$$

Evaluate the probability for each arrangement.

$$\textbf{A:}\quad \frac{1}{2}\left(\frac{1}{2}\right) + \frac{1}{2}\left(\frac{1}{2}\right) = \frac{1}{2}$$

$$\textbf{B:}\quad \frac{1}{2}(1) + \frac{1}{2}(0) = \frac{1}{2}$$

$$\textbf{C:}\quad \frac{1}{2}(1) + \frac{1}{2}\left(\frac{1}{3}\right) = \frac{2}{3}$$

$$\textbf{D:}\quad \frac{1}{2}(0) + \frac{1}{2}\left(\frac{2}{3}\right) = \frac{1}{3}$$

Since

$$\frac{2}{3} > \frac{1}{2} > \frac{1}{3},$$

Arrangement **C** gives the highest probability of selecting a white ball.

$$\boxed{\text{Option C}}$$

Key Points

- Use the law of total probability when the bin is selected randomly.
- Maximize the overall probability by comparing the probability of drawing a white ball for each arrangement.

Mistakes to be avoided

- Do not ignore that each bin is selected with probability $\frac{1}{2}$.
- Do not compare only the contents of one bin; evaluate the total probability for both bins.

Q6.09.1 MCQ 2009 2M Ans: B ● ○ ○

From the given joint PMF table,

$$P(X = 2, Y = 2) = 0.1$$

The marginal probability of $Y = 2$ is

$$P(Y = 2) = P(X = 1, Y = 2) + P(X = 2, Y = 2)$$

$$= 0.3 + 0.1 = 0.4$$

Therefore,

$$P(X = 2 \mid Y = 2) = \frac{P(X = 2, Y = 2)}{P(Y = 2)} = \frac{0.1}{0.4} = 0.25$$

(b) 0.25

Q6.08.1 MCQ 2008 1M Ans: A ● ○ ○

Since the normal distribution is symmetric about its mean $\mu = 100$,

$$P(X \geq 110) = \alpha$$

Hence,

$$P(X \leq 90) = \alpha$$

The required probability is

$$P(90 \leq X \leq 110) = 1 - P(X < 90) - P(X > 110)$$

$$= 1 - \alpha - \alpha$$

$$= 1 - 2\alpha$$

(a) $1 - 2\alpha$

Q6.08.2 MCQ 2008 2M Ans: C ● ○ ○

Let P_X be the probability that player X wins.
Player X can win on the first toss or after both players miss once:

$$P_X = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} P_X$$

$$P_X = \frac{1}{2} + \frac{1}{4} P_X$$

$$\frac{3}{4} P_X = \frac{1}{2}$$

$$P_X = \frac{2}{3}$$

$$\boxed{(c)\ \frac{2}{3}}$$

Q6.08.3 MCQ 2008 2M Ans: B ● ● ○

Arrival rate:

$$\lambda = 30 \text{ customers/hour}$$

Mean service time:

$$\frac{1}{\mu} = 100\,\text{s} = \frac{100}{3600} \text{ hour}$$

$$\mu = \frac{3600}{100} = 36 \text{ customers/hour}$$

For an single server queue model $(M/M/1)$, the average time spent in the system is

$$W = \frac{1}{\mu - \lambda}$$

$$W = \frac{1}{36 - 30} = \frac{1}{6}$$

$$W = 0.1667 \text{ hour}$$

$$\boxed{(b)\ 0.166}$$

Key Points

- For an sigle server queue model i.e $M/M/1$ queue:

$$W = \frac{1}{\mu - \lambda}$$

- Service and arrival rates must be expressed in the same units.

Q6.07.1 MCQ 2007 1M Ans: A ● ○ ○

With replacement, the two draws are independent.
Probability of drawing an ace in one draw:

$$P(\text{Ace}) = \frac{4}{52} = \frac{1}{13}$$

Therefore,

$$P(\text{both aces}) = \frac{1}{13} \times \frac{1}{13}$$

$$= \frac{1}{169}$$

$$\boxed{\text{(a) } \frac{1}{169}}$$

Q6.07.2 MCQ 2007 2M Ans: D ● ○ ○

First determine the constant K using the normalization condition:

$$\int_0^2 K(5x - 2x^2)\,dx = 1$$

$$K\left[\frac{5x^2}{2} - \frac{2x^3}{3}\right]_0^2 = 1$$

$$K\left(10 - \frac{16}{3}\right) = 1$$

$$K \cdot \frac{14}{3} = 1$$

$$K = \frac{3}{14}$$

Now,

$$P(X > 1) = \int_1^2 \frac{3}{14}(5x - 2x^2)\,dx$$

$$= \frac{3}{14}\left[\frac{5x^2}{2} - \frac{2x^3}{3}\right]_1^2$$

$$= \frac{3}{14}\left(\frac{14}{3} - \frac{11}{6}\right) = \frac{17}{28}$$

$$\boxed{\text{(d) } \frac{17}{28}}$$

Q6.07.3 MCQ 2007 2M Ans: A ● ○ ○

Since probabilities must sum to unity,

$$\frac{(2 + 5P) + (1 + 3P) + (1.5 + 2P)}{5} = 1$$

$$4.5 + 10P = 5$$

$$P = 0.05$$

The probabilities are

$$P(X = 1) = 0.45, \quad P(X = 2) = 0.23, \quad P(X = 3) = 0.32$$

Hence,

$$E[X] = 1(0.45) + 2(0.23) + 3(0.32)$$

$$= 0.45 + 0.46 + 0.96 = 1.87$$

(a) 0.05, 1.87

Q6.07.4 MCQ 2007 2M Ans: B ● ○ ○

For a Bernoulli random variable,

$$E[X_i] = p = 0.6$$

$$\text{Var}(X_i) = p(1-p) = 0.6 \times 0.4 = 0.24$$

Given

$$Y = \sum_{i=1}^{100} X_i$$

Mean of Y:

$$E[Y] = \sum_{i=1}^{100} E[X_i] = 100(0.6) = 60$$

Variance of Y:

$$\text{Var}(Y) = \sum_{i=1}^{100} \text{Var}(X_i) = 100(0.24) = 24$$

(b) 60 and 24

Q6.07.5 MCQ 2007 2M Ans: C ● ○ ○

Let X denote the number of accidents in a month.
Since accidents follow a Poisson distribution with mean

$$\lambda = 2$$

the probability of at least one accident is

$$P(X \geq 1) = 1 - P(X = 0)$$

Using the Poisson PMF,

$$P(X = 0) = e^{-2}$$

Therefore,

$$P(X \geq 1) = 1 - e^{-2}$$

$$= 1 - 0.1353$$

$$\approx 0.865$$

(c) 0.865

Q6.05.1 MCQ 2005 1M Ans: D ● ○ ○

When two dice are thrown, the total number of possible outcomes is

$$6 \times 6 = 36$$

The sum exceeds 8 for the following cases:

$$9,\ 10,\ 11,\ 12$$

Number of favorable outcomes:

Sum 9 : 4
Sum 10 : 3
Sum 11 : 2
Sum 12 : 1

Total favorable outcomes = 4 + 3 + 2 + 1 = 10

Therefore,

$$P(\text{sum} > 8) = \frac{10}{36} = \frac{5}{18}$$

(d) $\frac{10}{36}$

Q6.05.2 MCQ 2005 2M Ans: A ● ○ ○

Given the exponential PDF

$$f(t) = \alpha e^{-\alpha t}, \qquad 0 \le t < \infty$$

The required probability is

$$P(100 \le t \le 200) = \int_{100}^{200} \alpha e^{-\alpha t}\, dt$$

$$= \left[-e^{-\alpha t}\right]_{100}^{200}$$

$$= -e^{-200\alpha} + e^{-100\alpha}$$

$$= e^{-100\alpha} - e^{-200\alpha}$$

(a) $e^{-100\alpha} - e^{-200\alpha}$

Also in this Series

Every PrepFusion GATE title is built the same way: the real previous-year papers, nothing paraphrased or reconstructed, arranged so the pattern the exam repeats is visible on the page. Each questions volume has a companion solutions volume that reuses the same question identifiers, so you can move between the two without a lookup table.

NexusX	Network Theory, Analog Electronics, Digital Electronics, Control Systems and Signals & Systems — EC, EE, IN
SiliconX	Electronic Devices, Electromagnetic Field Theory and Communication Systems — EC
PowerX	Electrical Machines, Power Systems, Power Electronics, Measurements and Electromagnetic Field Theory — EE, IN
Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
General Aptitude	Verbal and quantitative aptitude, arranged year-wise and session-wise — all branches

Spotted a wrong answer key, a typo or a question we have mis-tagged? Report it on the PYQ portal at `pyq.prepfusion.in` — corrections go into the next printing, and you can request a video solution for any question from the same place.

PrepFusion